

Supplementary Information

Title: A greener and more scalable synthesis of biogenic polydisulfides from lipoic acid

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Table of Contents

Materials and Methods.....	S2
Synthetic Procedures.....	S3-S7
NMR Spectra for monomer and polymer.....	S9-S12
Catalyst Screening.....	S13-S20
Solvent Screening.....	S21-S36
ROP Thermodynamics.....	S37-S59
Thermomechanical Data.....	S60-S65
RCD.....	S66-S72
Life Cycle Analysis.....	S73-S80
One-pot ROP.....	S81-S82
Reduced Pressure ROP.....	S83-S84
One-pot RCD to ROP.....	S85-S86
Controlled CROP Experiments.....	S87-S92
Photographs.....	S93-S97

Materials and Methods

Materials

All compounds, unless otherwise indicated, were purchased from commercial sources and used as received. Lipoic acid was obtained from Bulk Supplements. Carbon tetrachloride (CCl₄) was washed with an aqueous workup and distilled before use to remove trace impurities.

Methods

Unless otherwise mentioned all experiments were performed in an ambient atmosphere using reagents and solvents as received with no prior purification or drying.

NMR spectroscopy experiments were performed at 298 K on a Agilent U4-DD2 operating at 400 MHz for ¹H-NMR. ¹³C-NMR were performed at 298 K on a Bruker Avance II operating at 100.57 MHz. spectra are referenced to residual protic solvents (CHCl₃ at δ = 7.26) and ¹³C NMR spectra are referenced to the residual solvent signal (CDCl₃ at δ = 77.16 ppm). The resonance multiplicities are described as s (singlet), d (doublet), t (triplet), q (quartet) or m (multiplet).

Gel permeation Chromatography (GPC) analyses were performed in THF on an Agilent 1260 Infinity II Multi-Detector GPC/SEC System fitted with RI, ultraviolet (UV, λ = 309 nm), and viscometer detectors. The polymers were eluted through an Agilent guard column (PLGel 5 μ M, 50 $\times$ 7.5 mm) and two Agilent mixed-C columns (PLGel 5 μ M, 300 $\times$ 7.5 mm) using THF (buffered with 200ppm BHT) as the mobile phase (flow rate = 1 mL·min⁻¹, 40 °C). Number average molecular weight (M_n), weight average molecular weight (M_w) and dispersity ($D_M = M_w/M_n$) were determined using Agilent GPC/SEC software (vA.02.01) against a 12-point calibration curve ($M_p = 162 - 6085000 \text{ g}\cdot\text{mol}^{-1}$) based on polystyrene standards (Easivial PS-M/H, Agilent).

Preparation of polymer thin films via hot-pressing. Thin films of polymer were fabricated using a heated press (Specac, UK). The instrument was first preheated to 55 °C. The polymer was then added into a stainless-steel mold (50 $\times$ 50 $\times$ 0.5 mm, L $\times$ W $\times$ thickness) between two sheets of

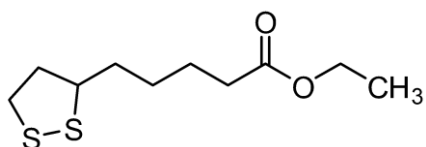
teflon and placed on the press. After 15 min of heating without pressure, the sample was then degassed (10×) to remove air bubbles. The sample was subsequently pressed for 20 min at 2 tons followed by 10 min at 10 tons. The sample was then cooled to ambient temperature at ~ 2 tons of pressure to prevent wrinkling of the film surface.

Dynamic mechanical thermal analysis (DMTA) data were obtained using a TA[®] instruments DMA 850. Film samples were cut into rectangular samples (L x W x thickness = 9.5 ± 1.0 mm x 6 mm x 1.5 mm) cooled to -90 °C and held isothermally for ca. 5 min. Storage and loss moduli, as well as the loss factor (ratio of E'' and E', tan δ) were probed as the temperature was ramped from -90 to 100 °C. Experiments were performed at 5 °C min⁻¹ ramp rate, 1 Hz strain rate and initial static load of 0.1 N.

Synthetic procedures

Synthesis of EtLp monomer

Synthesis of EtLp using Steglich coupling



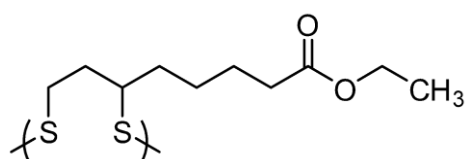
A 250 mL round bottom flask was filled with dichloromethane (DCM, 50 mL), lipoic acid (10 g, 48.5 mmol, 1.0 equiv.), 4-(N,N-dimethylamino)pyridine (DMAP, 5.9 g, 48 mmol, 1.0 equiv.), and ethanol (6.7 g, 145 mmol, 3.0 equiv.). The reaction mixture was then stirred until homogeneous. Then the flask was cooled in an ice-bath and *N*-(3-dimethylamino-propyl)-*N'*-ethylcarbodiimide hydrochloride (EDC, 9.3 g, 48mmol, 1.0 equiv.) was added in portions over several minutes. After the addition of EDC, the reaction mixture was stirred overnight (*ca.* 16 h) slowly warming to ambient. The reaction mixture was transferred to a separatory funnel and the organic layer was washed with 1 M HCl (3 × 25 mL), 1 M NaHCO₃ (3 × 25 mL), and brine (1 × 25 mL). The organic phase was separated and dried with 4Å molecular sieves overnight. The anhydrous organic phase was then gravity filtered and concentrated in vacuo until most of the DCM was removed affording the product (EtLp) (yield = 8.92 g, 78.6%). The product was stored in an amber bottle to protect from light and dried on the bench top for several days before characterization. ¹H-NMR (400 MHz, CDCl₃) δ 4.09 (q, J = 7.1 Hz, 2H), 3.53 (dq, J = 8.4, 6.4 Hz, 1H), 3.38 – 2.85 (m, 2H), 2.42 (dtd, J = 13.0, 6.6, 5.4 Hz, 1H), 2.27 (t, J = 7.4 Hz, 2H), 1.87 (dq, J = 12.7, 7.0 Hz, 1H), 1.77 – 1.54 (m, 4H), 1.54 – 1.31 (m, 2H), 1.22 (t, J = 7.1 Hz, 3H). ¹³C-NMR (CDCl₃, 100.57 MHz), δ =173.38, 60.19, 56.28, 40.16, 38.45, 34.54, 34.03, 28.69, 24.63, 14.23.

Synthesis of EtLp using Fischer esterification

A 1 L two-neck round bottom flask was charged with lipoic acid (100 g, 484.7 mmol, 1.0 equiv.) and ethanol (432.9 g, 548.6 mL, 9395 mmol, 19.5 equiv.) and stirred until homogenous. A condensing column was then attached to the central joint, and the other joint was closed using a cap. The reaction was placed in an oil bath heated to 90 °C. Once the solution started to reflux a catalytic amount of sulfuric acid (0.5 g, 0.3 mL, 5.3 mmol, 0.011 equiv.) was added through the side joint, then the joint was resealed. The reaction refluxed overnight (*ca.* 16 h). Sodium bicarbonate was then carefully added (1 g, 11.9 mmol, 0.025 equiv.) to the reaction mixture while in the bath at 90 °C. The mixture was stirred for 1 h, then the oil bath was turned off and the solution cooled to approximately 40 °C. The reaction mixture was then filtered through cotton wool into a 1 L round bottom flask and concentrated in vacuo. Ethanol was distilled until salts precipitated from solution. The vacuum was then turned off and the concentrated solution was

diluted in ethyl acetate (450.2 g, 499 mL, 5109 mmol, 10.5 equiv.). This dilute solution was filtered through cotton wool and transferred to a 1 L separatory funnel. The organic phase was washed with 1 M sodium bicarbonate solution (2 × 50 mL), DI water (1 × 50 mL) and brine (1 × 50 mL). The organic phase was then collected and dried over 4Å molecular sieves overnight. The solution was gravity filtered into a 1 L round bottom flask, additional ethyl acetate (30 mL) was used to wash the filter funnel and achieve quantitative transfer. The solution was then concentrated in vacuo, transferred into an amber jar covered in aluminum foil, and then dried overnight on the benchtop to afford the product EtLp (yield = 85.11 g, 75.02%). The monomer was subsequently stored at 4 °C after characterization to prevent accidental polymerization. ¹H-NMR (400 MHz, CDCl₃) δ 4.09 (q, J = 7.1 Hz, 2H), 3.53 (dq, J = 8.4, 6.4 Hz, 1H), 3.38 – 2.85 (m, 2H), 2.42 (dtd, J = 13.0, 6.6, 5.4 Hz, 1H), 2.27 (t, J = 7.4 Hz, 2H), 1.87 (dq, J = 12.7, 7.0 Hz, 1H), 1.77 – 1.54 (m, 4H), 1.54 – 1.31 (m, 2H), 1.22 (t, J = 7.1 Hz, 3H).. ¹³C-NMR (CDCl₃, 100.57 MHz), δ = 173.38, 60.19, 56.28, 40.16, 38.45, 34.54, 34.03, 28.69, 24.63, 14.23.

General method for the synthesis of polyEtLp by Cationic Ring Opening Polymerization (CROP)



EtLp monomer was dissolved in solvent (acetone) to a known concentration (0.5–4.0 M) and stirred until homogenous. The monomer solution was placed in a temperature-controlled bath (5–40 °C). Cationic initiator (1 –10 mol%) was dissolved in a known amount of solvent and stirred until homogenous. The initiator was placed in the same temperature-controlled bath. Once both solutions reached thermal equilibrium, the initiator solution was added to the monomer solution and vortexed until homogeneous. Then the monomer solution was immediately added back into the temperature-controlled bath. The initial concentration of monomer ([EtLp]₀) was based on the final volume of solvent (including solvent from the initiator solution). After the desired amount of time, a small amount of polymer gel was placed in a vial and diluted with CDCl₃ containing a small quantity of NEt₃ (pH ~ 8) for ¹H-NMR analysis to assess monomer conversion. Another sample of gel was diluted in THF containing a small quantity of NEt₃ (pH ~ 8) for a final polymer concentration of ~10 mg mL⁻¹. The crude polymer was then purified by stirring in a large excess (~10×) of diethyl ether to remove monomer and initiator. Multiple washes with ether were used to ensure that all monomer and cationic initiator were removed. The crude polymer was dried on the benchtop overnight and then dried in a vacuum oven at 30 °C overnight to obtain the final polymer. ¹H-NMR (400 MHz, CDCl₃, 25 °C) δ 4.13 (s, 2H), 2.77 (s, 3H), 2.32 (s, 2H), 1.96 (s, 2H), 1.66 (s, 8H), 1.26 (s, 3H).

The ether fractions that were used to wash the crude polymer were also collected and placed in a separatory funnel. The organic phase was washed with 1 M sodium bicarbonate solution (50 mL × 2), DI water (50 mL × 1) and brine (50 mL × 1) before drying overnight with 4Å molecular sieves. The organic phase was then filtered and solution concentrated in vacuo to recover unreacted EtLp.

General Method for CROP Initiator Screening

EtLp monomer (1 g, 4.272 mmol, 1.0 equiv) was placed in a 6-dram vial and dissolved in DCM (0.535 mL, 0.712 g, 8.378 mmol, 1.96 equiv.). The monomer solution was placed in an oil bath set to 30 °C. The cationic initiator (0.05 equiv.) was placed in a 2-dram vial and dissolved in DCM (0.535 mL, 0.712 g, 8.378 mmol, 1.96 equiv.). The initiator solution was also placed in an oil bath set to 30 °C. The solutions were both equilibrated for 30 min to reach thermal equilibrium. Then the initiator solution was quickly added to the monomer solution and subsequently vortexed until

homogenous. The reaction mixture was immediately capped and added back into the oil bath. At various time points, a small amount of polymer gel was placed in a vial and diluted with CDCl_3 containing a small quantity of NEt_3 ($\text{pH} \sim 8$) for $^1\text{H-NMR}$ analysis to assess conversion.

General Method for CROP Solvent Screening with DPP

EtLp monomer (2.000 g, 8.544 mmol, 1.0 equiv.) was weighed using a 4-digit analytical balance and placed in a 6-dram vial. The vial was then placed on metal heating block at $30\text{ }^\circ\text{C}$. In a separate 2-dram vial, solvent was weighed using known densities to obtain a volume of 2.136 mL. Then, diphenyl phosphate (DPP) (0.1068 g, 0.427 mmol, 0.05 equiv.) was weighed using a 4-digit analytical balance and added to the solvent vial. The initiator solution was heated to $30\text{ }^\circ\text{C}$ in an oven. Both vials were equilibrated for 30 min to reach thermal equilibrium. Then, the initiator solution (DPP) was quickly added to the monomer solution and vortexed until homogenous. The reaction mixture was then immediately resealed and placed back on the heating block. At various time points, a small amount of polymer gel was placed in a vial and diluted with CDCl_3 containing a small quantity of NEt_3 ($\text{pH} \sim 8$) for $^1\text{H-NMR}$ analysis to assess conversion. Samples were measured in triplicate (3 independent samples) for each time point.

General Method for CROP Monomer Concentration Screening with DPP

EtLp monomer (100.0 mg, 0.427 mmol, 1.0 equiv.) was weighed using a 4-digit analytical balance and placed into a 2-dram vial. The vial was placed on a metal heating block at $30\text{ }^\circ\text{C}$ and equilibrated for 30 min. A separate initiator stock solution was prepared by dissolving DPP (1.1008 g, 4.400 mmol) in 22 mL of acetone ($[\text{DPP}] = 200\text{ mM}$). The vial was immediately sealed using a rubber septum. The stock solution was heated to the desired temperature on the heating block and equilibrated for 30 min. Stock solution (106.8 μL , 0.05 equiv. DPP) was then added to the vial containing monomer ($[\text{EtLp}]_0 = 4.00\text{ M}$) and vortexed until homogenous. *At this stage, additional solvent was added to achieve desired concentration for different experiments, i.e. $[\text{EtLp}]_0 = 0.5\text{ M}$, 1.00 M , 2.00 M .* The reaction mixture was then immediately resealed and placed back on the heating block. At each time point, a small amount of polymer gel was extracted and placed in a vial and diluted with CDCl_3 containing a small quantity of NEt_3 ($[\text{NEt}_3] = 28\text{ mM}$) for $^1\text{H-NMR}$ analysis to assess conversion. Samples were measured in triplicate (3 independent samples) for each time point.

General Method for CROP with Various Amounts of DPP

EtLp monomer (100.0 mg, 0.427 mmol, 1.0 equiv.) was weighed using a 4-digit analytical balance. The monomer was placed on a heating block at $30\text{ }^\circ\text{C}$ for 30 minutes to reach equilibrium. In a 6-dram vial, an initiator stock solution was made by adding DPP (0.2202 g, 0.8800 mmol for 0.01 equiv; 1.1008 g, 4.400 mmol for 0.05 equiv; 2.2016 g, 8.800 mmol for 0.10 equiv.) and 22 mL of acetone ($[\text{DPP}] = 200\text{ mM}$). The vial was sealed with a rubber septum and heated to $30\text{ }^\circ\text{C}$ for 30 min to reach thermal equilibrium. 106.8 μL of the stock solution was added to the EtLp monomer vial (0.01, 0.05 or 0.10 equiv. DPP respectively, $[\text{EtLp}]_0 = 4.00\text{ M}$). The solution was quickly vortexed until homogeneous, sealed, and then returned to the temperature-controlled block. At each time point, the reaction was sacrificed for analysis by adding 1 mL of $\text{CDCl}_3/\text{NEt}_3$ stock solution ($[\text{NEt}_3] = 5.6$ for 0.01 equiv., 28 mM for 0.05 equiv.), 56 mM for 0.10 equiv.) to the polymer gel for $^1\text{H-NMR}$ analysis to assess conversion. Samples were measured in triplicate (3 independent samples) for each time point.

General Method for CROP Equilibrium Screening with DPP

EtLp monomer (100.0 mg, 0.427 mmol, 1.0 equiv.) was weighed using a 4-digit analytical balance and placed into a 2-dram vial. The monomer was then placed in a temperature-controlled bath for experiments below ambient temperature, or a heating block for temperatures above ambient temperature. An initiator stock solution was prepared by placing DPP (1.1008 g, 4.400 mmol) and 22 mL of acetone ([DPP] = 200 mM) in a septum-sealed vial. The stock solution was cooled or heated to the desired temperature and equilibrated for 30 min to reach thermal equilibrium. 106.8 μ L of stock solution was added to the EtLp monomer vial (0.05 equiv. DPP, [EtLp]₀ = 4.00 M). The solution was quickly vortexed until homogeneous, sealed, and then returned to the temperature-controlled bath or block. After 24 h, 1 mL of CDCl₃/NEt₃ stock solution ([NEt₃] = 28 mM) was added to the polymer gel for ¹H-NMR analysis to assess conversion. Twelve independent samples were analyzed for each isotherm, outliers ≥ 5 % error were removed to obtain a more accurate value. Conversion data was used to obtain equilibrium constants.

General Method for CROP Control Experiments

EtLp (5.1835 g, 0.02 mol) and acetone (4 mL) were placed in a 6-dram vial. In a separate 6-dram vial, DPP (277 mg, 5 mol%) was dissolved in acetone (1.5 mL). Both solutions were stored over 3Å molecular sieves overnight. The EtLp solution was then transferred into an oven-dried 50 mL 3-neck round bottom flask and was purged with nitrogen gas for 10 min. The flask was heated to 30 °C in an oil bath and equilibrated for 30 min. The DPP solution was added to the reaction flask using a syringe while the monomer solution was under vigorous stirring. After 24 h, the flask was cooled to ambient temperature and opened to atmosphere. A small amount of polymer gel was placed in a vial and diluted with CDCl₃ containing a small quantity of NEt₃ (pH ~ 8) for ¹H-NMR analysis. Another sample of gel was diluted in THF containing a small quantity of NEt₃ (pH ~ 8) for a final polymer concentration of ~10 mg mL⁻¹. The crude polymer was purified by stirring in a large excess (~10 $\times$) of diethyl ether to remove monomer and initiator. Multiple washes with ether were used to ensure that all monomer and cationic initiator were removed. The crude polymer was then dried on the bench top overnight before drying in a vacuum oven at 30 °C overnight to obtain the final polymer which was assessed by ¹H-NMR and GPC.

Method for the Cationic Ring Closing Depolymerization (RCD) of polyEtLp

PolyEtLp (5 g, 21.36 mmol, 1.0 equiv.) and chloroform (400 mL, 593 g, 4969, 232 equiv.) ([polyEtLp]₀ = 60 mM) were added to a 1 L 2-neck round bottom flask. A condensing column was added to the center joint and the side joint was sealed with a septum. The round bottom flask was placed in an oil bath at 75 °C. Once the solution started to reflux a catalytic amount of sulfuric acid (20 mg, 0.01 mL, 0.21 mmol, 0.01 equiv.) was added through the side joint and then the joint was resealed. The reaction was refluxed for 2 h. While the reaction flask was still in the bath at 75 °C, 1 M sodium bicarbonate solution (50 mL) was carefully added. The joint was resealed, and the reaction flask was stirred vigorously to ensure adequate mixing between the organic and aqueous phases. After 1 h, the mixture was cooled and transferred to a separatory funnel. The aqueous layer was removed, and the organic phase was subsequently washed with DI water (3 $\times$ 50 mL) and brine (1 $\times$ 50 mL). The organic phase was collected and dried over 4Å molecular sieves overnight. The solution was then gravity filtered into a 1 L round bottom flask; additional chloroform was used to achieve quantitative transfer. The solution was concentrated in vacuo,

transferred to a 6-dram vial covered in aluminum foil and then dried overnight on the benchtop to afford the regenerated monomer (yield = 4.199 g, 83.9%).

Method for the One-Pot Synthesis of polyEtLp

Lipoic acid (25g, 121.175 mmol, 1.0 equiv.) and ethanol (468 g, 600 mL, 10158.9 mmol, 83.83 equiv.) were added to a 1 L 2-neck round bottom flask and the mixture stirred until homogenous. A condensing column was added to the center joint and the side joint was sealed with a septum. The round bottom flask was placed in an oil bath at 90 °C. Once the solution started to reflux a catalytic amount of sulfuric acid (20 mg, 0.01 mL, 0.21 mmol, 0.01 equiv.) was added through the side joint and then the joint was resealed. The reaction was refluxed overnight (ca. 16 h). The oil bath was then heated to 100 °C to distill ethanol (398.45 g, 505 mL, 8640 mmol). The oil bath was turned off and the solution was cooled to ~30 °C while stirring. NEt₃ (1.0122 g, 1.39 mL, 10 mmol, 0.08 equiv) was then added. The solution was then filtered to separate the crude polymer and recover the ethanol and unreacted ethyl lipoate. The crude polymer was characterized via ¹H NMR (to assess monomer conversion) and GPC and then purified by stirring in a large excess (~10×) of diethyl ether to remove monomer and initiator. The sample was dried in a vacuum oven at 30 °C overnight to obtain the final polymer (yield = 9.49 g, 33%) which was characterized by ¹H-NMR and GPC.

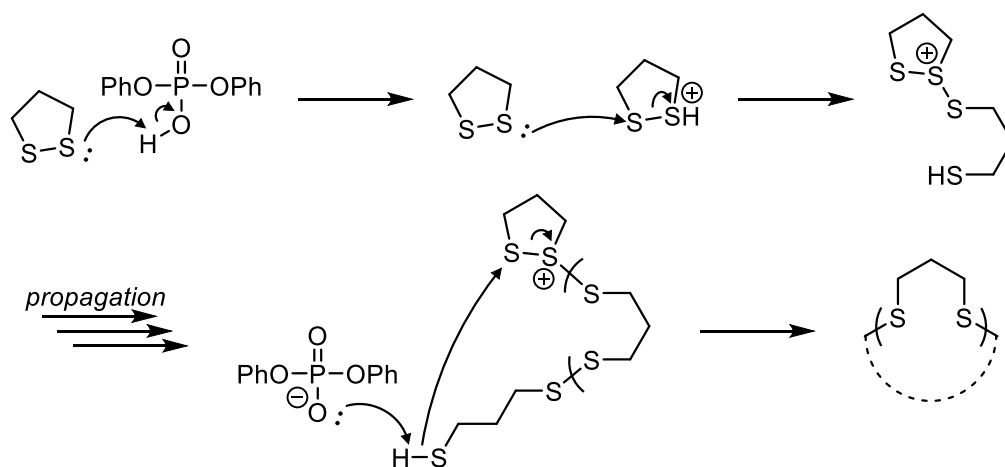
The recovered solution of ethyl lipoate in ethanol was diluted with ethyl acetate (150 mL) and transferred to a separatory funnel. The organic phase was subsequently washed with 1 M HCl (1 × 25 mL), 1M NaHCO₃ (3 × 25 mL) and brine (1 × 25 mL). The organic phase was collected and dried over 4Å molecular sieves overnight. The solution was then gravity filtered into a round bottom flask and concentrated in vacuo. The product was then transferred into an amber bottle and dried on the benchtop for several days to afford spectroscopically pure monomer (2.57 g).

Method for the Rotary Evaporator CROP of EtLp

EtLp monomer (5.00 g, 21.36 mmol, 1.0 equiv.) and acetone (20 mL) were added to a 6-dram vial. In another vial, DPP (267.2 mg, 1.068 mmol, 0.05 equiv.) and acetone were added (5 mL). The two solutions were then transferred to another vial and vortexed until homogenous. The reaction mixture was attached to a rotary evaporator and cooled to 5 °C in a circulating bath. The vacuum was then turned on and lowered until acetone started to distill over. The vacuum was turned off after acetone stopped distilling over (ca. 1 h) and polymer precipitated during this process. The crude polymer was characterized via ¹H NMR (to assess monomer conversion) and GPC. Purification and isolation of the crude polymer was not performed.

Method for the RCD into ROP of polyEtLp

PolyEtLp (3.7879 g, 16.182 mmol, 1.0 equiv.) and chloroform (302 mL, 593 g, 3754, 232 equiv.) were added ([polyEtLp]₀ = 60 mM) to a 2-neck round bottom flask. A condensing column was added to the center joint and the side joint was sealed with a septum. The round bottom flask was placed in an oil bath at 75 °C. Once the solution started to reflux a catalytic amount of sulfuric acid (0.043 mL, 0.21 mmol, 0.01 equiv.) was added through the side joint and then the joint was resealed. The reaction was refluxed for 2 h. The reaction mixture was then filtered into a 500 mL round bottom flask and attached to a rotary evaporator equipped with a 30 °C bath to concentrate in vacuo. The crude polymer was characterized via ¹H NMR (to assess monomer conversion) and GPC. Purification and isolation of the crude polymer was not performed.



Scheme S1.

Proposed Mechanism for the CROP of 1,2-dithiolane with DPP. Adapted from a prior study on the CROP of 1,2-dithiolane with TFA.¹

NMR Spectra for monomer and polymer

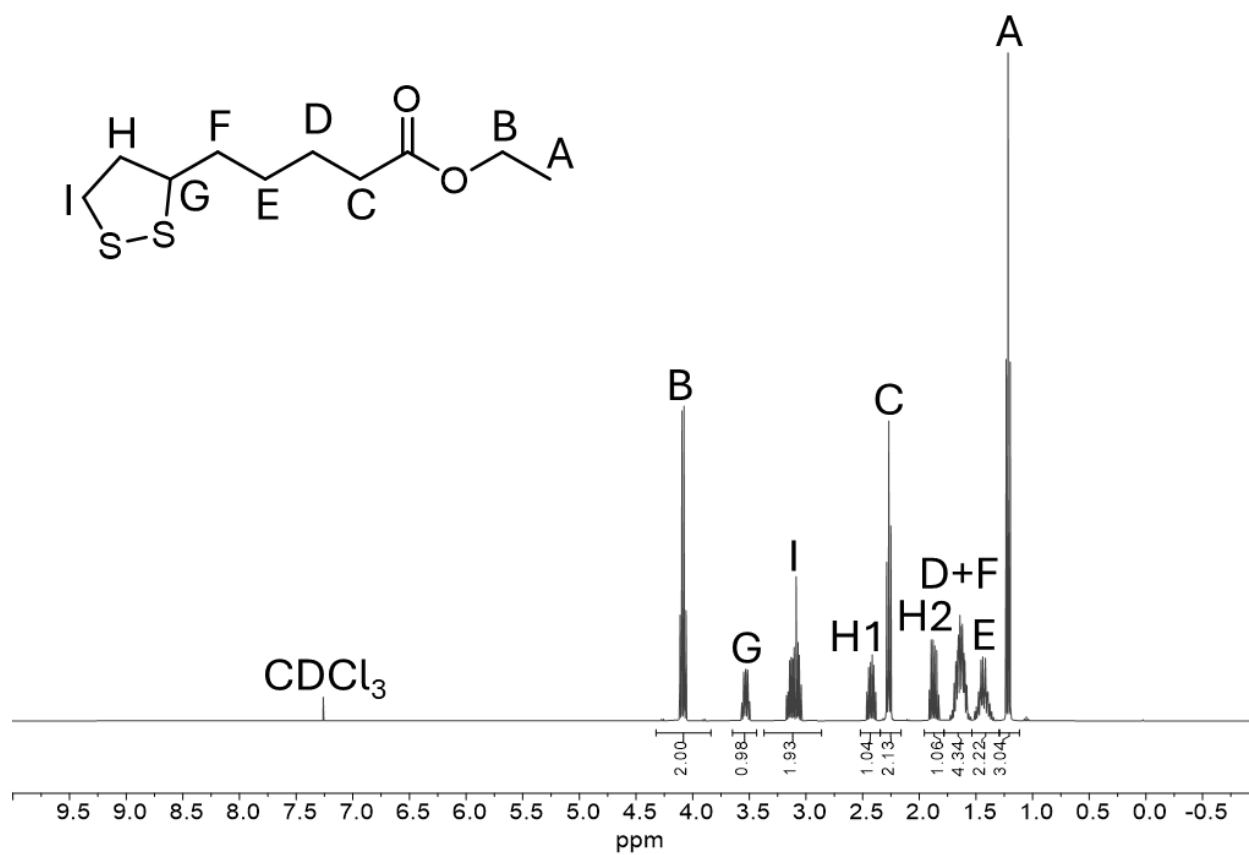


Figure S1.

EtLp ¹H-NMR spectrum – 400 MHz, 298 K, CDCl₃.

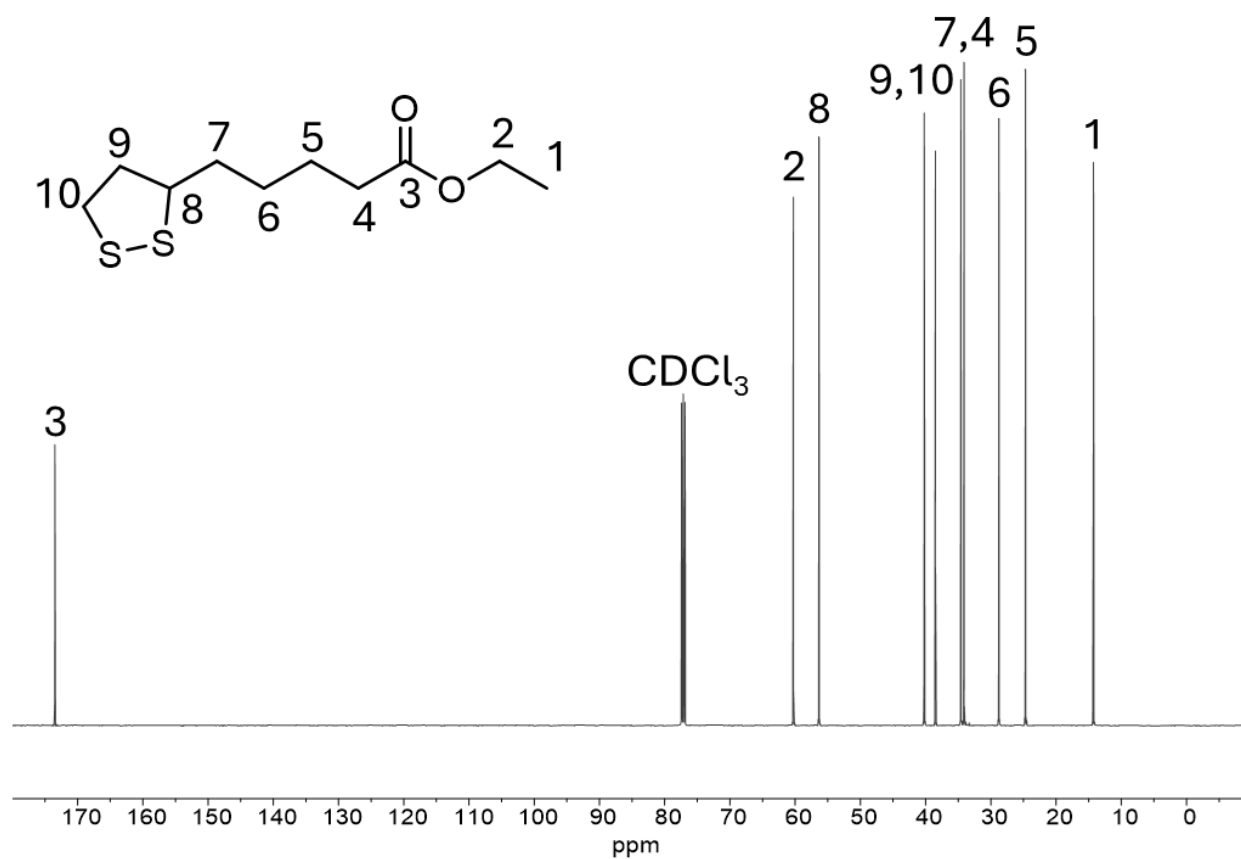


Figure S2.
EtLp ^{13}C -NMR spectrum – 100.57 MHz, 298 K, CDCl_3 .

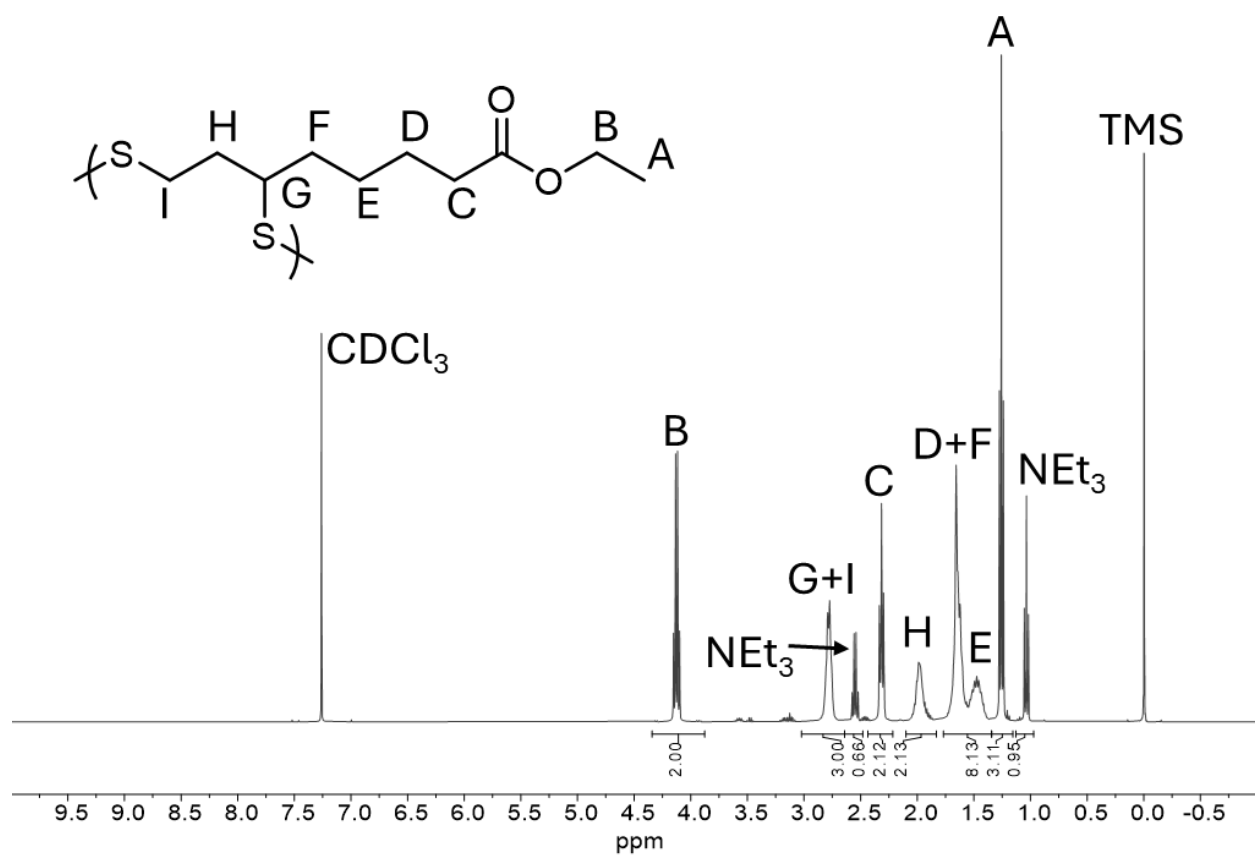


Figure S3.
polyEtLp $^1\text{H-NMR}$ spectrum – 400 MHz, 298 K, CDCl_3 .

$$\text{Conversion(\%)} = \frac{\frac{g'+i'}{3}}{\frac{g'+i'}{3} + \frac{i}{2}} 100$$

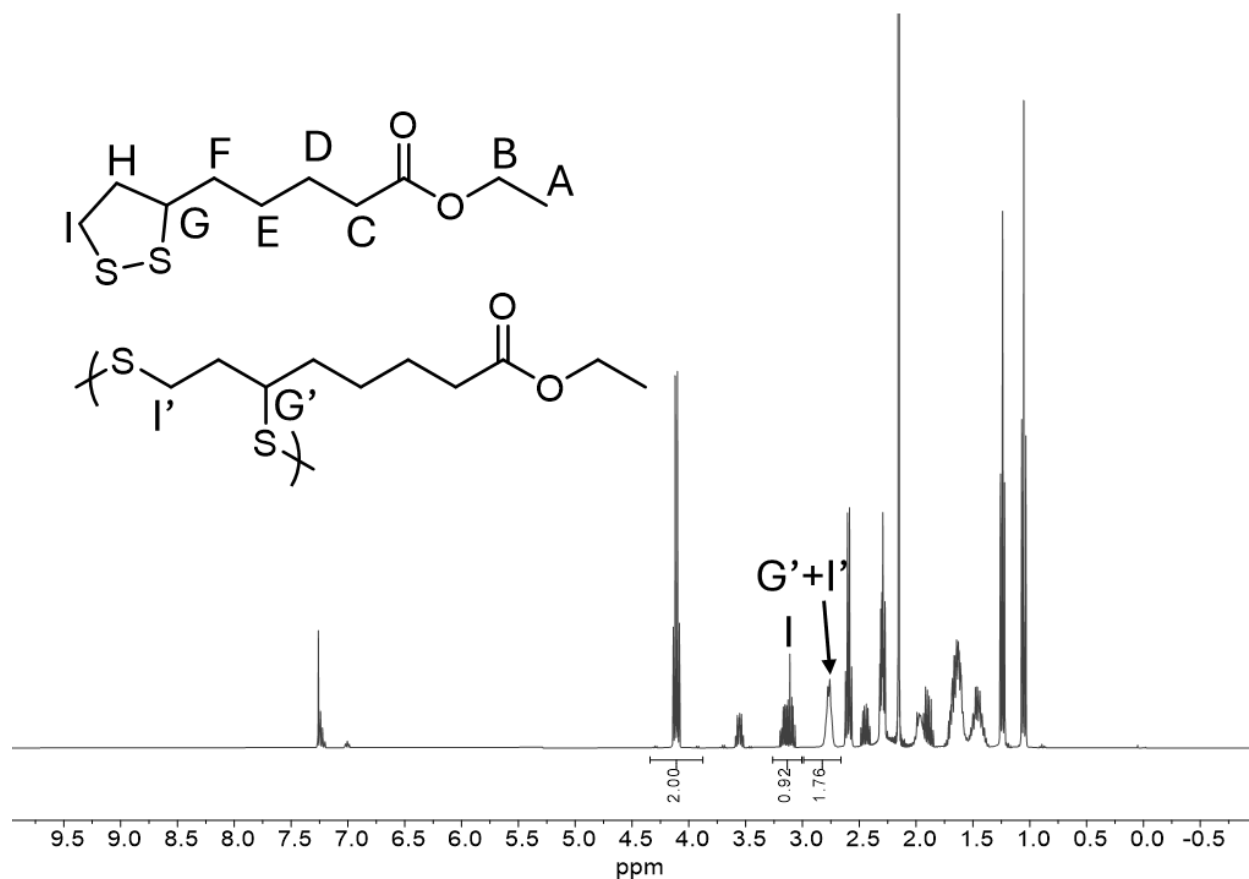


Figure S4.

Crude mixture of EtLp and polyEtLp ¹H-NMR spectrum – 400 MHz, 298 K, CDCl₃. Conversion equation is shown with assigned signals used for calculation.

Catalyst Screening

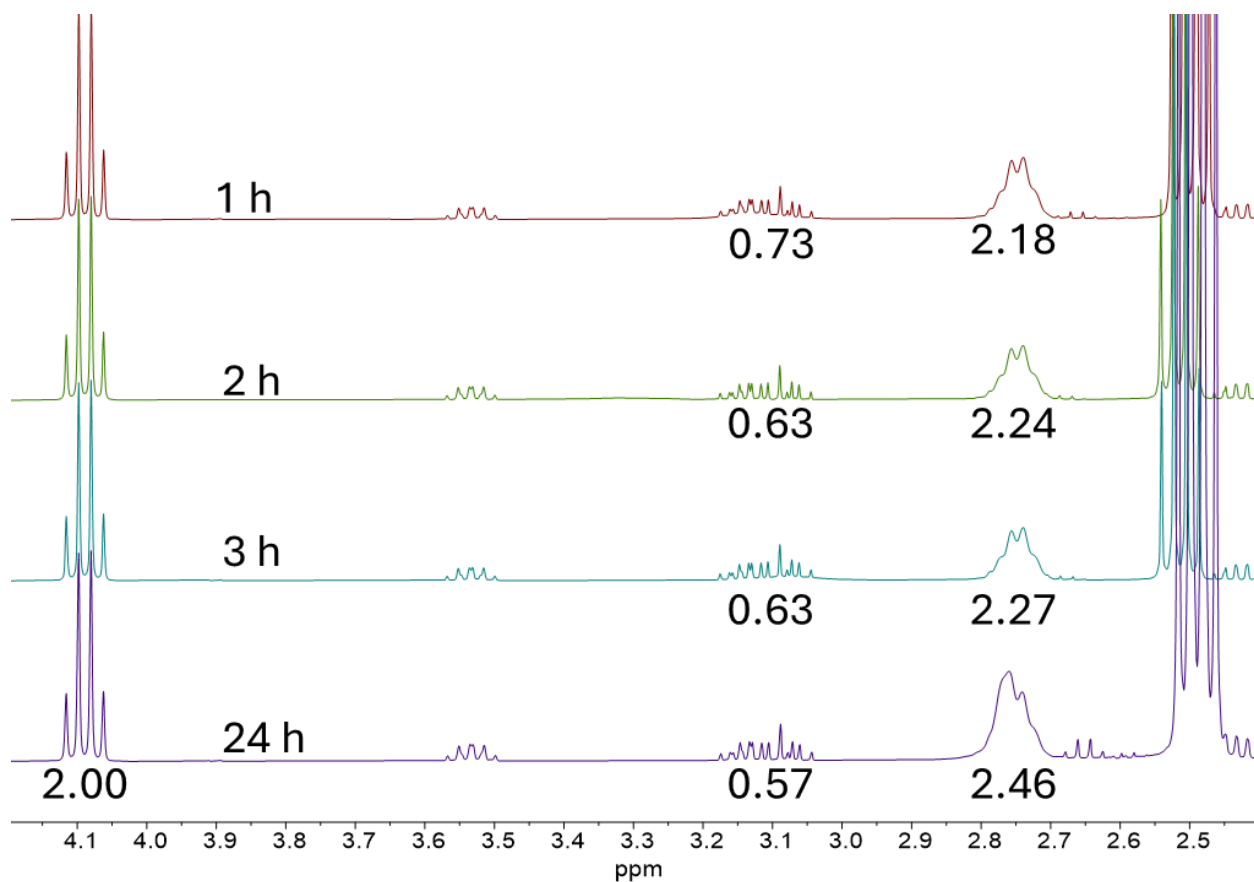


Figure S5.

Crude ¹H-NMR of CROP of EtLp with trifluoroacetic acid (TFA). [EtLp]₀ = 4.0M, 0.05 equiv. TFA, DCM, 30 °C. Integration values are shown at different time stamps. ¹H-NMR spectrum – 400 MHz, 298 K, CDCl₃.

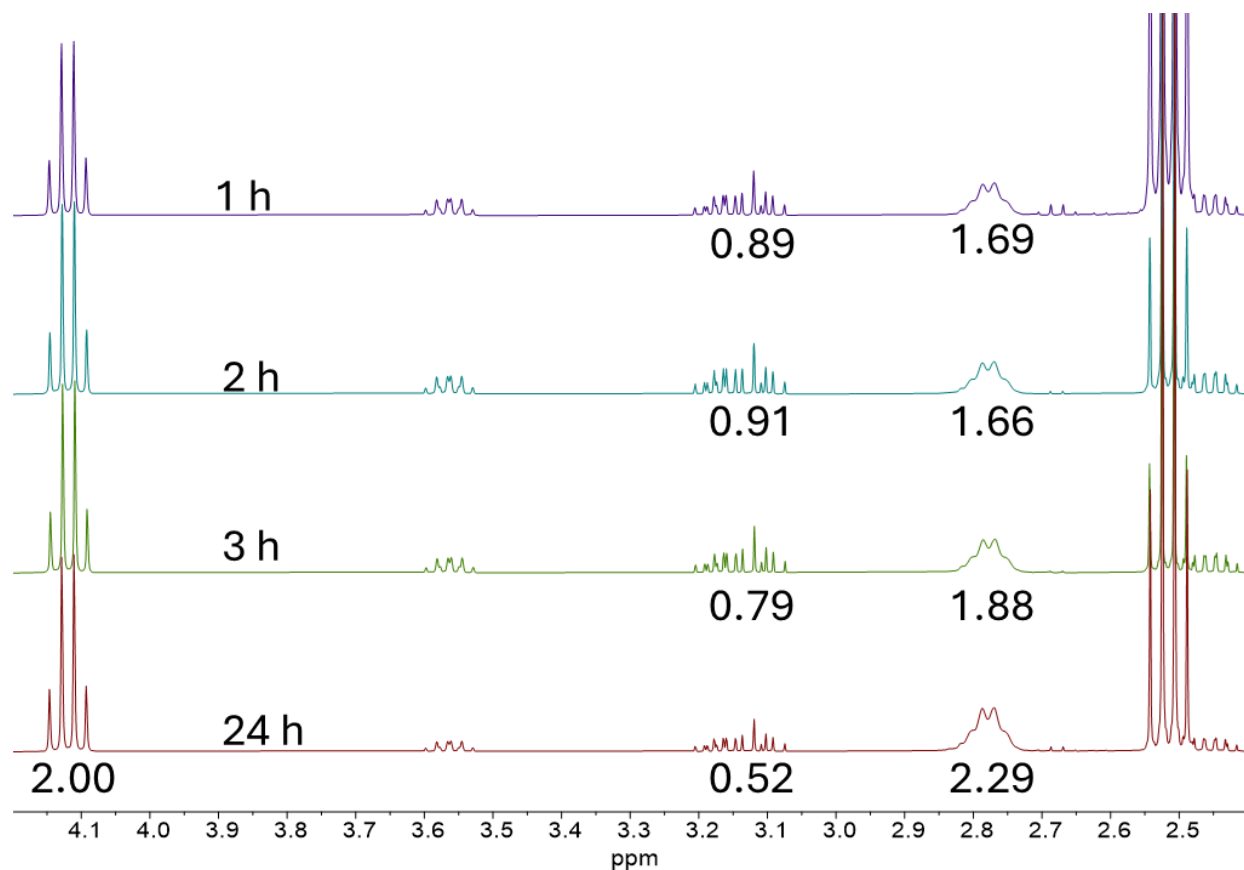


Figure S6.

Crude ^1H -NMR of CROP of EtLp with HCl (12.1 M). $[\text{EtLp}]_0 = 4.0\text{M}$, 0.05 equiv. HCl, DCM, 30 °C. Integration values are shown at different time stamps. ^1H -NMR spectrum – 400 MHz, 298 K, CDCl_3 .

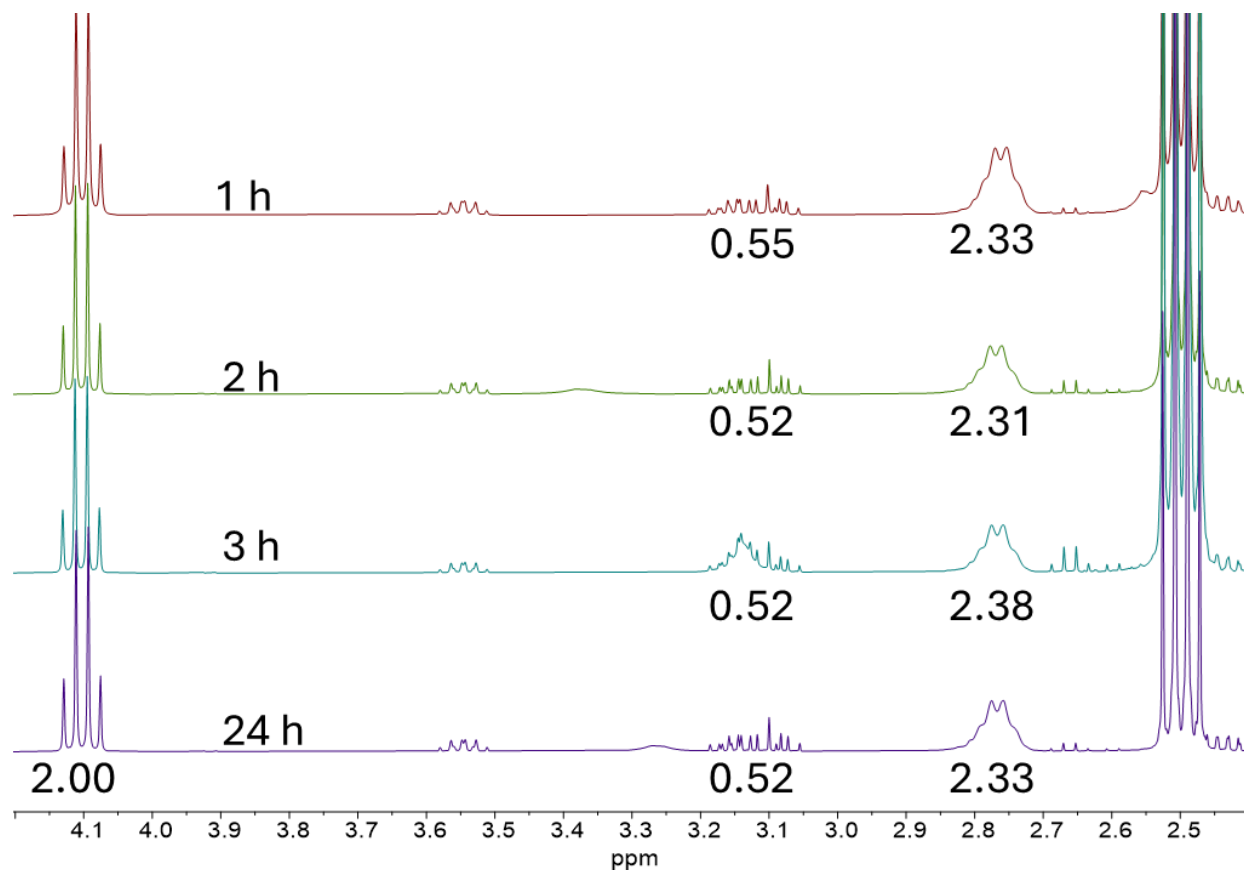


Figure S7.

Crude ^1H -NMR of CROP of EtLp with HClO_4 (70 wt% solution). $[\text{EtLp}]_0 = 4.0\text{M}$, 0.05 equiv. HClO_4 , DCM, 30 $^\circ\text{C}$. Integration values are shown at different time stamps. ^1H -NMR spectrum – 400 MHz, 298 K, CDCl_3 .

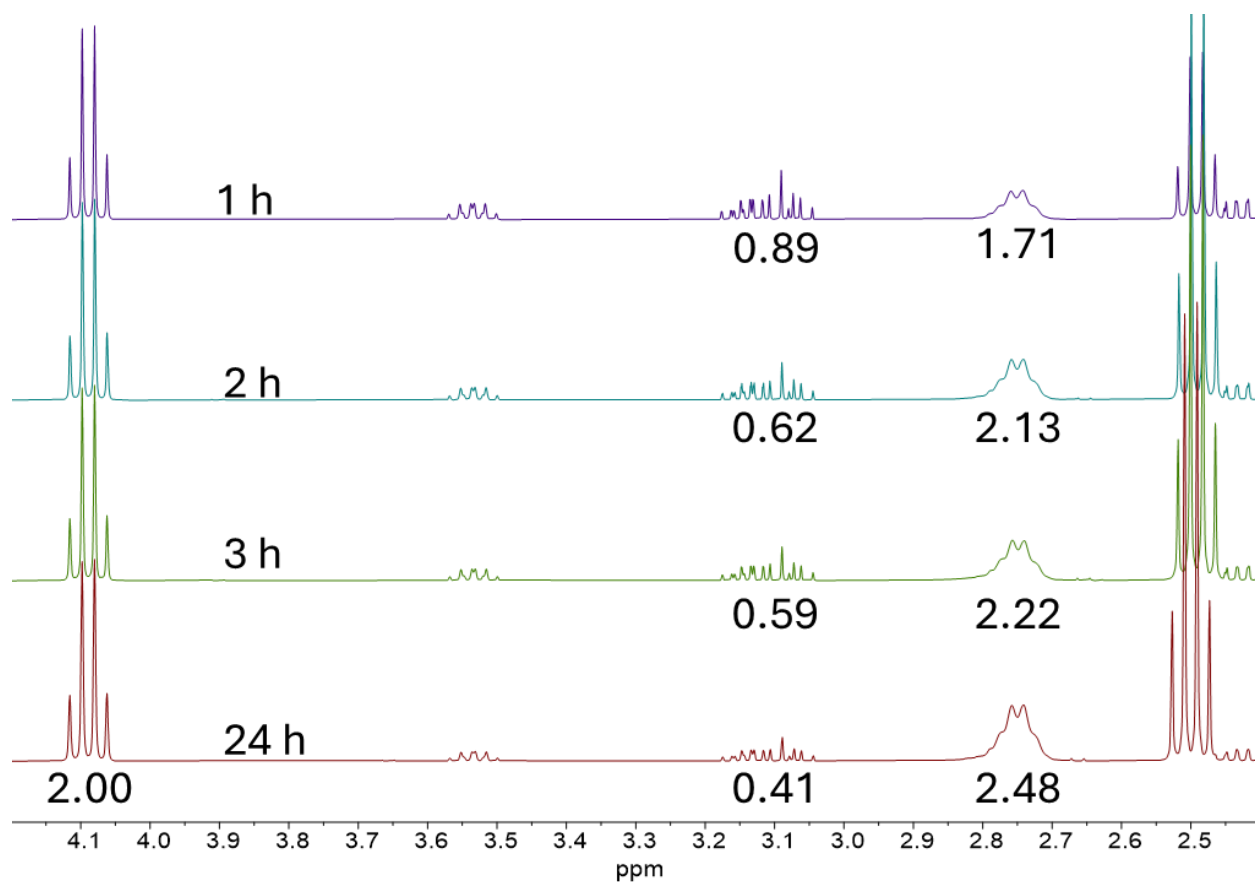


Figure S8.

Crude ¹H-NMR of CROP of EtLp with H₂SO₄ (18 M). [EtLp]₀ = 4.0M, 0.05 equiv. H₂SO₄, DCM, 30 °C. Integration values are shown at different time stamps. ¹H-NMR spectrum – 400 MHz, 298 K, CDCl₃.

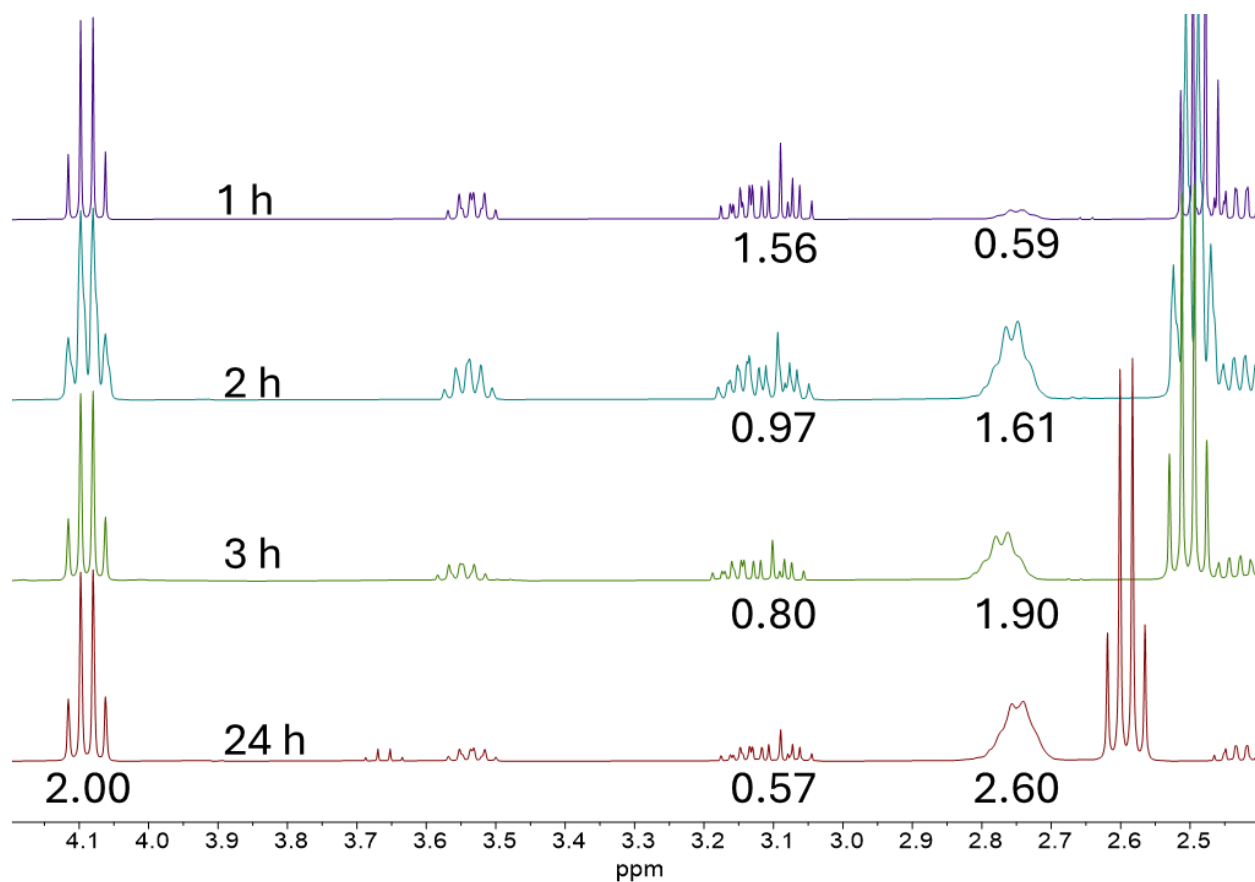


Figure S9.

Crude ^1H -NMR of CROP of EtLp with *p*-toluenesulfonic acid (p-TSA). $[\text{EtLp}]_0 = 4.0\text{M}$, 0.05 equiv. p-TSA, DCM, 30 °C. Integration values are shown at different time stamps. ^1H -NMR spectrum – 400 MHz, 298 K, CDCl_3 .

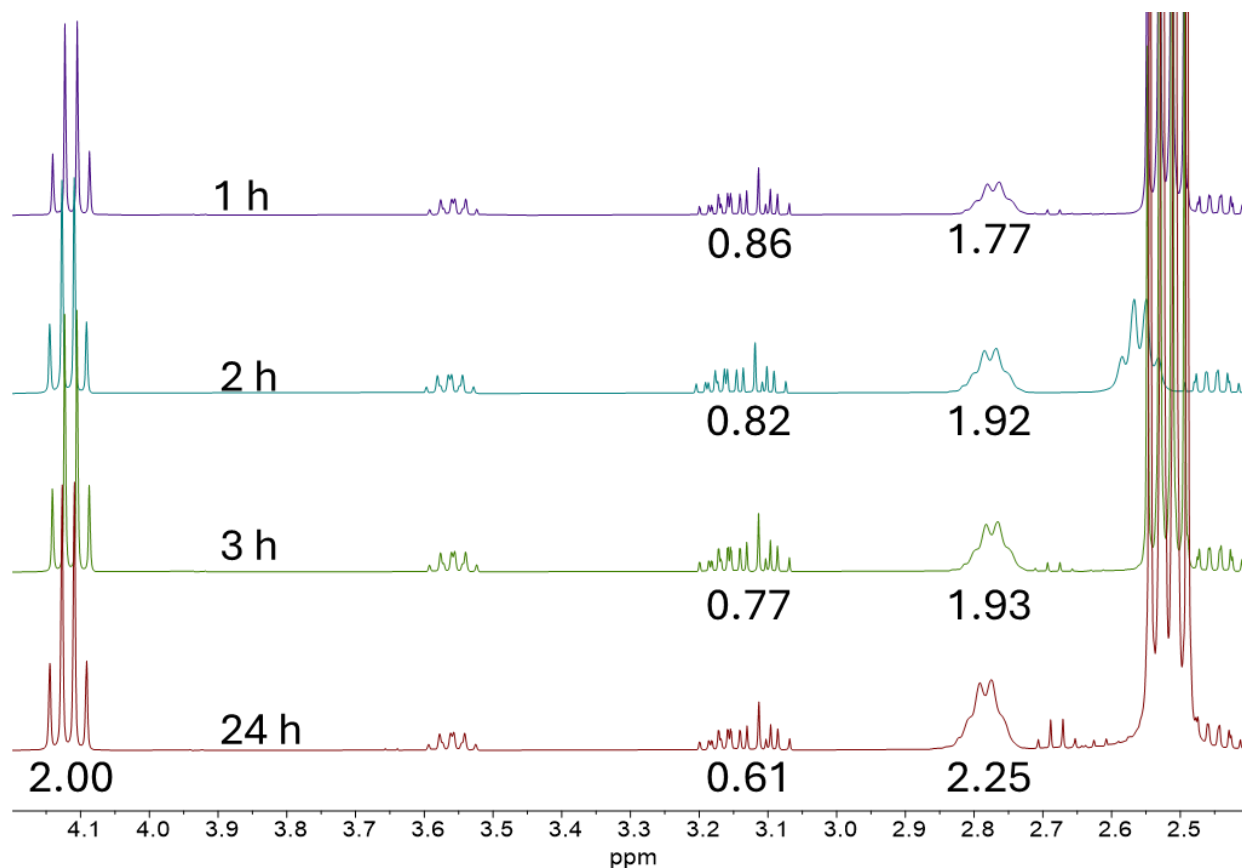


Figure S10.

Crude $^1\text{H-NMR}$ of CROP of EtLp with DPP. $[\text{EtLp}]_0 = 4.0\text{M}$, 0.05 equiv. DPP, DCM, 30 °C. Integration values are shown at different time stamps. $^1\text{H-NMR}$ spectrum – 400 MHz, 298 K, CDCl_3 .

Table S1.

Summary of conversion data for cationic initiator screening for the ROP of EtLp.

Initiator	1 h	2 h	3 h	24 h
TFA	63 ± 3	65 ± 4	67 ± 2	73.3 ± 0.2
HCl	49 ± 5	49 ± 3	57 ± 3	72 ± 2
HClO_4	75 ± 3	72 ± 4	76 ± 2	76.2 ± 0.5
H_2SO_4	60 ± 2	66 ± 2	66 ± 3	77 ± 2
pTSA	23 ± 2	50 ± 2	59 ± 2	74 ± 2
DPP	56.0 ± 0.8	59 ± 1	61 ± 1	73 ± 1

$n = 3$ for all initiators at each time point; $[\text{EtLp}]_0 = 4.0\text{ M}$, 0.05equiv. cationic initiator, all reactions were in DCM, Temp. = 30 °C. Conversion (%) was calculated using the equation in Figure S4.

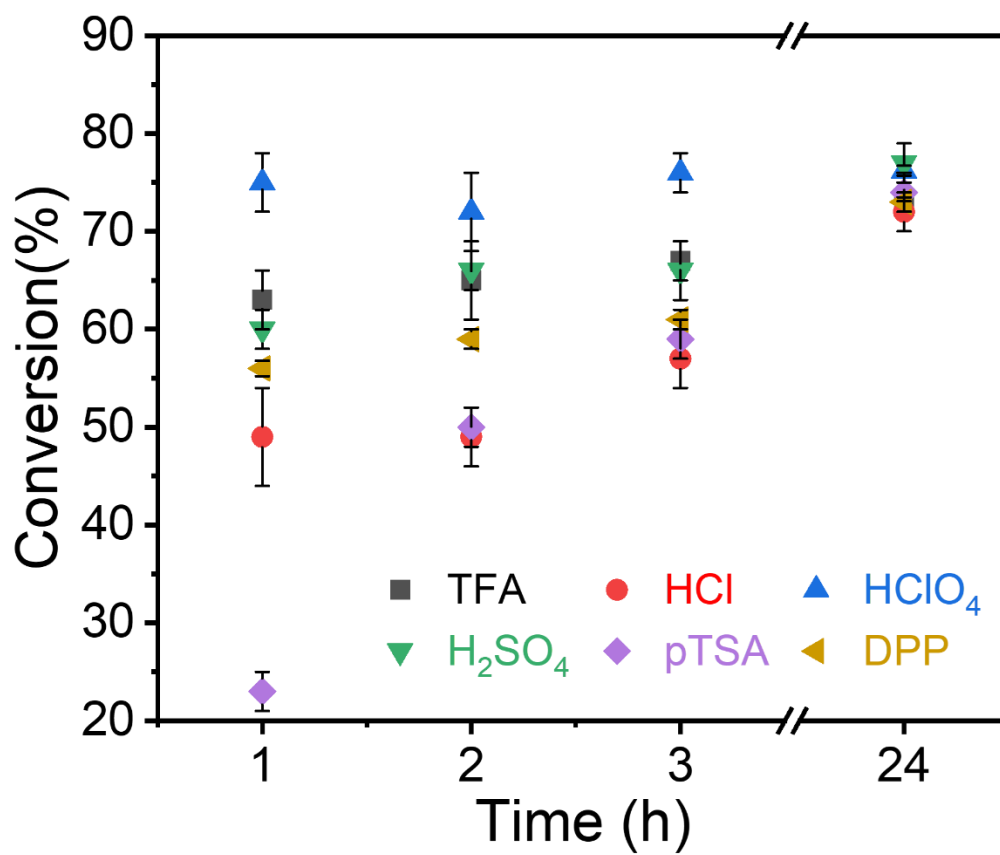


Figure S11.

Scatter Plot of conversion data for cationic initiator screening for the ROP of EtLp. Data summarized in Table S1.

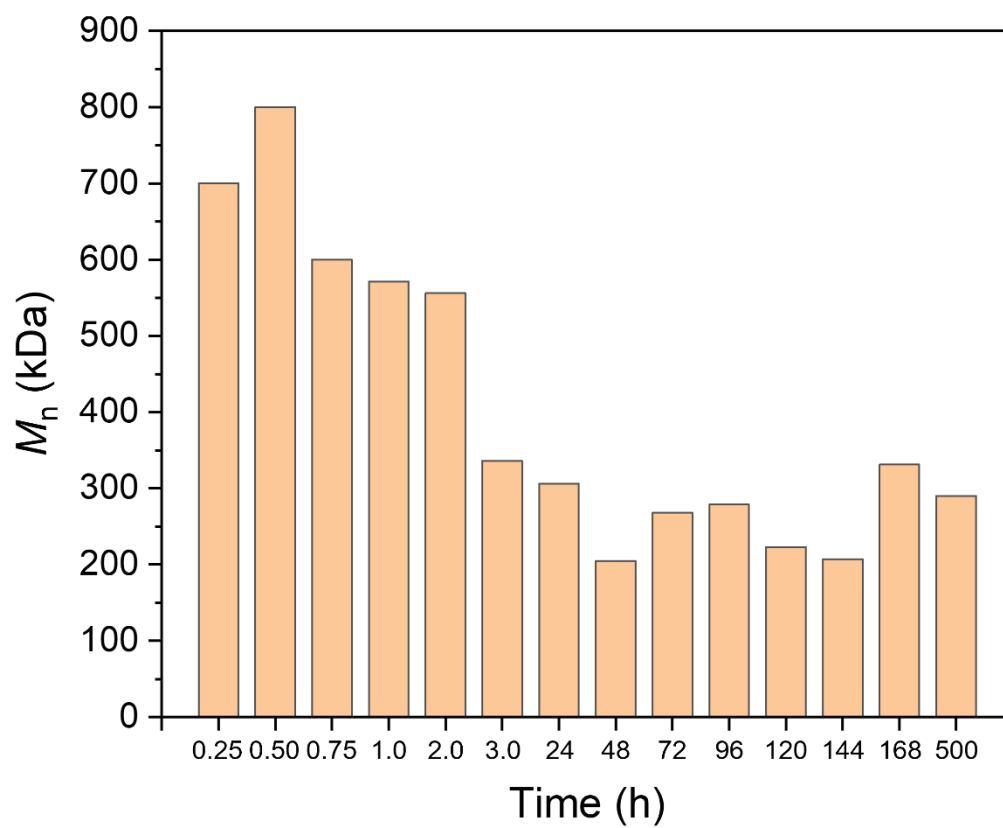


Figure S12.

Bar chart of M_n vs time for CROP of EtLp in DCM. $[\text{EtLp}]_0 = 4.0\text{M}$, 0.05 equiv. DPP, DCM, 30 °C. Values were obtained using GPC.

Solvent Screening

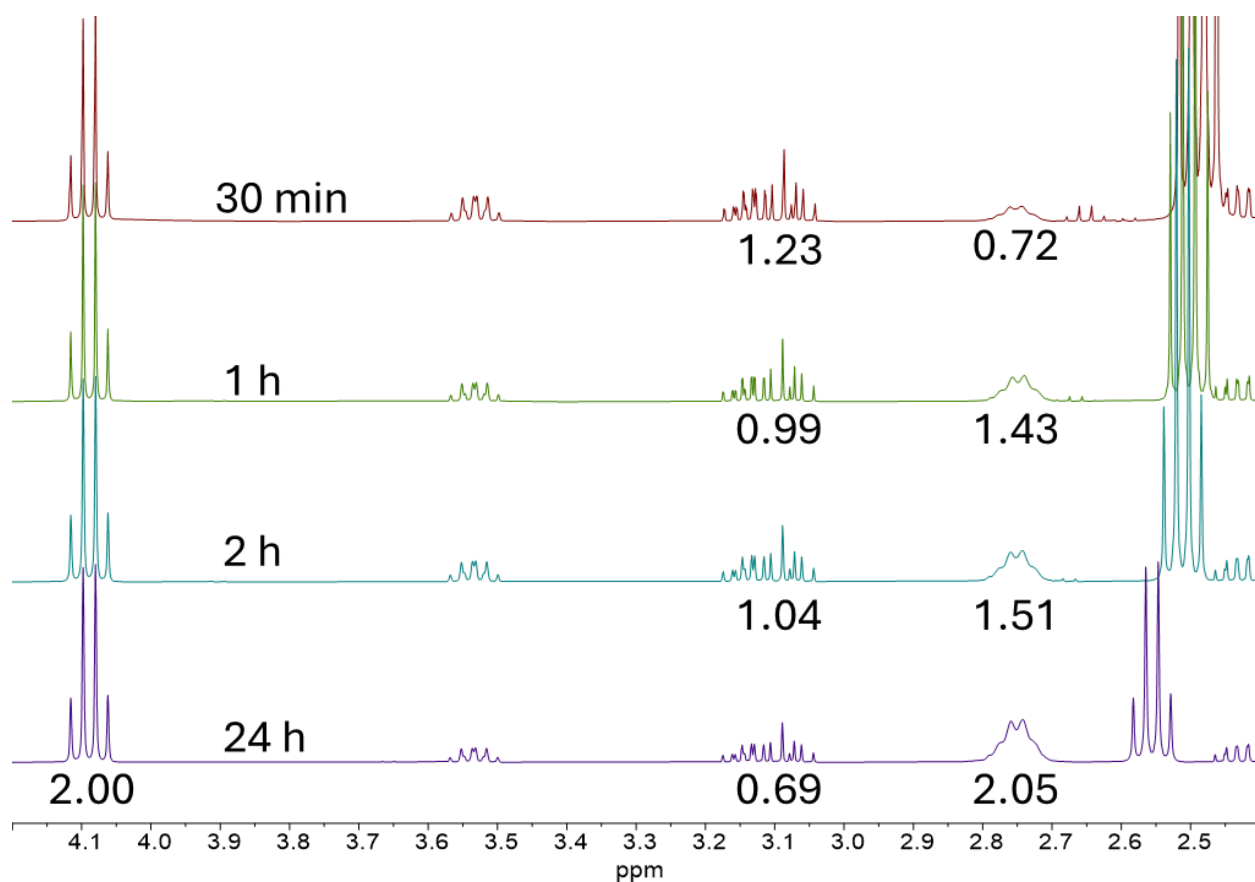


Figure S13.

Crude ^1H -NMR of CROP of EtLp in DCM. $[\text{EtLp}]_0 = 4.0\text{M}$, 0.05 equiv. DPP, DCM, 30 °C. Integration values are shown at different time stamps. ^1H -NMR spectrum – 400 MHz, 298 K, CDCl_3 .

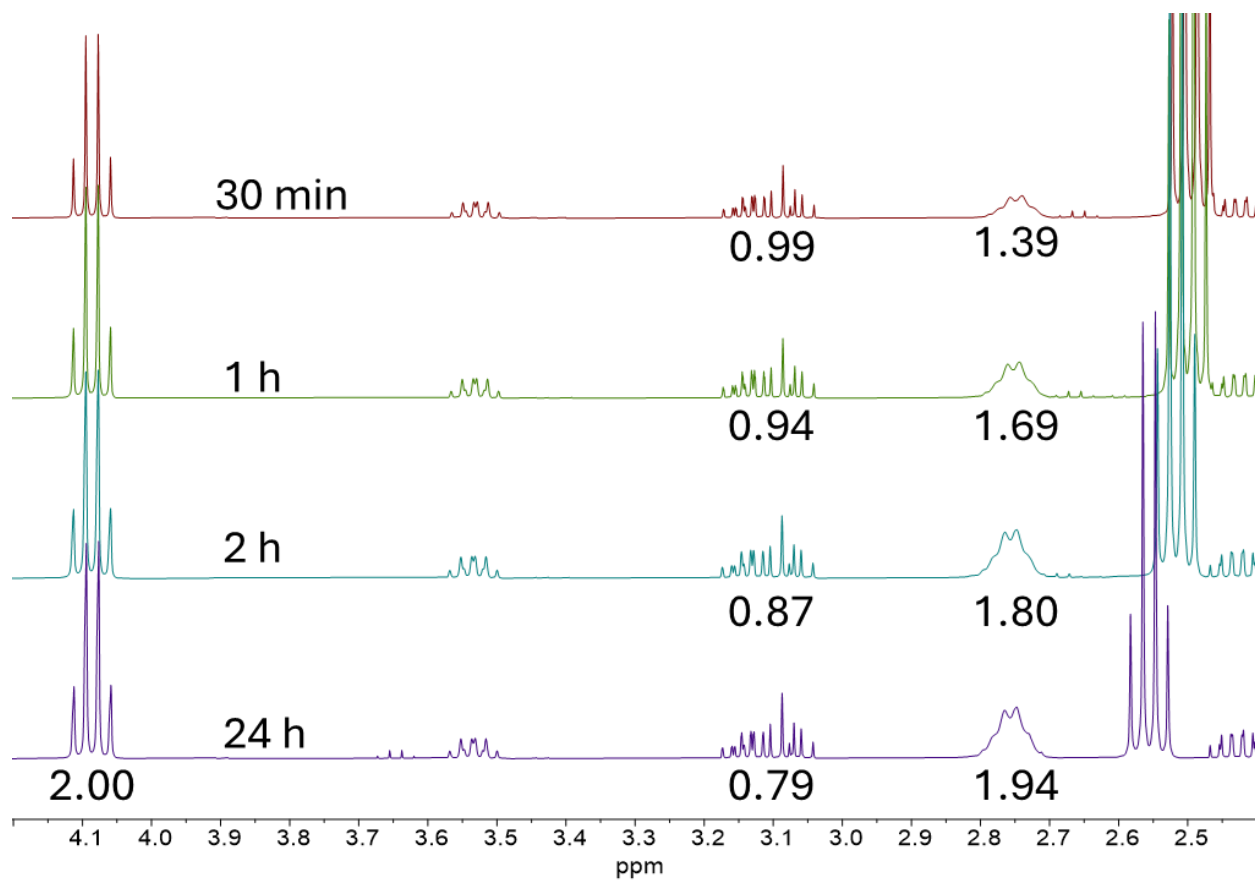


Figure S14.

Crude ^1H -NMR of CROP of EtLp in acetone. $[\text{EtLp}]_0 = 4.0\text{M}$, 0.05 equiv. DPP, acetone, 30 °C. Integration values are shown at different time stamps. ^1H -NMR spectrum – 400 MHz, 298 K, CDCl_3 .

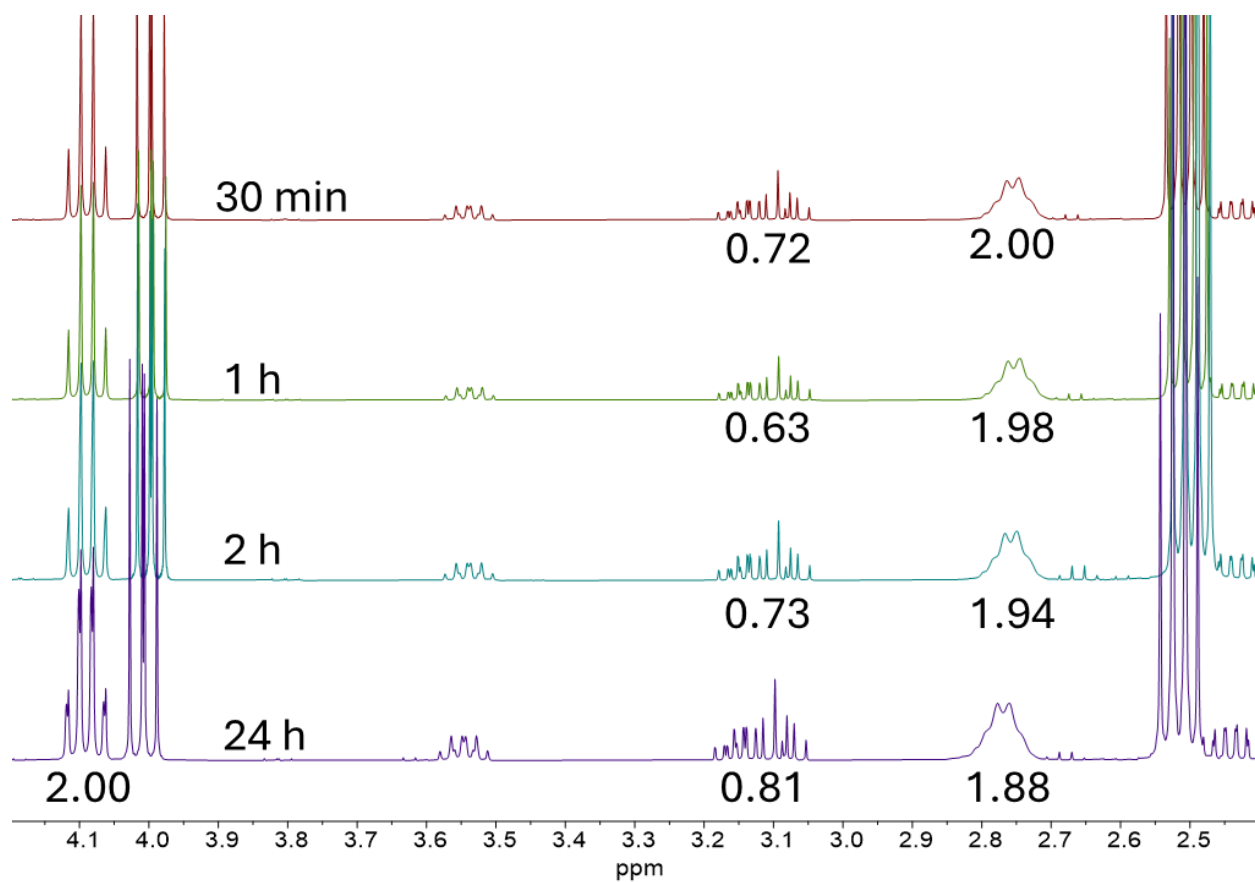


Figure S15.

Crude ¹H-NMR of CROP of EtLp in propylene carbonate (PC). [EtLp]₀ = 4.0M, 0.05 equiv. DPP, PC, 30 °C. Integration values are shown at different time stamps. ¹H-NMR spectrum – 400 MHz, 298 K, CDCl₃.

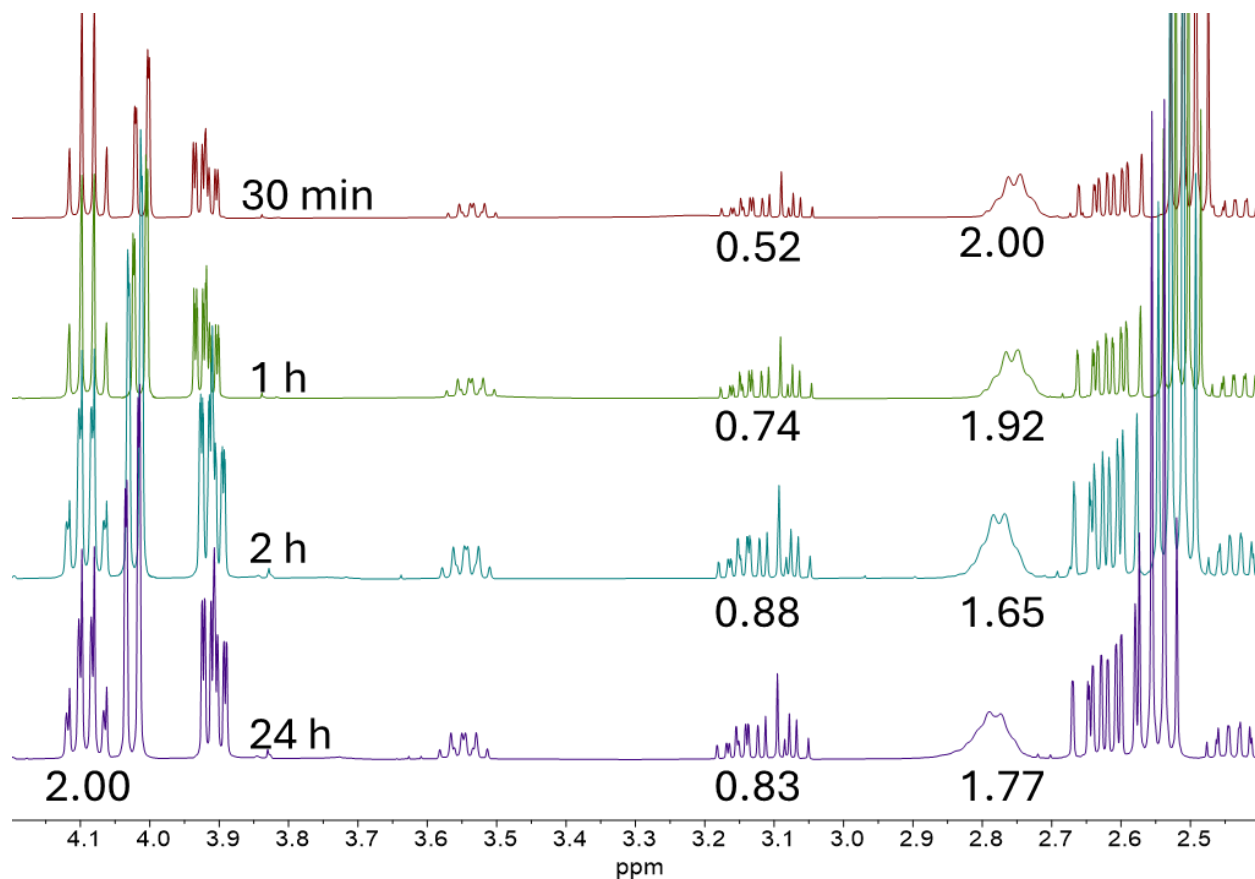


Figure S16.

Crude ¹H-NMR of CROP of EtLp in cyrene. [EtLp]₀ = 4.0M, 0.05 equiv. DPP, cyrene, 30 °C. Integration values are shown at different time stamps. ¹H-NMR spectrum – 400 MHz, 298 K, CDCl₃.

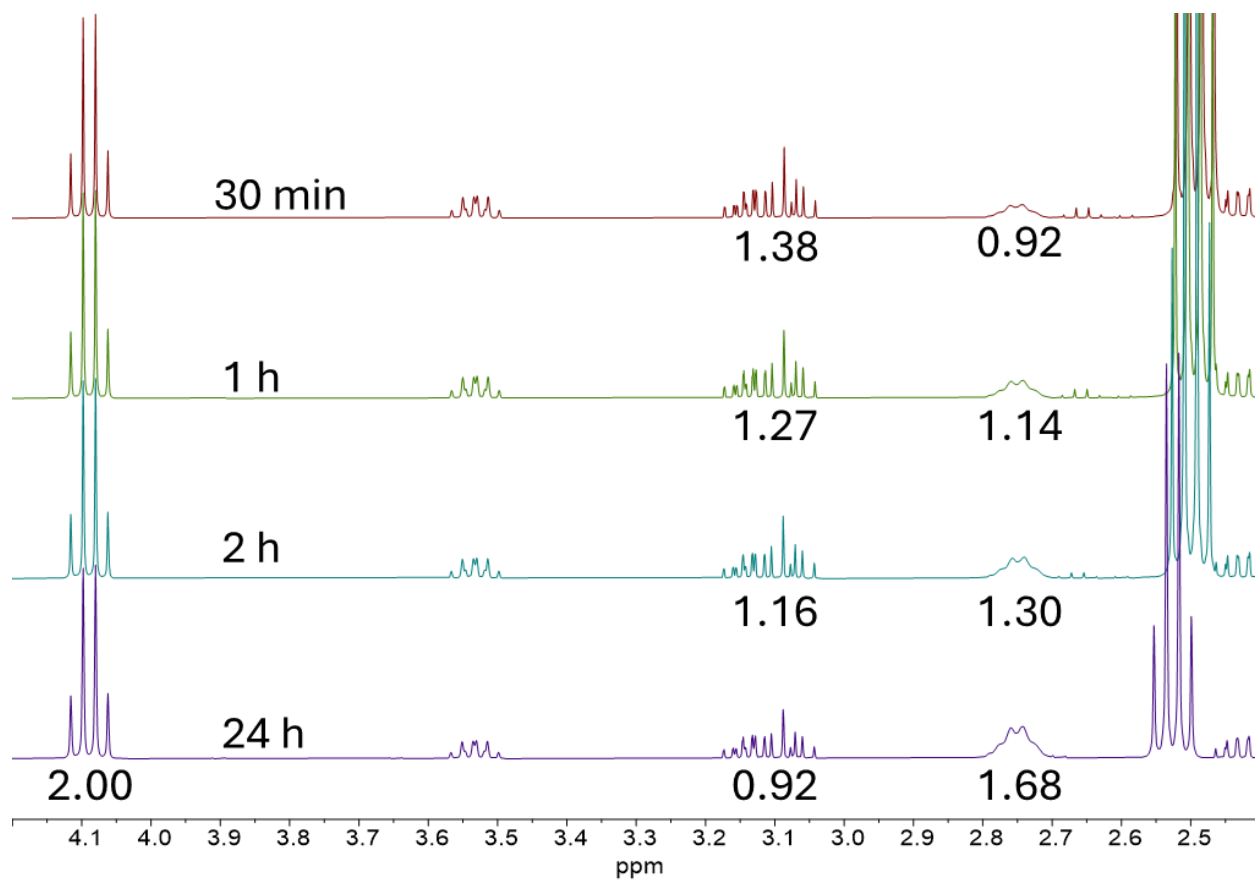


Figure S17.

Crude ¹H-NMR of CROP of EtLp in CHCl₃. [EtLp]₀ = 4.0M, 0.05 equiv. DPP, CHCl₃, 30 °C. Integration values are shown at different time stamps. ¹H-NMR spectrum – 400 MHz, 298 K, CDCl₃.

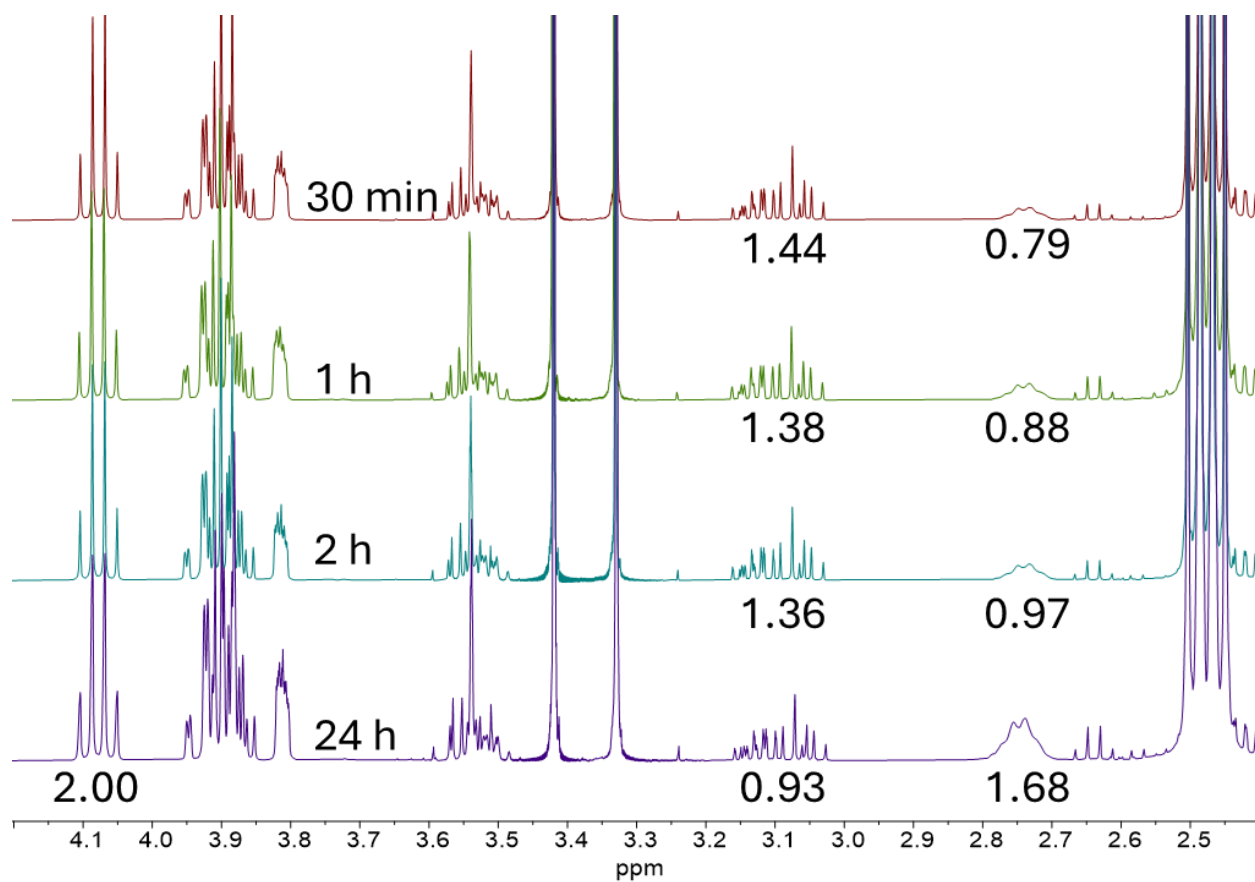


Figure S18.

Crude ¹H-NMR of CROP of EtLp in dimethyl isosorbide (DMI). [EtLp]₀ = 4.0M, 0.05 equiv. DPP, DMI, 30 °C. Integration values are shown at different time stamps. ¹H-NMR spectrum – 400 MHz, 298 K, CDCl₃.

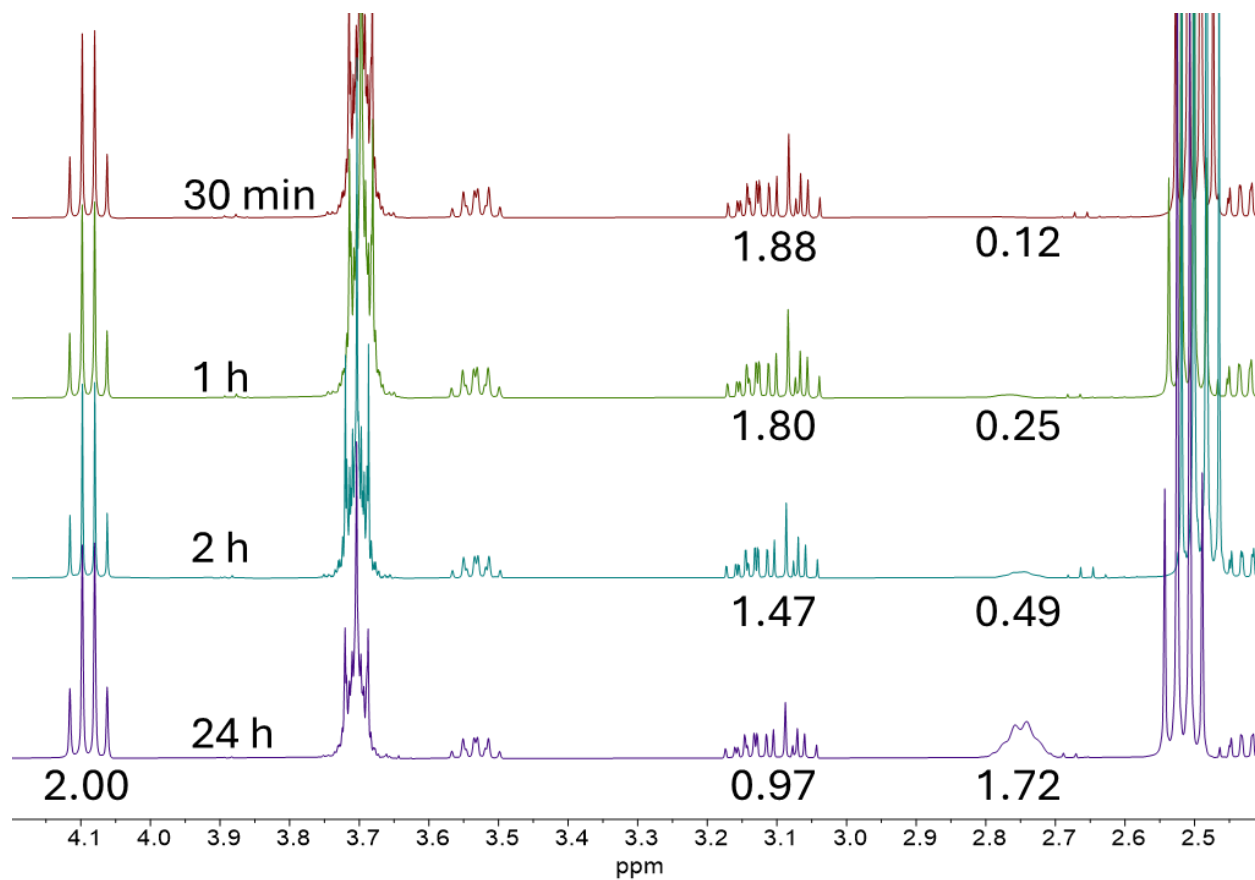


Figure S19.

Crude ^1H -NMR of CROP of EtLp in THF. $[\text{EtLp}]_0 = 4.0\text{M}$, 0.05 equiv. DPP, THF, 30 °C. Integration values are shown at different time stamps. ^1H -NMR spectrum – 400 MHz, 298 K, CDCl_3 .

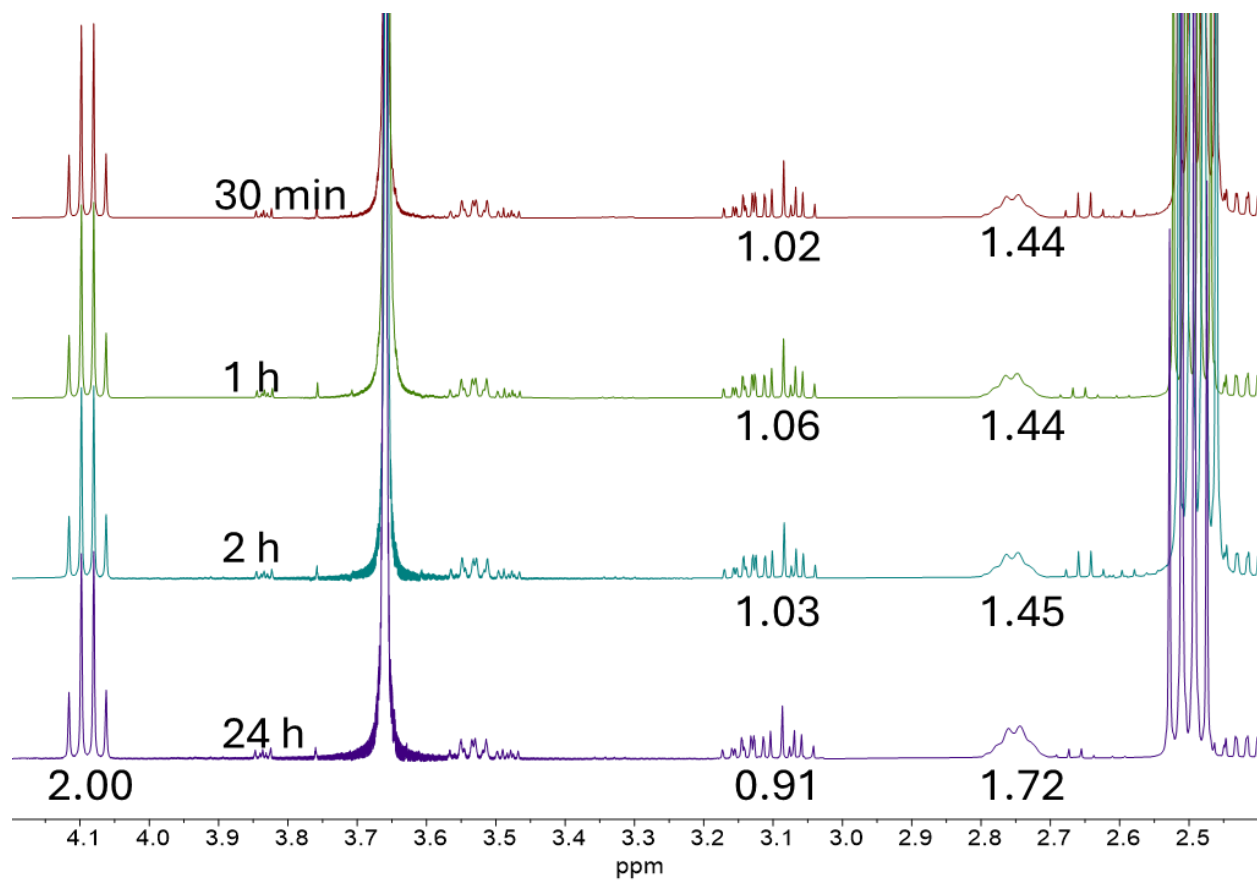


Figure S20.

Crude ¹H-NMR of CROP of EtLp in 1,4-dioxane (1,4-diOx). [EtLp]₀ = 4.0M, 0.05 equiv. DPP, 1,4-diOx, 30 °C. Integration values are shown at different time stamps. ¹H-NMR spectrum – 400 MHz, 298 K, CDCl₃.

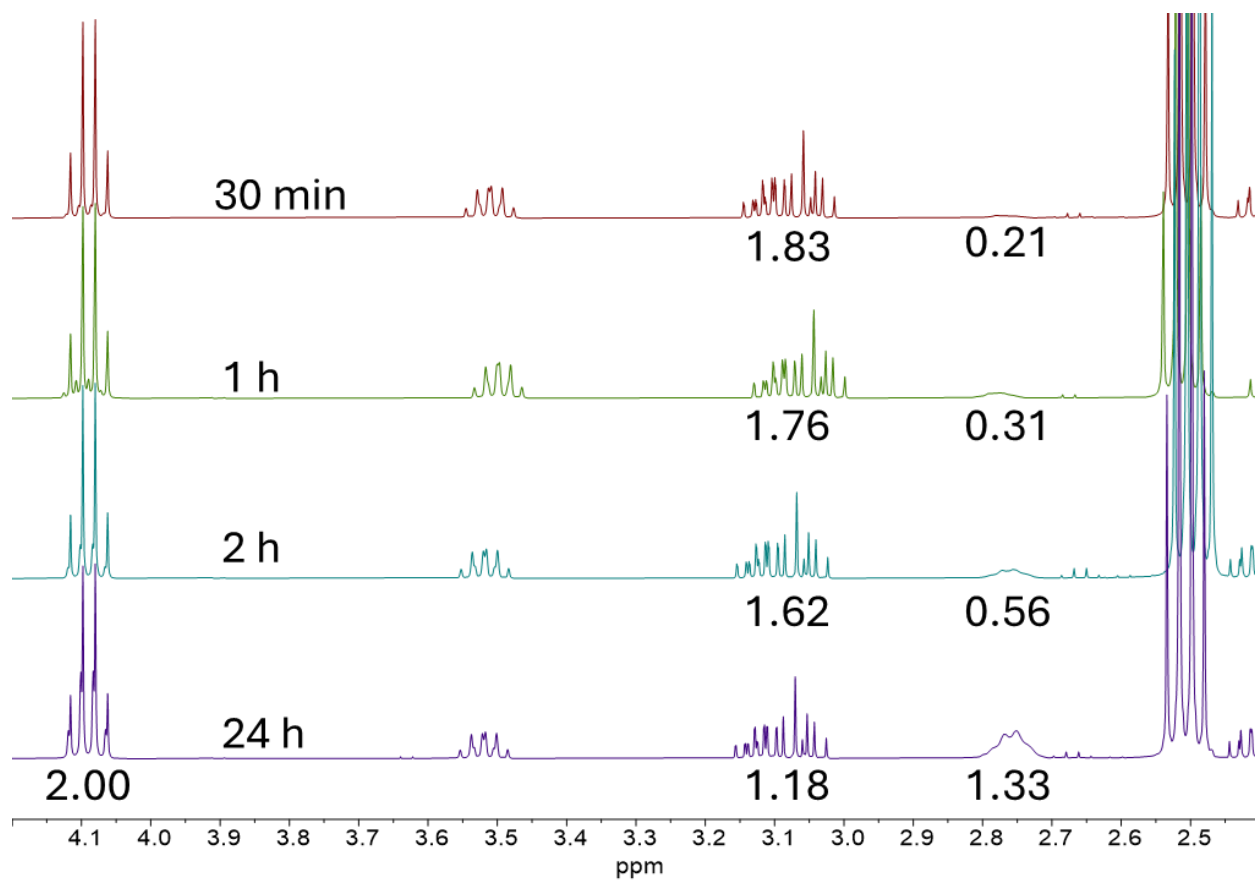


Figure S21.

Crude ¹H-NMR of CROP of EtLp in cyclopentyl methyl ether (CPME). [EtLp]₀ = 4.0M, 0.05 equiv. DPP, CPME, 30 °C. Integration values are shown at different time stamps. ¹H-NMR spectrum – 400 MHz, 298 K, CDCl₃.

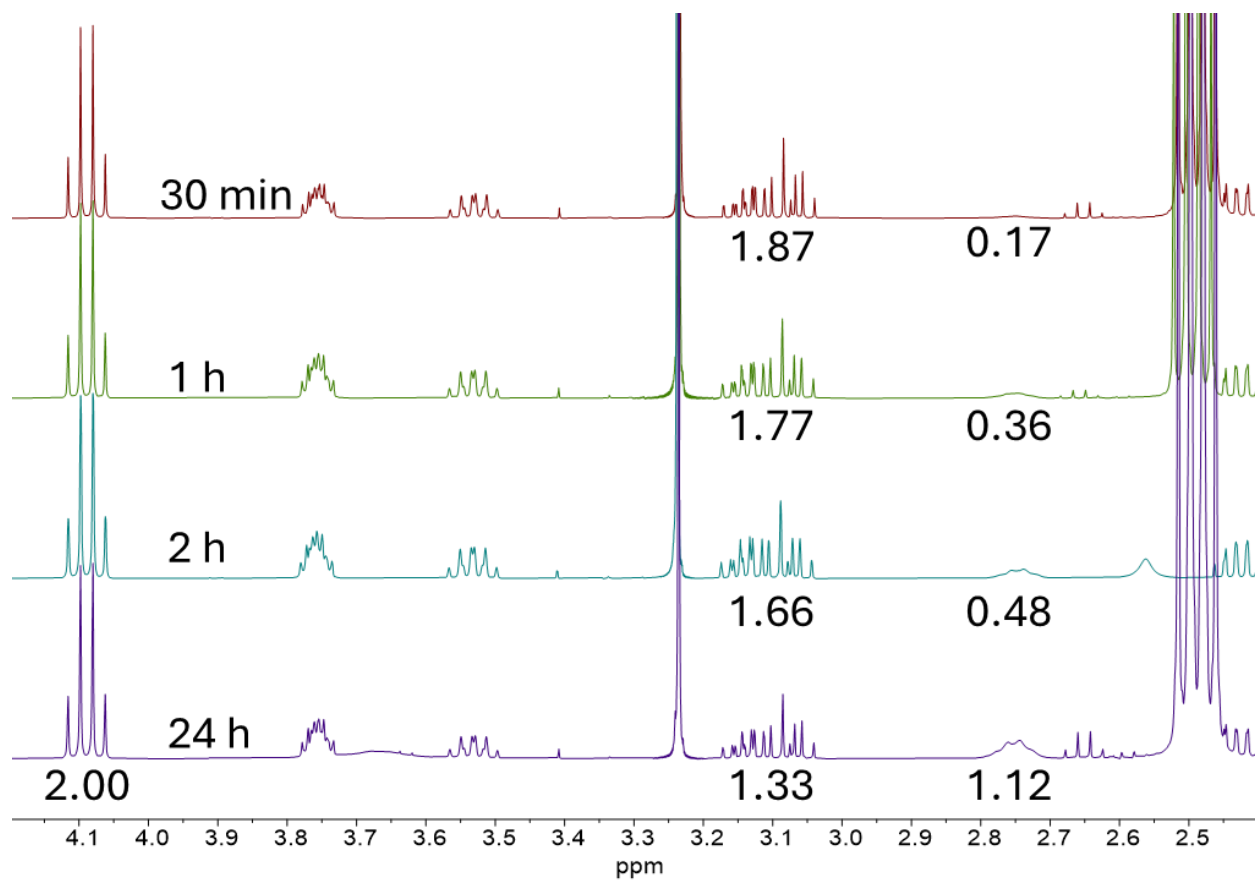


Figure S22.

Crude ¹H-NMR of CROP of EtLp in 2-methyltetrahydrofuran (2-MeTHF). [EtLp]₀ = 4.0M, 0.05 equiv. DPP, 2-MeTHF, 30 °C. Integration values are shown at different time stamps. ¹H-NMR spectrum – 400 MHz, 298 K, CDCl₃.

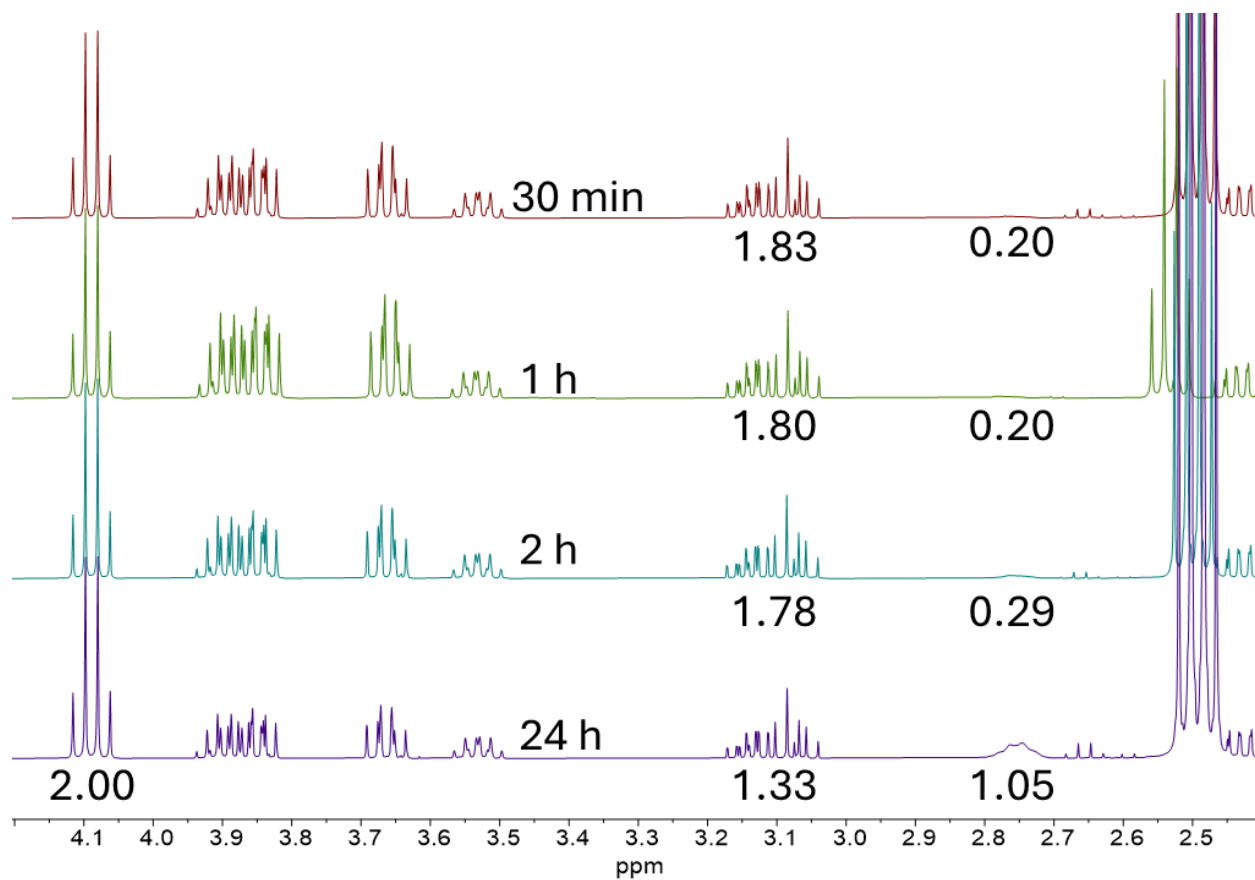


Figure S23.

Crude ^1H -NMR of CROP of EtLp in CCl_4 . $[\text{EtLp}]_0 = 4.0\text{M}$, 0.05 equiv. DPP, CCl_4 , 30 °C. Integration values are shown at different time stamps. ^1H -NMR spectrum – 400 MHz, 298 K, CDCl_3 .

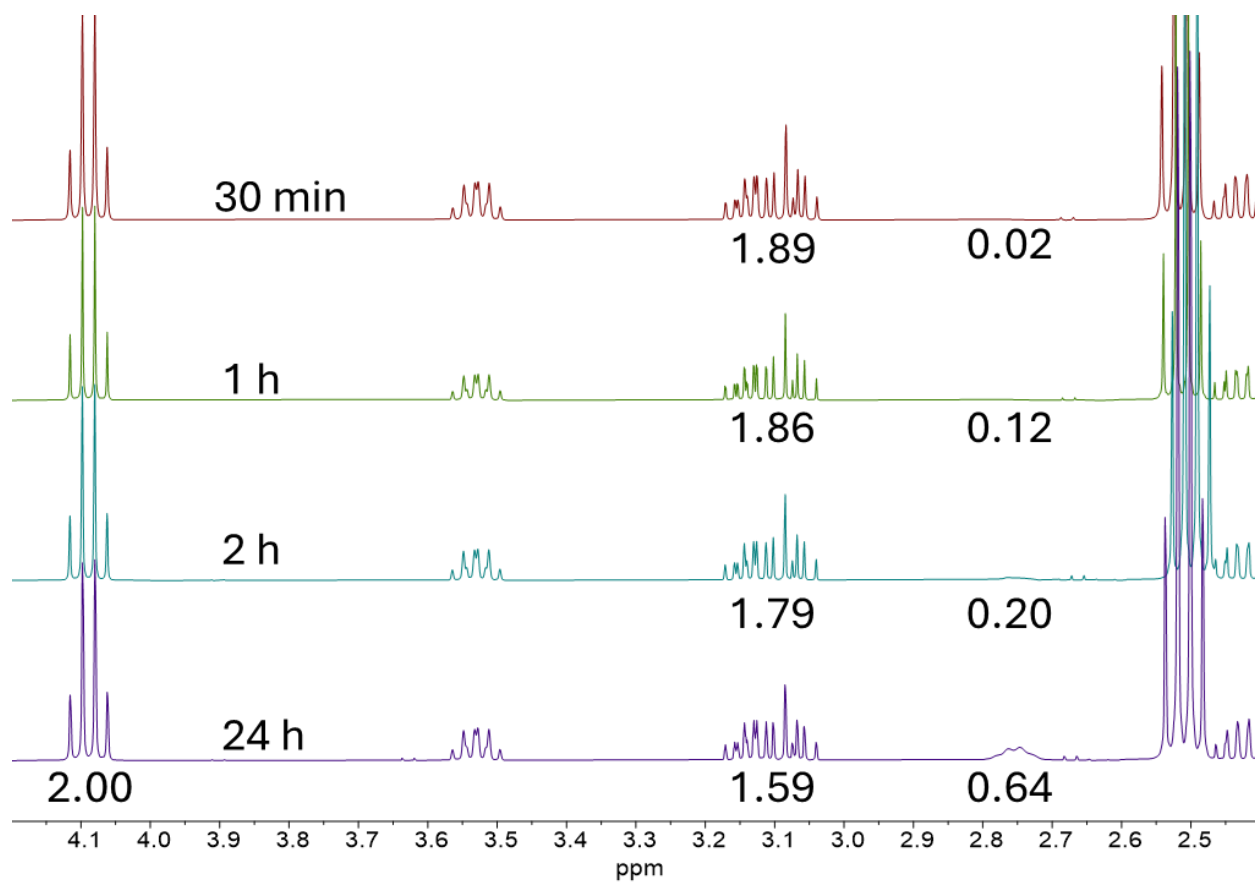


Figure S24.

Crude ¹H-NMR of CROP of EtLp in Rodiasolv® polar clean. [EtLp]₀ = 4.0M, 0.05 equiv. DPP, Rodiasolv®, 30 °C. Integration values are shown at different time stamps. ¹H-NMR spectrum – 400 MHz, 298 K, CDCl₃.

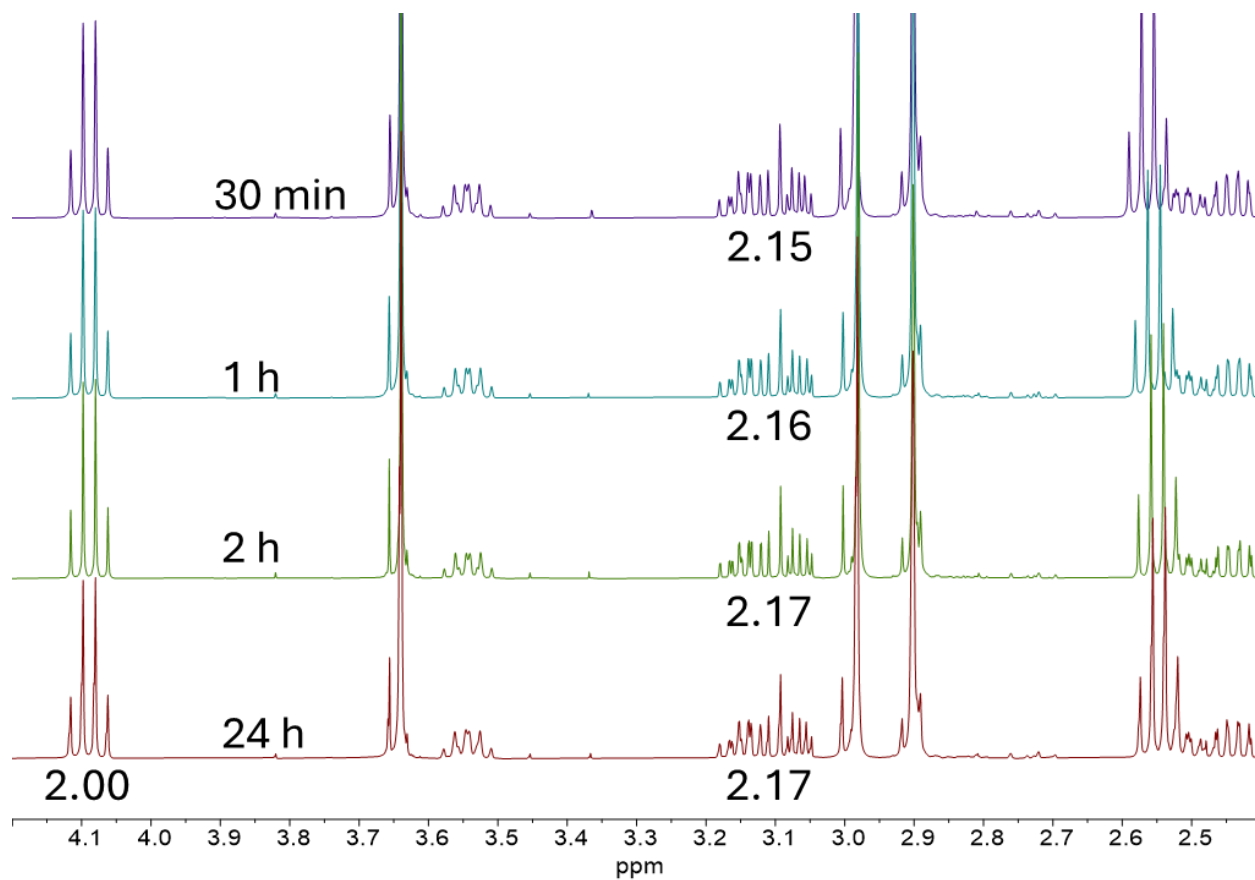


Figure S25.

Crude ^1H -NMR of CROP of EtLp in DMF. $[\text{EtLp}]_0 = 4.0\text{M}$, 0.05 equiv. DPP, DMF, 30 °C. Integration values are shown at different time stamps. ^1H -NMR spectrum – 400 MHz, 298 K, CDCl_3 .

Table S2.

Summary of conversion data for solvent screening for the CROP of EtLp.

Solvent	0.5 h	1 h	2 h	24 h
DCM	45 ± 7	50 ± 3	53 ± 3	64 ± 4
Acetone	47 ± 1	54 ± 2	59 ± 2	61.1 ± 0.7
PC	64 ± 1	66 ± 3	62 ± 2	60 ± 2
Cyrene	57 ± 3	63 ± 2	56 ± 8	56 ± 3
Chloroform	31 ± 1	37 ± 2	42 ± 2	55 ± 1
DMI	25 ± 1	28 ± 2	31 ± 1	55 ± 1
THF	4.3 ± 0.2	9 ± 1	19 ± 4	54.2 ± 0.4
1,4-diOx	48.2 ± 0.8	48 ± 1	49 ± 1	53 ± 2
Toluene	7.3 ± 0.5	13 ± 1	20 ± 1	43 ± 1
CPME	7 ± 1	11 ± 1	16 ± 1	37 ± 3
2-MeTHF	6.0 ± 0.7	6.5 ± 0.1	9.0 ± 0.3	31 ± 3
CCl ₄	2.9 ± 0.4	4.3 ± 0.4	6.9 ± 0.1	25 ± 3
Rodiasolv®	1.3 ± 0.6	0.3 ± 0.6	0.7 ± 0.6	2 ± 2
DMF	2 ± 2	0.3 ± 0.6	0 ± 0	0 ± 0

n = 3 for all solvents at each time point; [EtLp]₀ = 4.0 M, 0.05equiv. DPP, T = 30 °C. Conversion(%) was calculated using the equation in Figure S4.

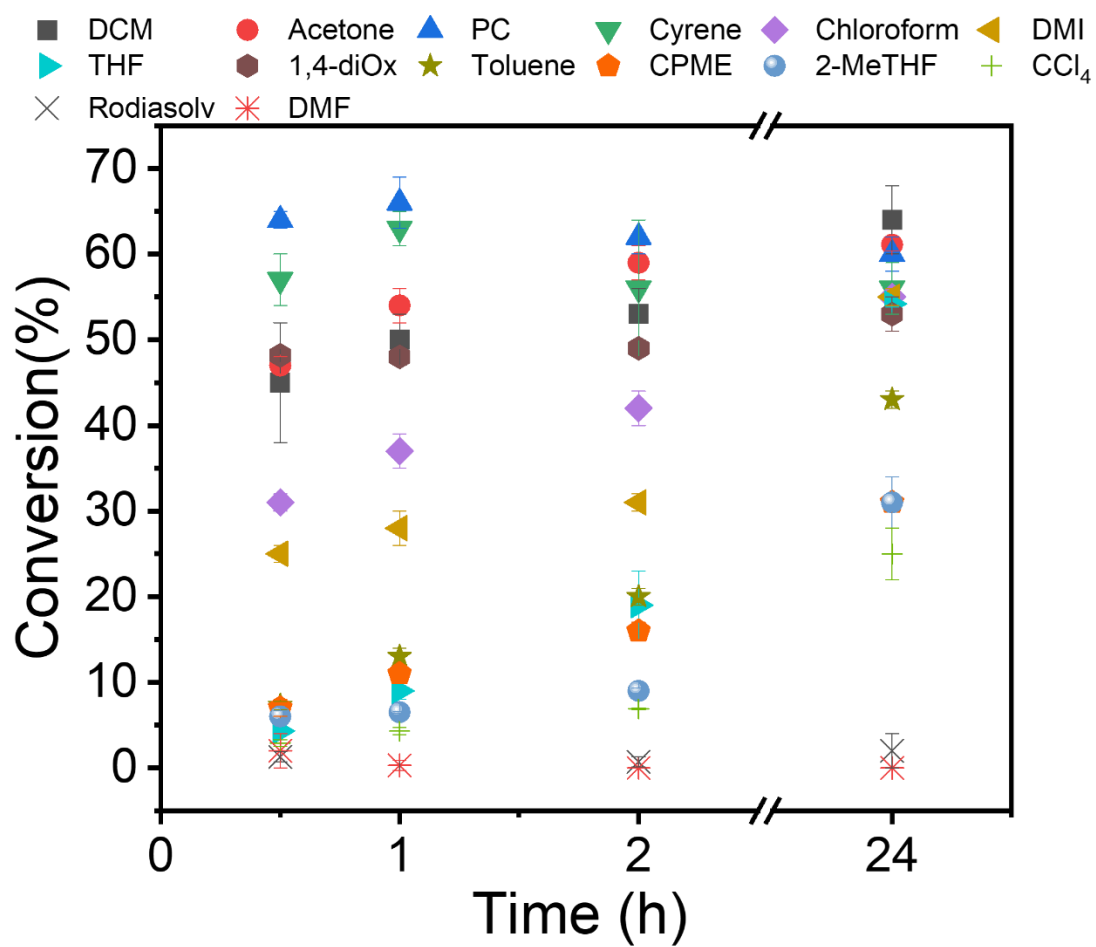


Figure S26.

Scatter plot of the conversion data for solvent screening for the CROP of EtLp. Data is summarized in Table S2.

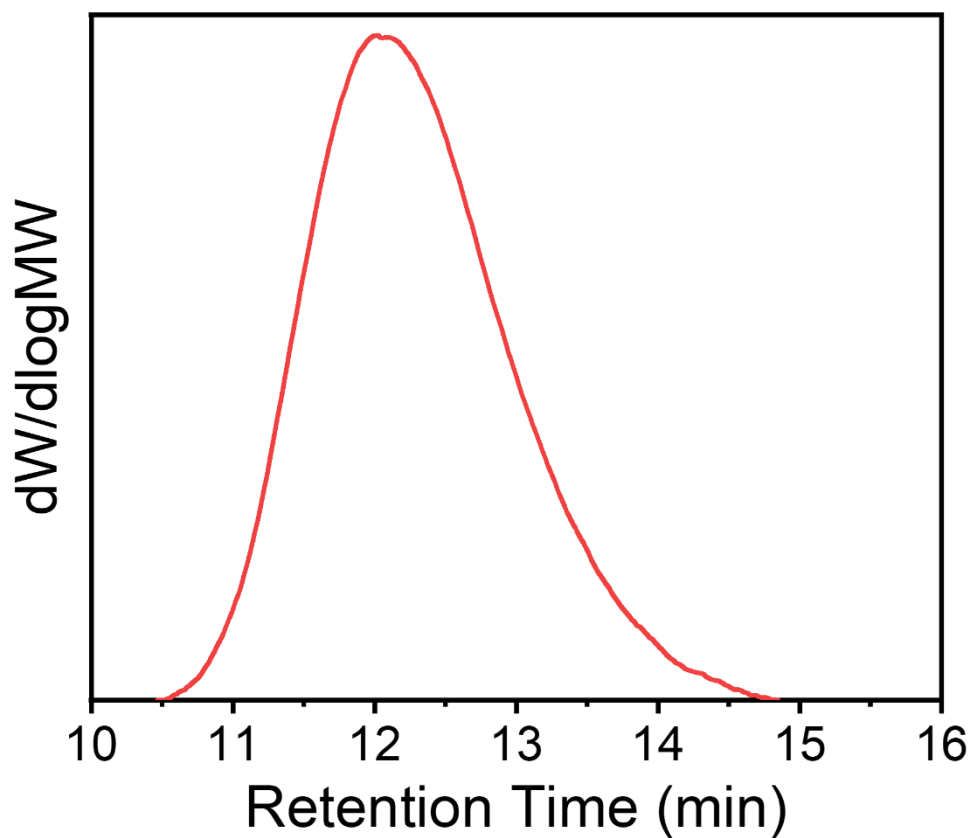


Figure S27.

GPC chromatogram of crude polymer from the CROP of EtLp in DCM. $[EtLp]_0 = 4.0$ M, 0.05 equiv. DPP, DCM, 30 °C, 24 h. Molecular weight data listed in Table S5 entry 1. *Elution of residual monomer not shown but it appears at ~20 min retention time.

ROP Thermodynamics

Concentration effects

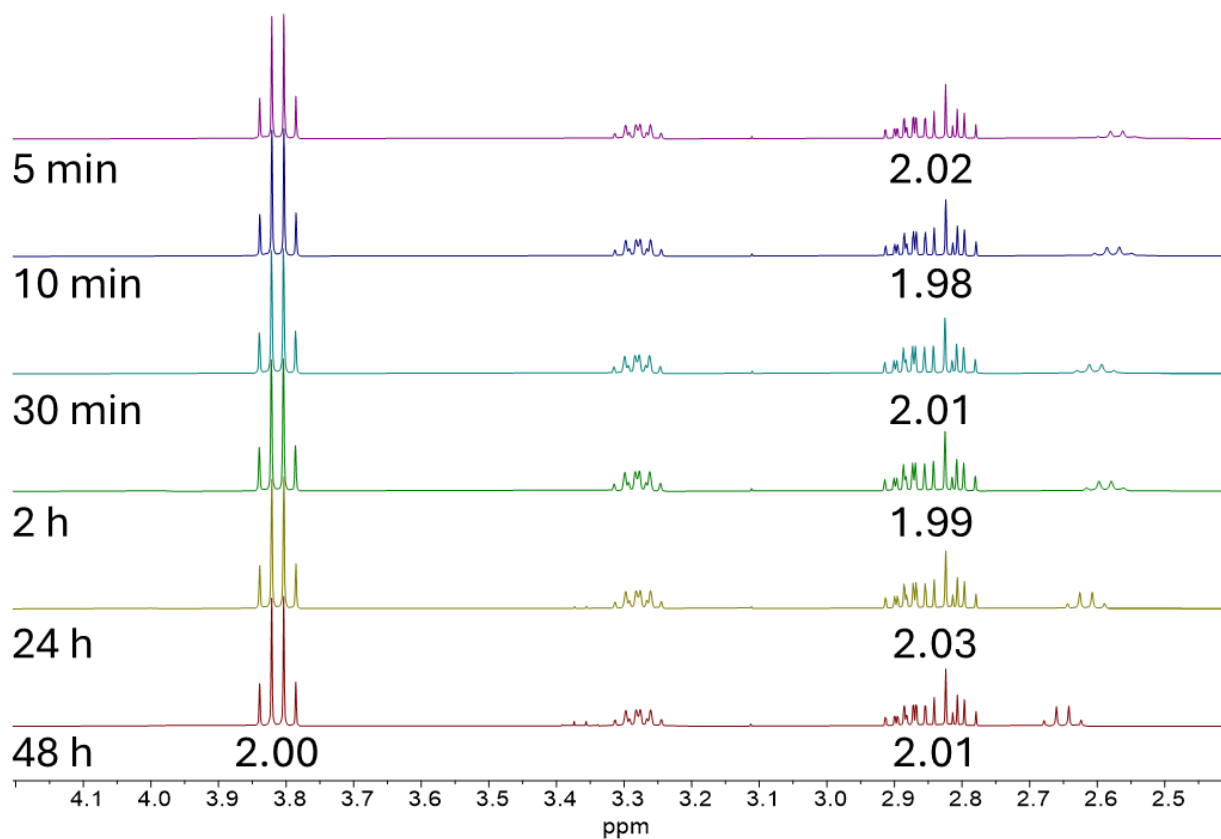


Figure S28.

Crude ¹H-NMR of CROP of EtLp at an initial monomer concentration of 0.5 M. [EtLp]₀ = 0.5 M, 0.05 equiv. DPP, acetone, 30 °C. Integration values are shown at different time stamps. ¹H-NMR spectrum – 400 MHz, 298 K, CDCl₃.

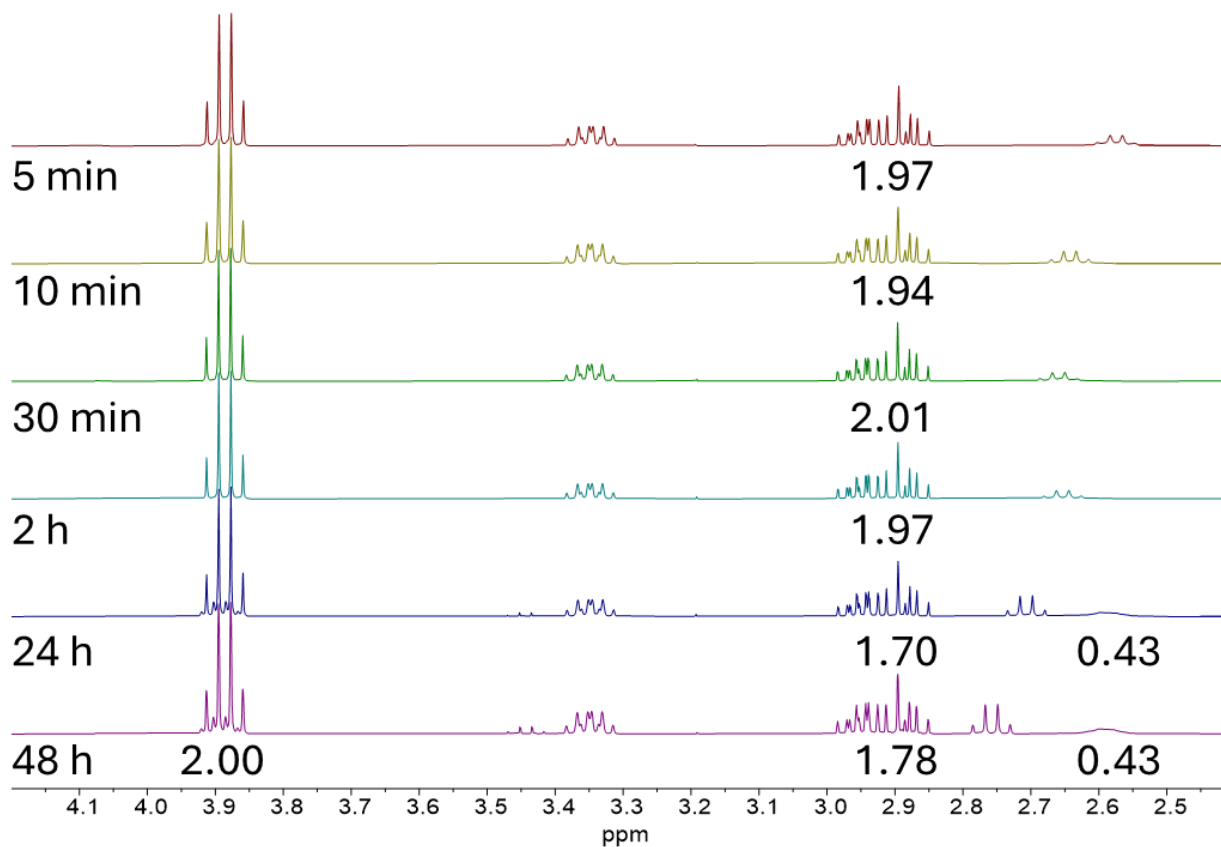


Figure S29.

Crude ^1H -NMR of CROP of EtLp at initial monomer concentration of 1.0 M. $[\text{EtLp}]_0 = 1.0$ M, 0.05 equiv. DPP, acetone, 30 °C. Integration values are shown at different time stamps. ^1H -NMR spectrum – 400 MHz, 298 K, CDCl_3 .

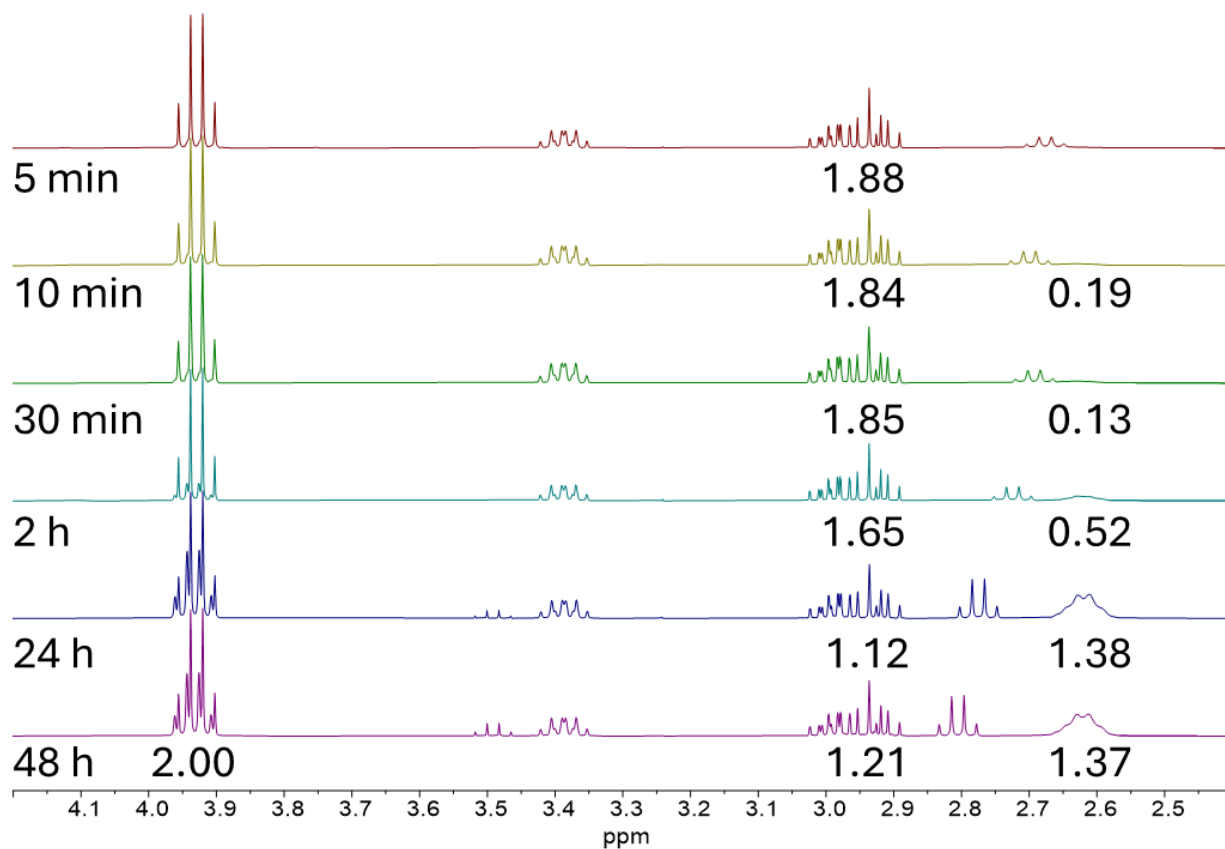


Figure S30.

Crude ¹H-NMR of CROP of EtLp at initial monomer concentration of 2.0 M. [EtLp]₀ = 2.0 M, 0.05 equiv. DPP, acetone, 30 °C. Integration values are shown at different time stamps. ¹H-NMR spectrum – 400 MHz, 298 K, CDCl₃.

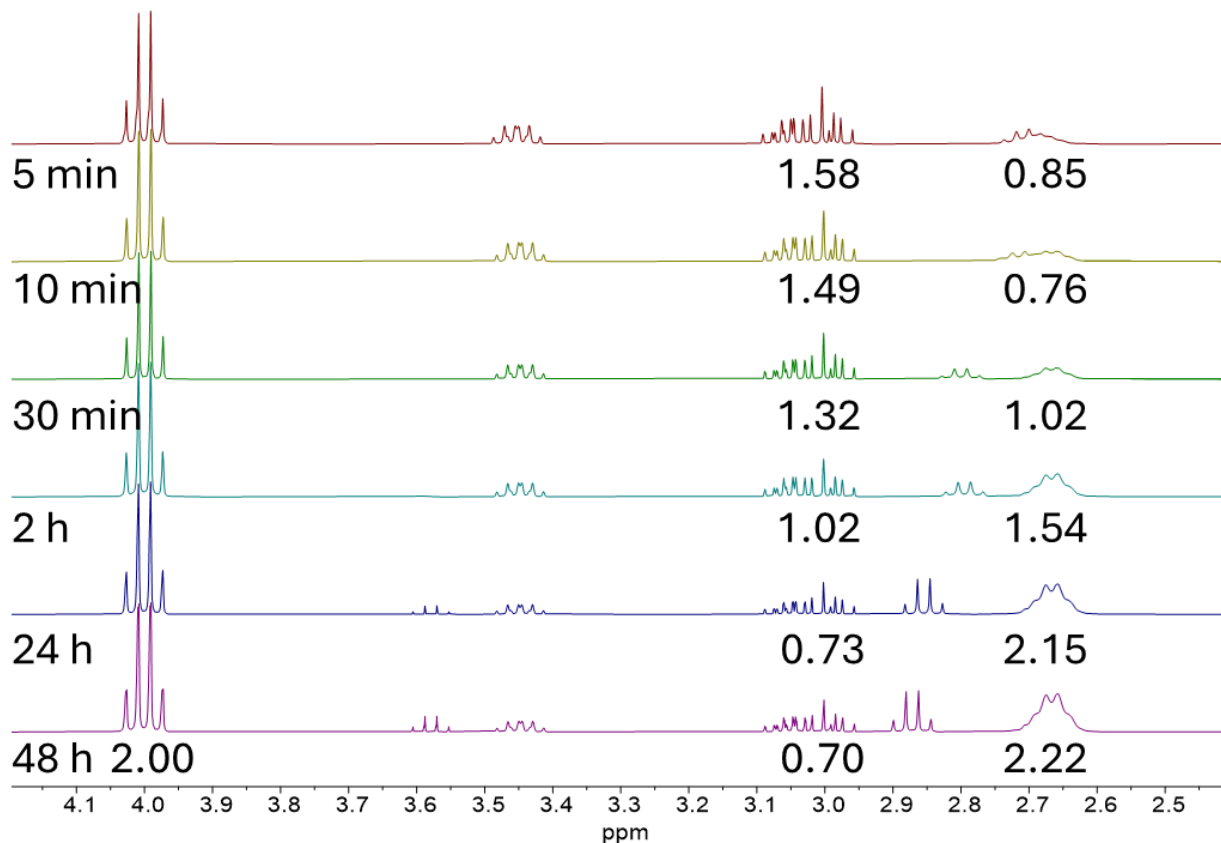


Figure S31.

Crude ^1H -NMR of CROP of EtLp at initial monomer concentration of 4.0 M. $[\text{EtLp}]_0 = 4.0 \text{ M}$, 0.05 equiv. DPP, acetone, 30°C . Integration values are shown at different time stamps. ^1H -NMR spectrum – 400 MHz, 298 K, CDCl_3 .

Table S3.

Summary of conversion data for the CROP of EtLp at different initial monomer concentrations.

Time	0.5 M	1 M	2 M	4 M
5 min	3 ± 1	3.8 ± 0.1	3 ± 2	19.9 ± 0.4
10 min	1 ± 1	6 ± 3	6.7 ± 0.7	31.2 ± 0.4
30 min	0 ± 0	0 ± 0	6 ± 1	34.0 ± 0.1
2 h	0 ± 0	0 ± 0	17.0 ± 0.8	49.8 ± 0.3
24 h	0 ± 0	12 ± 1	44.5 ± 0.1	65.5 ± 0.7
48 h	0 ± 0	13.1 ± 0.9	41 ± 2	67.7 ± 0.3

$n = 3$ for all solvents at each time point; 0.05equiv. DPP, acetone, $T = 30^\circ\text{C}$. Conversion (%) was calculated using the equation in Figure S4.

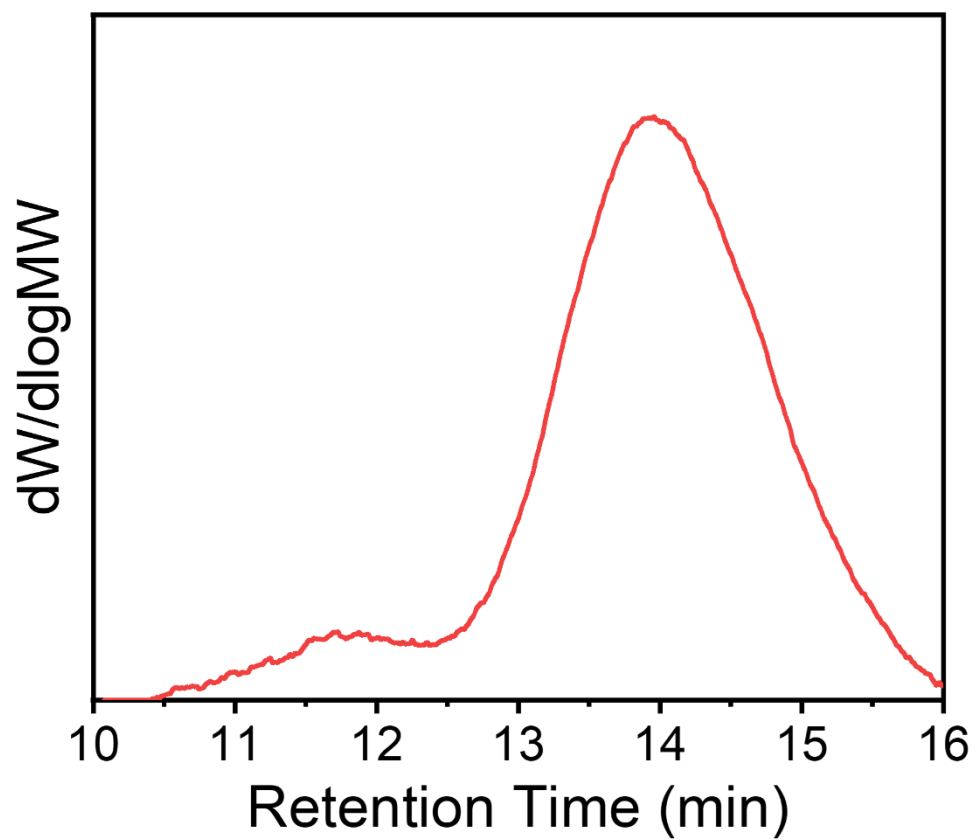


Figure S32.

GPC chromatogram of crude polymer from the CROP of EtLp at initial monomer concentration of 1.0 M. $[\text{EtLp}]_0 = 1.0 \text{ M}$, 0.05 equiv. DPP, acetone, 30 °C, 24 h. Molecular weight data listed in Table S5 entry 2.

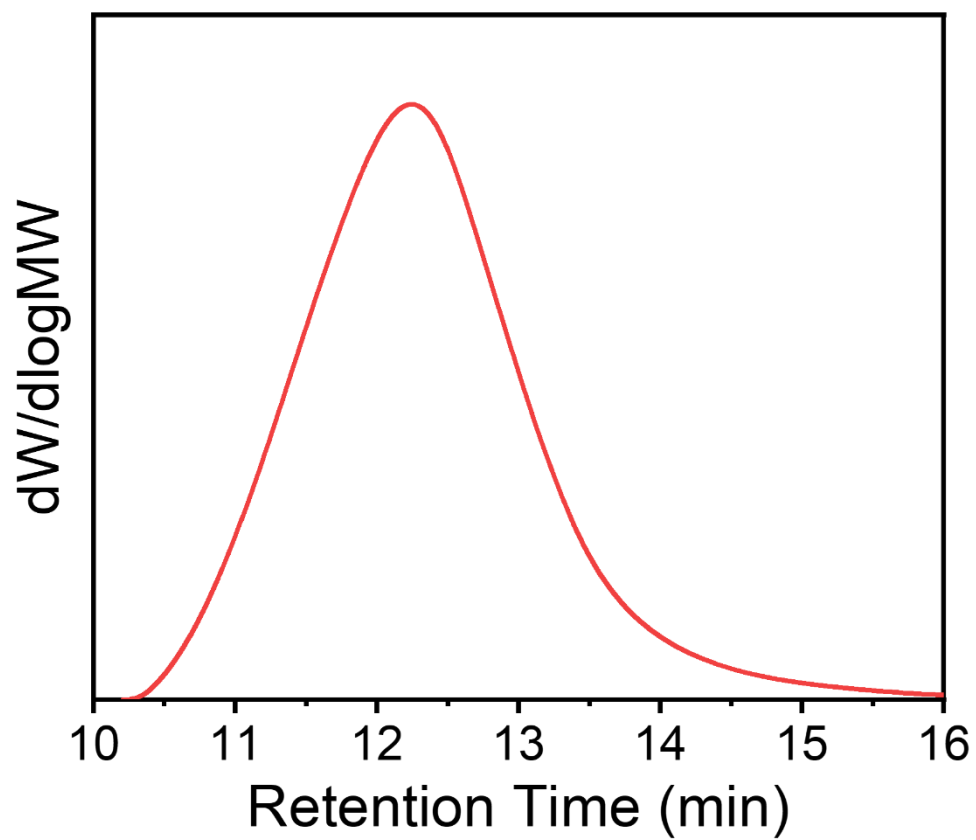


Figure S33.

GPC chromatogram of crude polymer from the CROP of EtLp at initial monomer concentration of 4.0 M. $[EtLp]_0 = 4.0$ M, 0.05 equiv. DPP, acetone, 30 °C, 24 h. Molecular weight data listed in Table S5 entry 4.

Catalyst Loading Effects

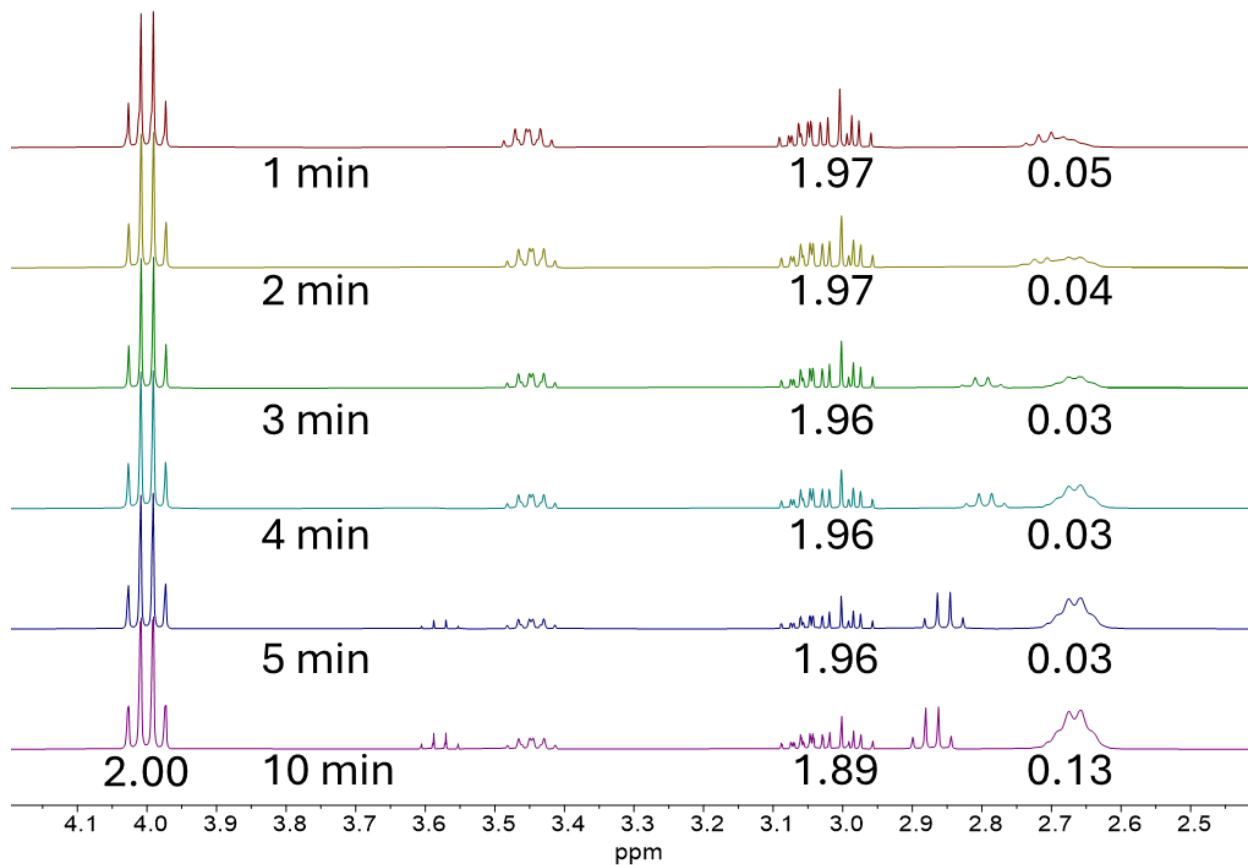


Figure S34.

Crude ¹H-NMR of polymer from the CROP of EtLp with 1 mol(%) DPP initiator at different time stamps. Integration values are shown. [EtLp]₀ = 4.0 M, 0.01 equiv. DPP, acetone, 30 °C. ¹H-NMR spectrum – 400 MHz, 298 K, CDCl₃.

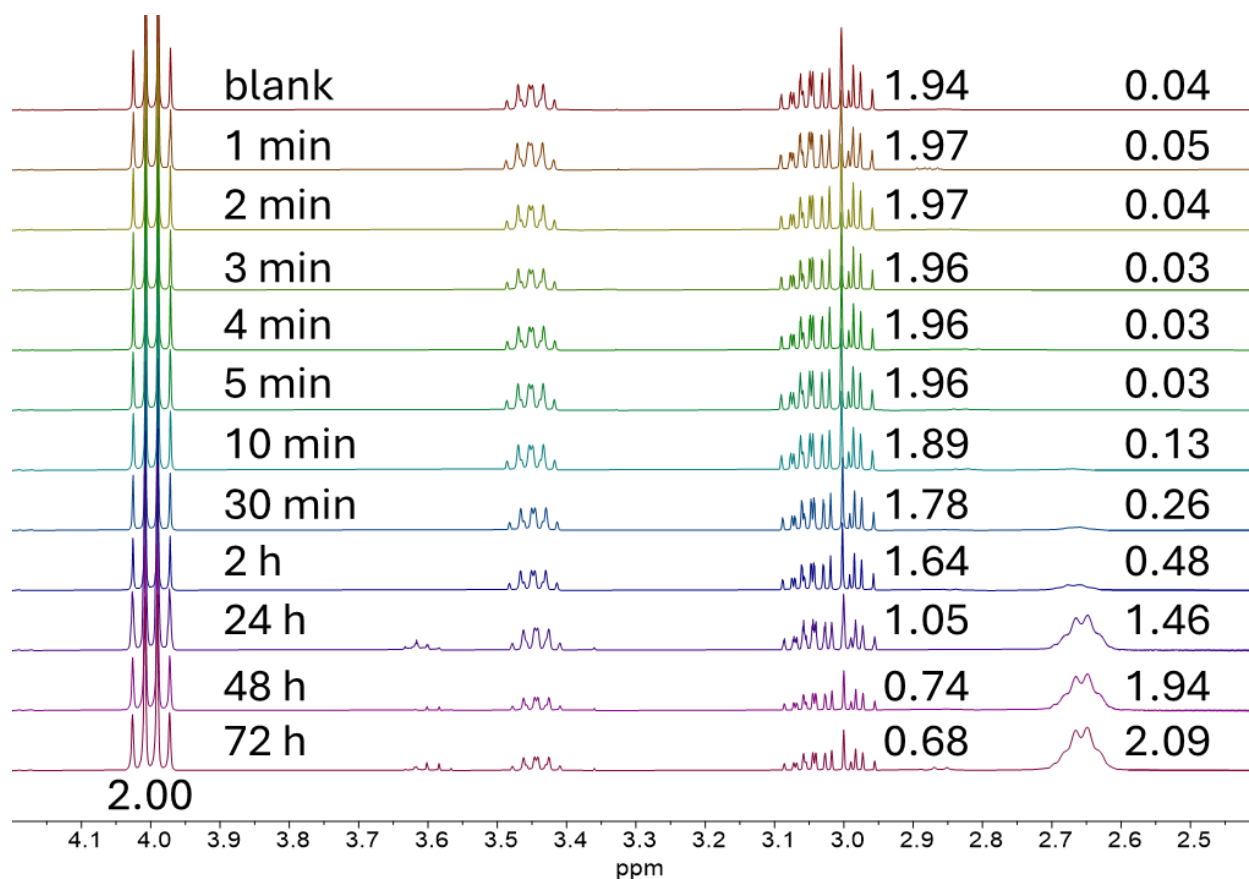


Figure S35.

Crude ^1H -NMR of polymer from the CROP of EtLp with 1 mol(%) DPP initiator at different time stamps. Integration values are shown. $[\text{EtLp}]_0 = 4.0 \text{ M}$, 0.01 equiv. DPP, acetone, 30 °C. ^1H -NMR spectrum – 400 MHz, 298 K, CDCl_3 .

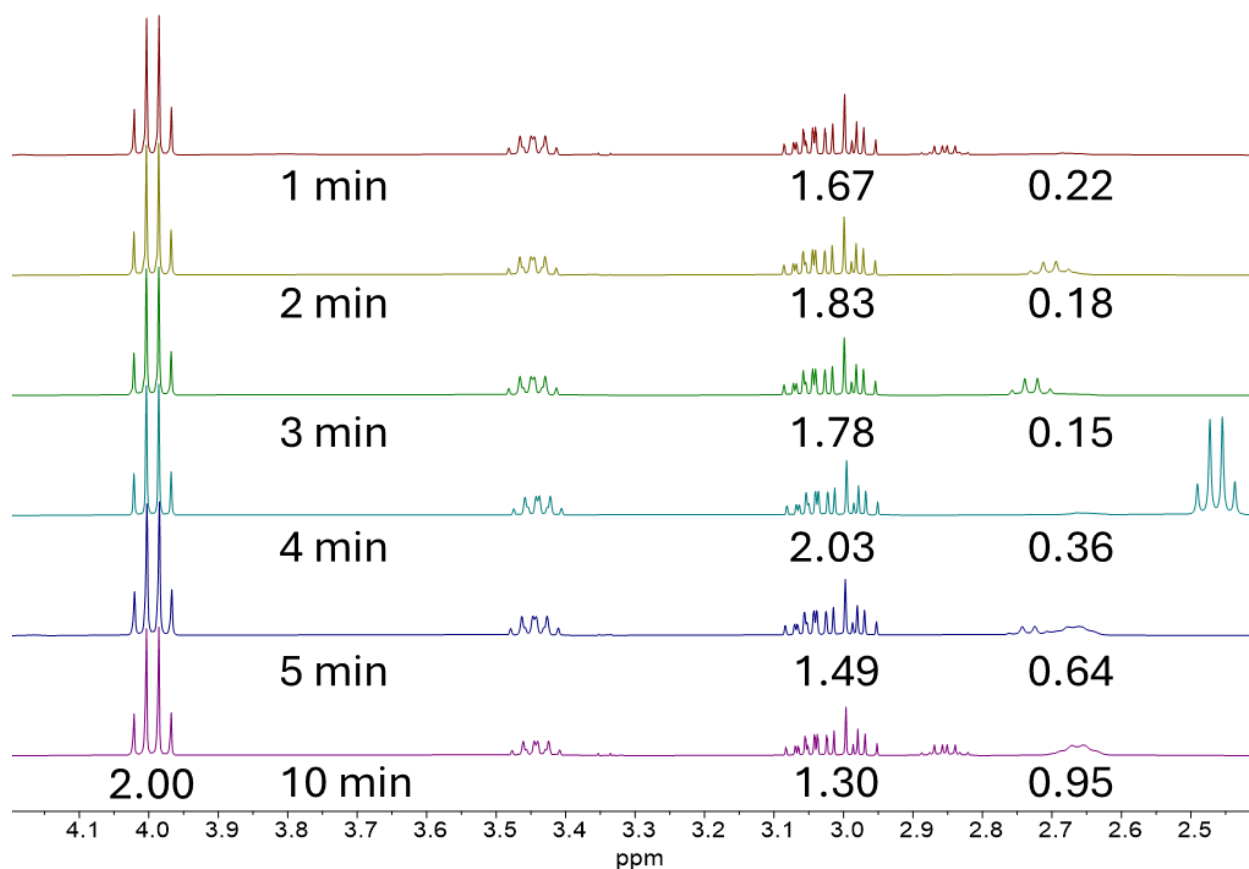


Figure S36.

Crude ¹H-NMR of polymer from the CROP of EtLp with 5 mol(%) DPP initiator at different time stamps. Integration values are shown. [EtLp]₀ = 4.0 M, 0.05 equiv. DPP, acetone, 30 °C. ¹H-NMR spectrum – 400 MHz, 298 K, CDCl₃.

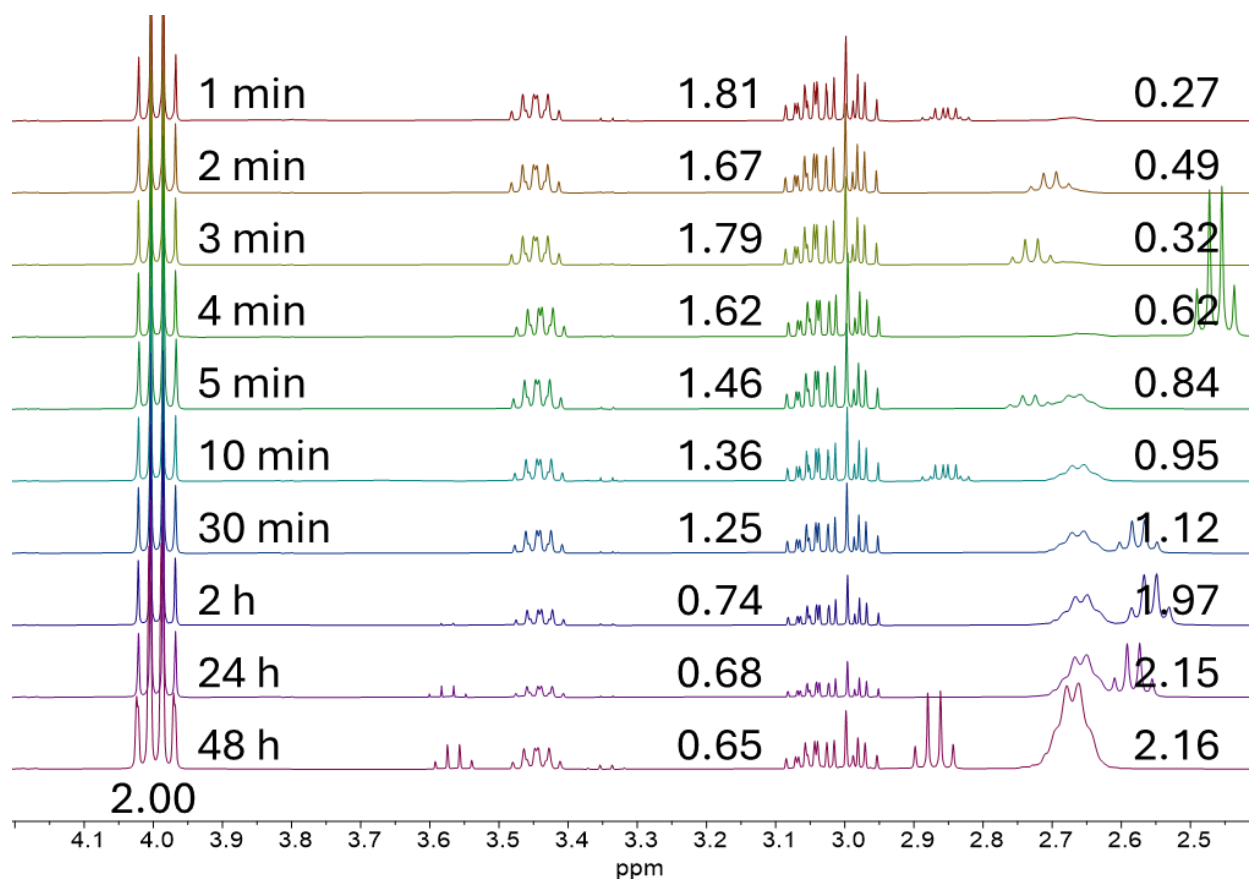


Figure S37.

Crude ^1H -NMR of polymer from the CROP of EtLp with 5 mol(%) DPP initiator at different time stamps. Integration values are shown. $[\text{EtLp}]_0 = 4.0 \text{ M}$, 0.05 equiv. DPP, acetone, 30 °C. ^1H -NMR spectrum – 400 MHz, 298 K, CDCl_3 .

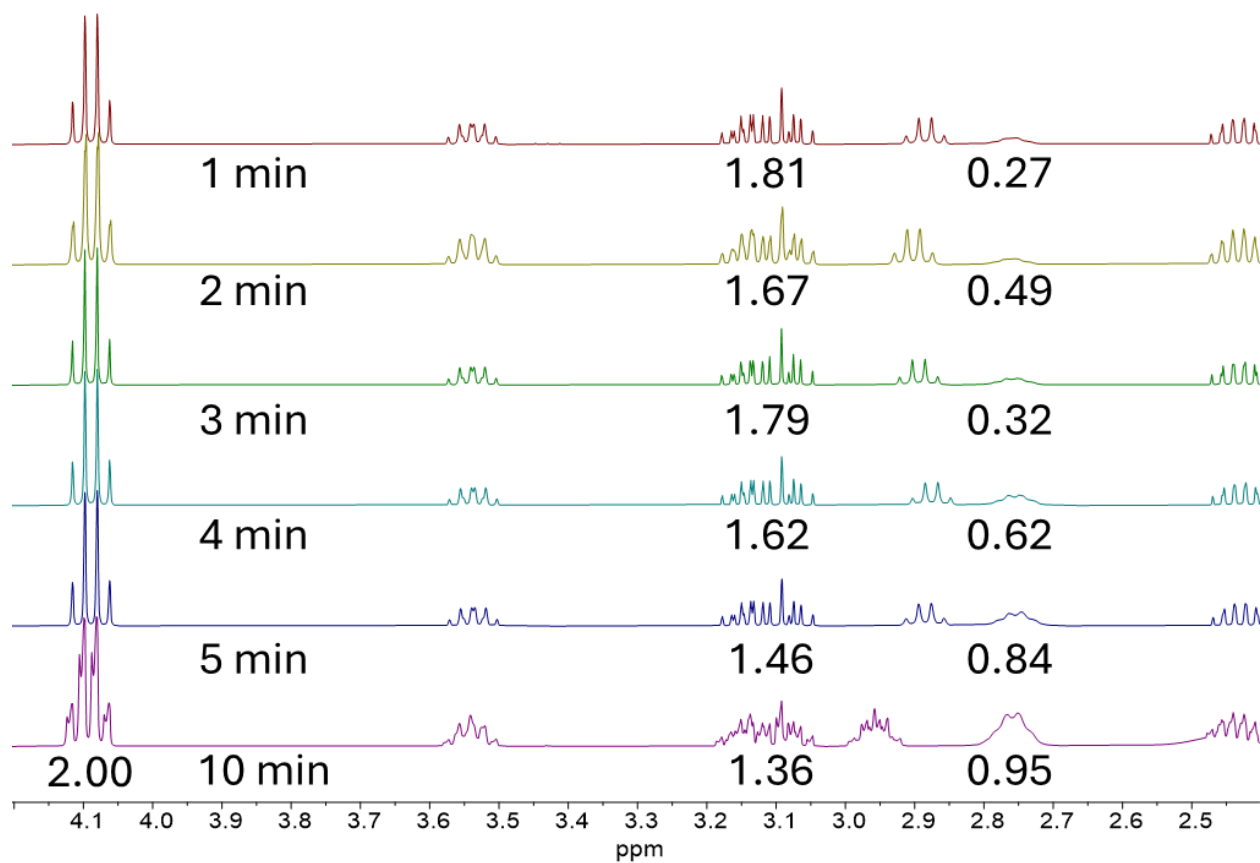


Figure S38.

Crude ¹H-NMR of polymer from the CROP of EtLp with 10 mol(%) DPP initiator at different time stamps. Integration values are shown. [EtLp]₀ = 4.0 M, 0.1 equiv. DPP, acetone, 30 °C. ¹H-NMR spectrum – 400 MHz, 298 K, CDCl₃.

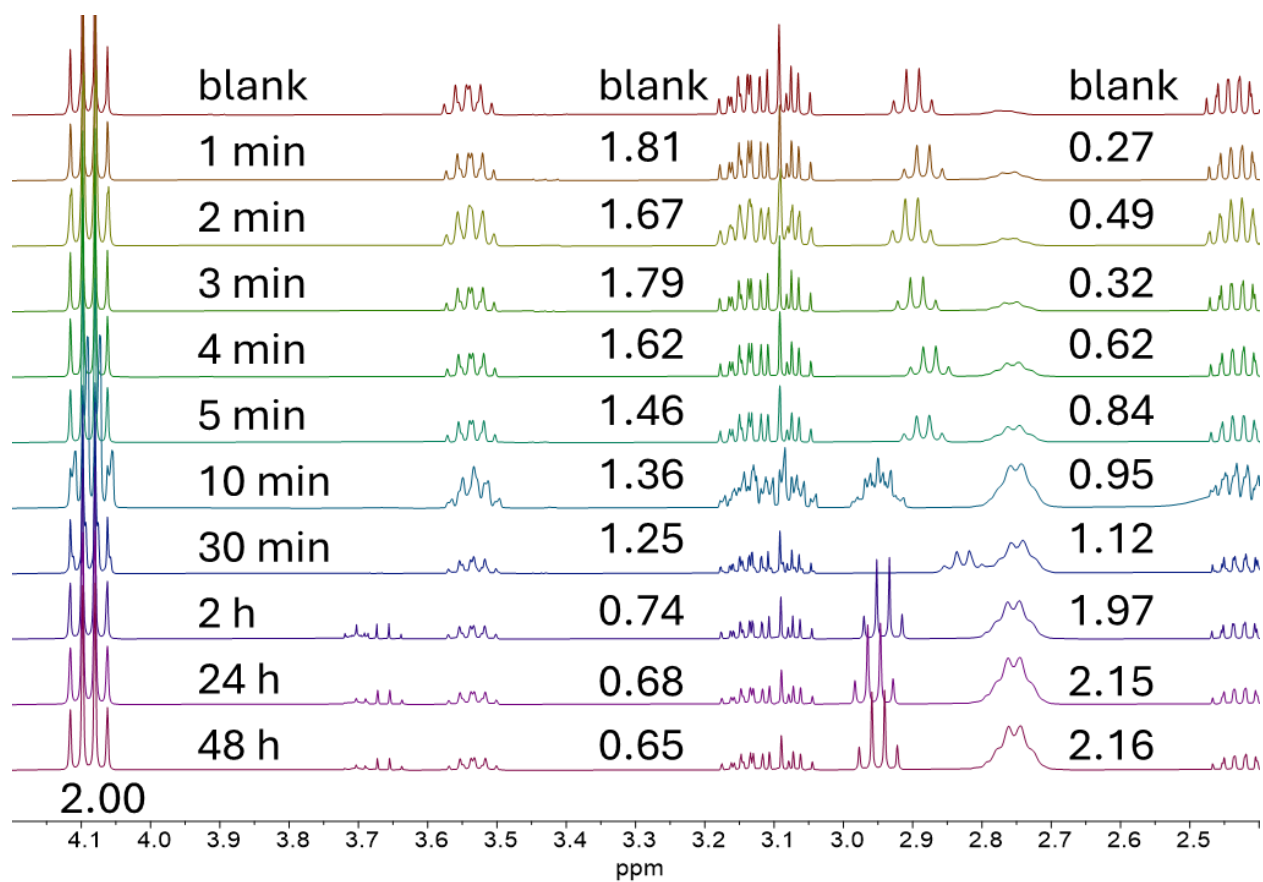


Figure S39.

Crude ^1H -NMR of polymer from the CROP of EtLp with 10 mol(%) DPP initiator at different time stamps. Integration values are shown. $[\text{EtLp}]_0 = 4.0 \text{ M}$, 0.1 equiv. DPP, acetone, 30°C . ^1H -NMR spectrum – 400 MHz, 298 K, CDCl_3 .

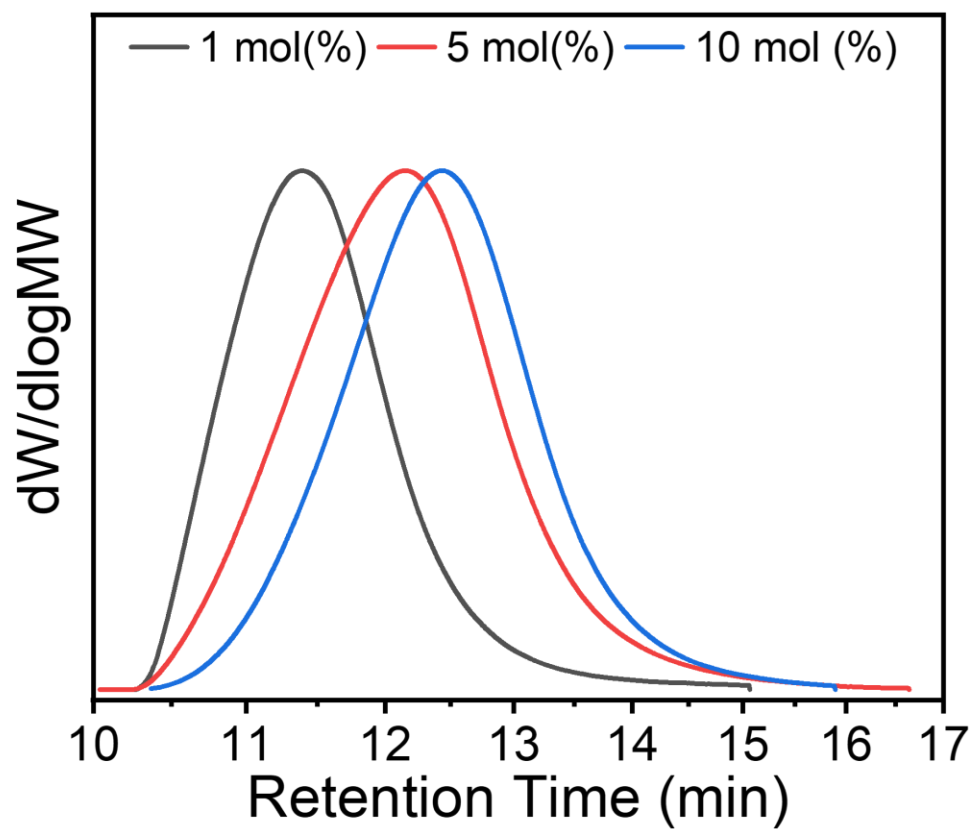


Figure S40.

GPC chromatogram of crude polymer from the CROP of EtLp at different initiator concentrations. $[EtLp]_0 = 4.0$ M, acetone, 30 °C, 24 h. Molecular weight data listed in Table S5 entries 3-5.

Temperature Effects

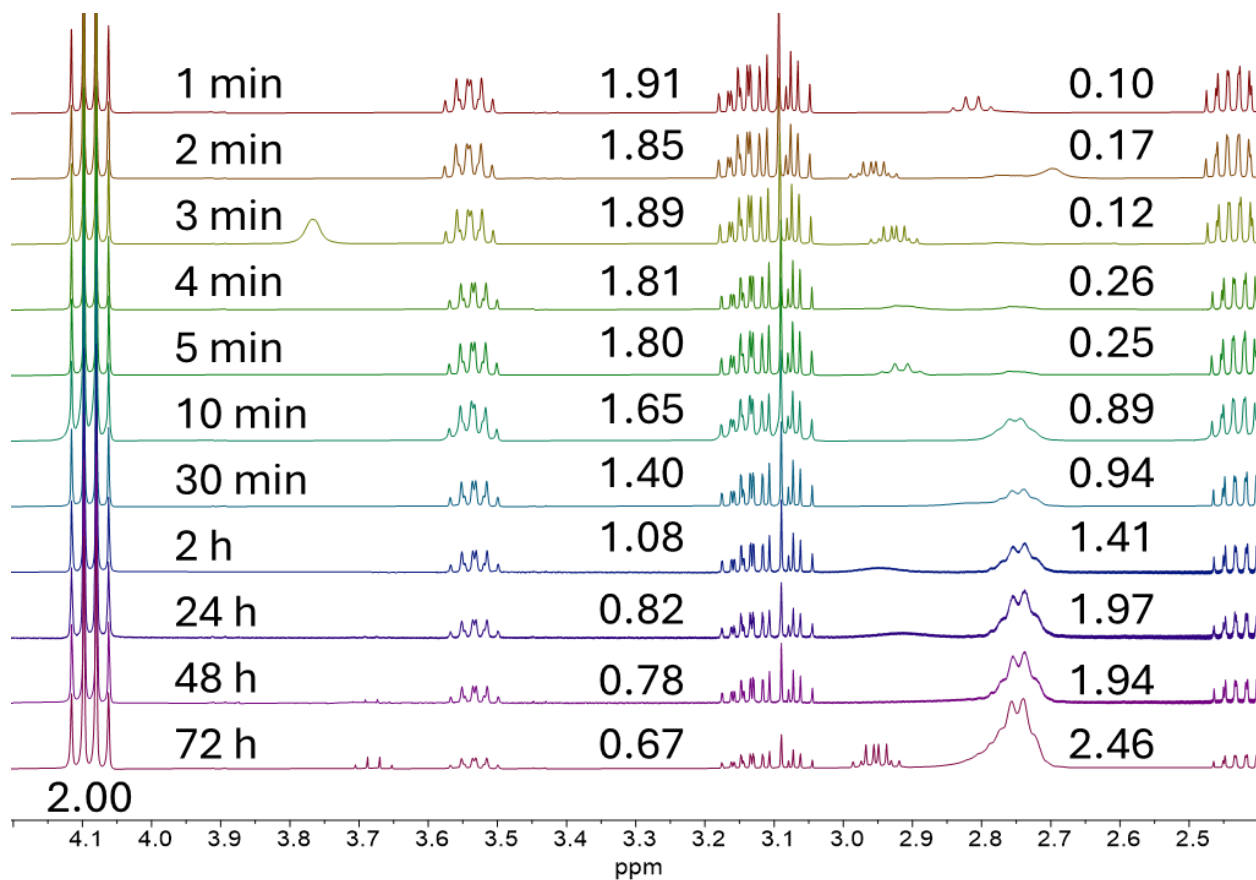


Figure S41.

Crude ^1H -NMR of polymer from the CROP of EtLp at 5 °C at different time stamps. Integration values are shown. $[\text{EtLp}]_0 = 4.0 \text{ M}$, 0.05 equiv. DPP, acetone, 5 °C. ^1H -NMR spectrum – 400 MHz, 298 K, CDCl_3 .

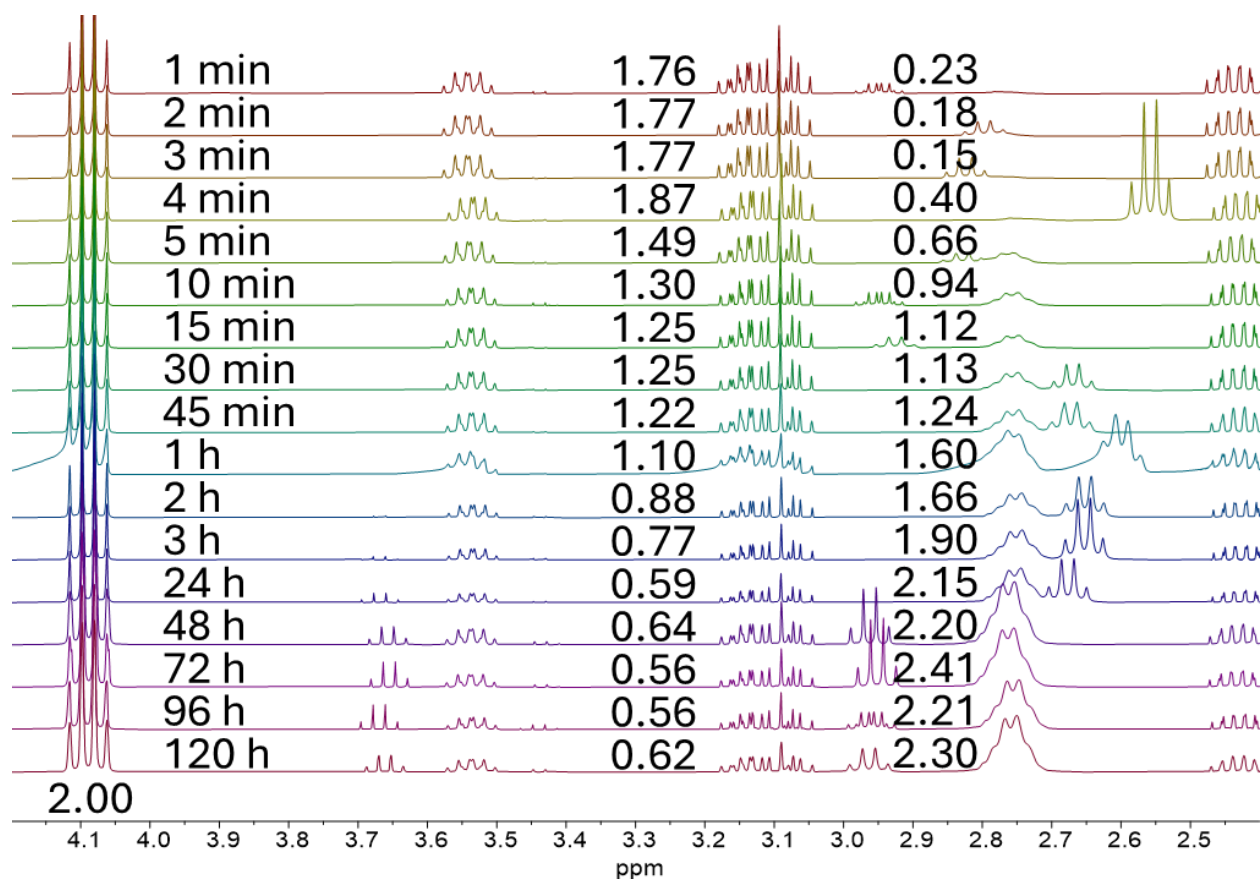


Figure S42.

Crude ^1H -NMR of polymer from the CROP of EtLp at 30 °C at different time stamps. Integration values are shown. $[\text{EtLp}]_0 = 4.0 \text{ M}$, 0.05 equiv. DPP, acetone, 30 °C. ^1H -NMR spectrum – 400 MHz, 298 K, CDCl_3 .

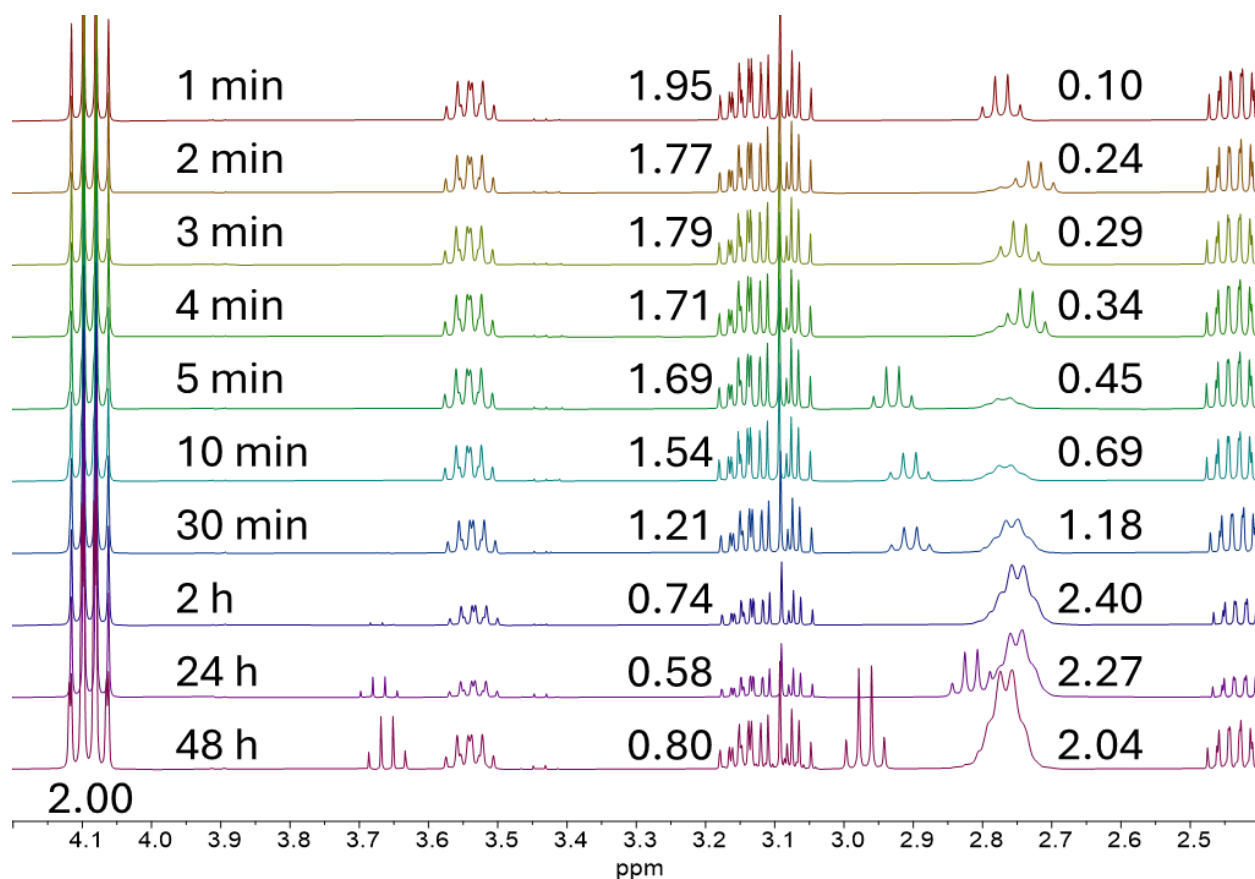


Figure S43.

Crude ^1H -NMR of polymer from the CROP of EtLp at 40 °C at different time stamps. Integration values are shown. $[\text{EtLp}]_0 = 4.0 \text{ M}$, 0.05 equiv. DPP, acetone, 40 °C. ^1H -NMR spectrum – 400 MHz, 298 K, CDCl_3 .

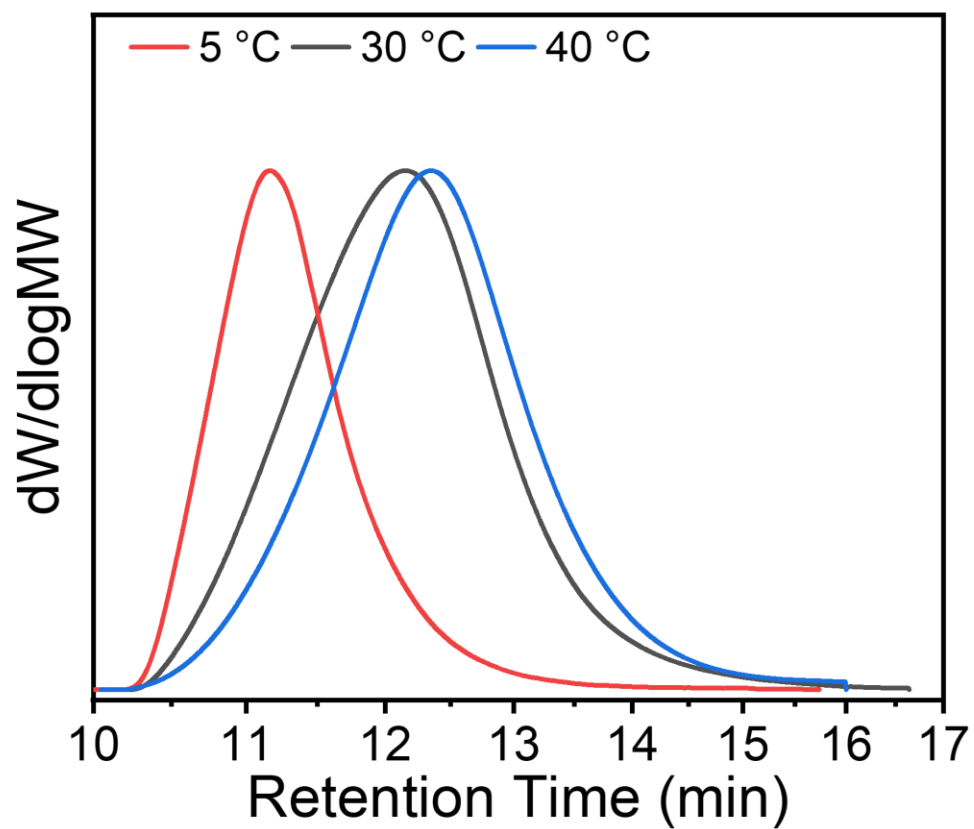


Figure S44.

GPC chromatogram of crude polymer from the CROP of EtLp at different temperatures. $[\text{EtLp}]_0 = 4.0 \text{ M}$, 0.05 equiv. DPP, acetone, 24 h. Molecular weight data listed in Table S5 entries 4, 6-7.

Equilibrium Thermodynamics

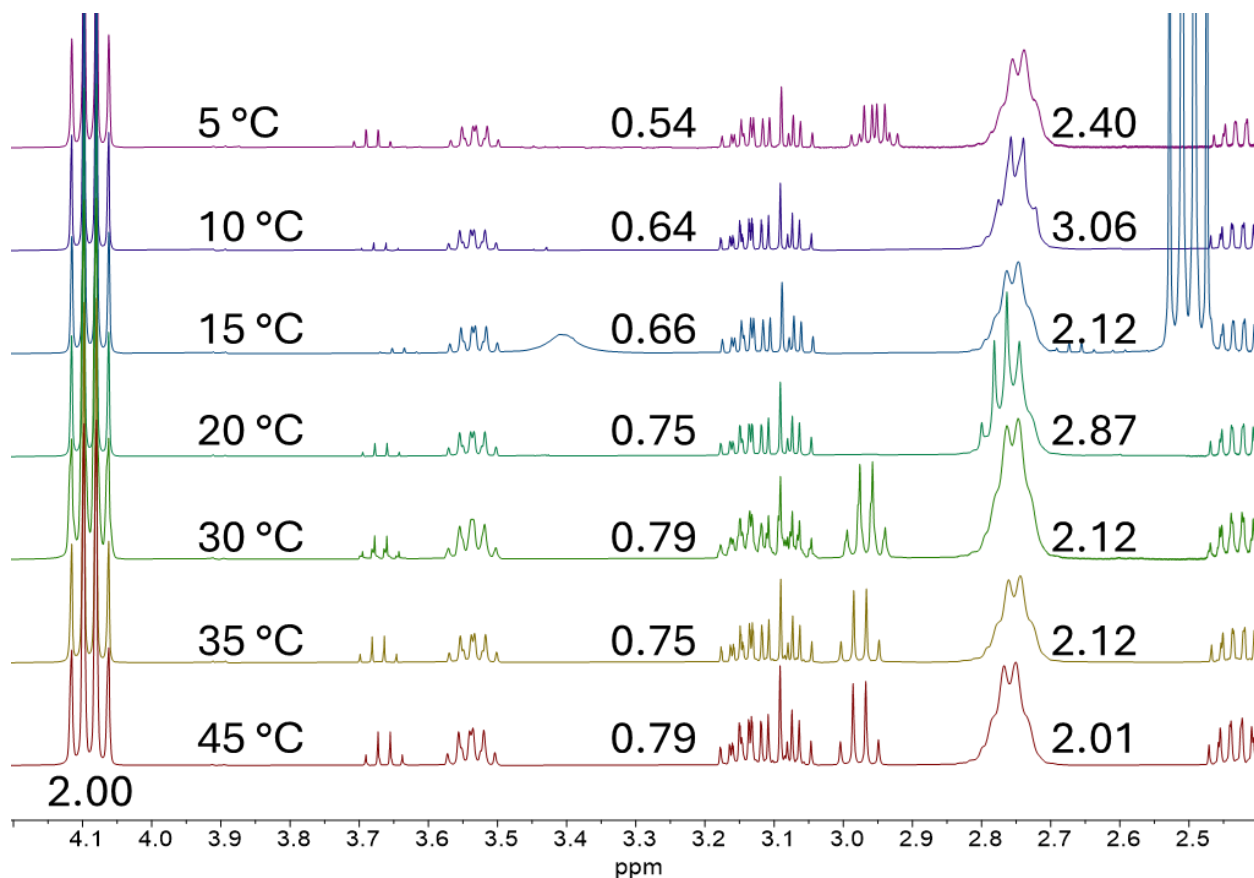


Figure S45.

Crude ^1H -NMR of polymer(s) from the CROP of EtLp at different temperatures after 24 h. Integration values are shown. $[\text{EtLp}]_0 = 4.0 \text{ M}$, 0.05 equiv. DPP, acetone. ^1H -NMR spectrum – 400 MHz, 298 K, CDCl_3 .

Table S4.

Summary of Equilibrium for the CROP of EtLp with DPP in acetone.

Temperature (K)	Conversion (%)	$[\text{EtLp}]_{\text{eq}} \text{ (M)}$	K_{eq}
278	78 ± 2	0.88	3.65 ± 0.11
283	76.2 ± 0.7	0.95	3.21 ± 0.04
288 ^a	67.2 ± 0.8	1.31	2.05 ± 0.03
293	69 ± 2	1.24	2.20 ± 0.09
303	65 ± 1	1.40	1.87 ± 0.05
308	62 ± 1	1.52	1.67 ± 0.05
313	59 ± 1	1.64	1.45 ± 0.05

$$\Delta H^\circ = -18 \pm 1 \text{ kJ mol}^{-1}, \Delta S^\circ = 57 \pm 3 \text{ J mol}^{-1} \text{ K}^{-1}$$

$n \geq 8$ for each temperature; conversion was calculated using the equation in Figure S4. K_{eq} was calculated by taking the ratio of polyEtLp to EtLp at equilibrium, thermodynamics constants were obtained using the Van't Hoff equation. ^a sample was determined to be an outlier and omitted from Van't Hoff analysis.

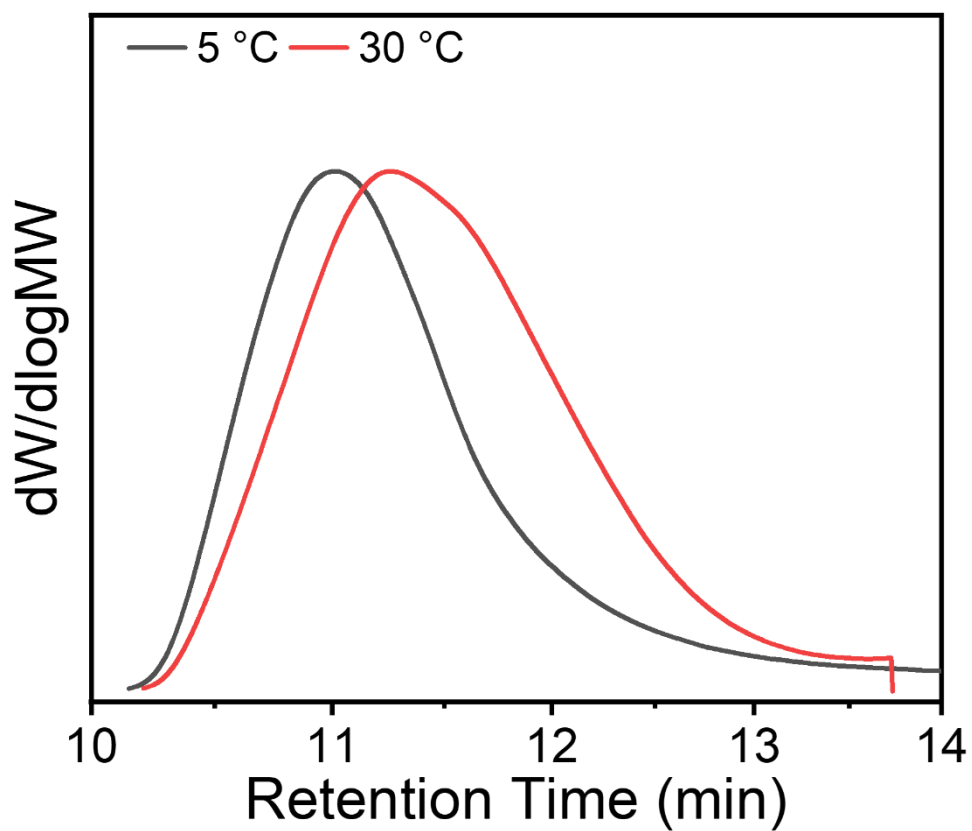


Figure S46.

GPC chromatogram of pure polymer from the control CROP of EtLp at different temperatures. The reaction was performed under a nitrogen atmosphere with dried reagents. $[EtLp]_0 = 4.0\text{ M}$, 0.05 equiv. DPP, acetone, 24 h, 1 atm N_2 . Molecular weight data listed in Table S5 entries 8, 9.

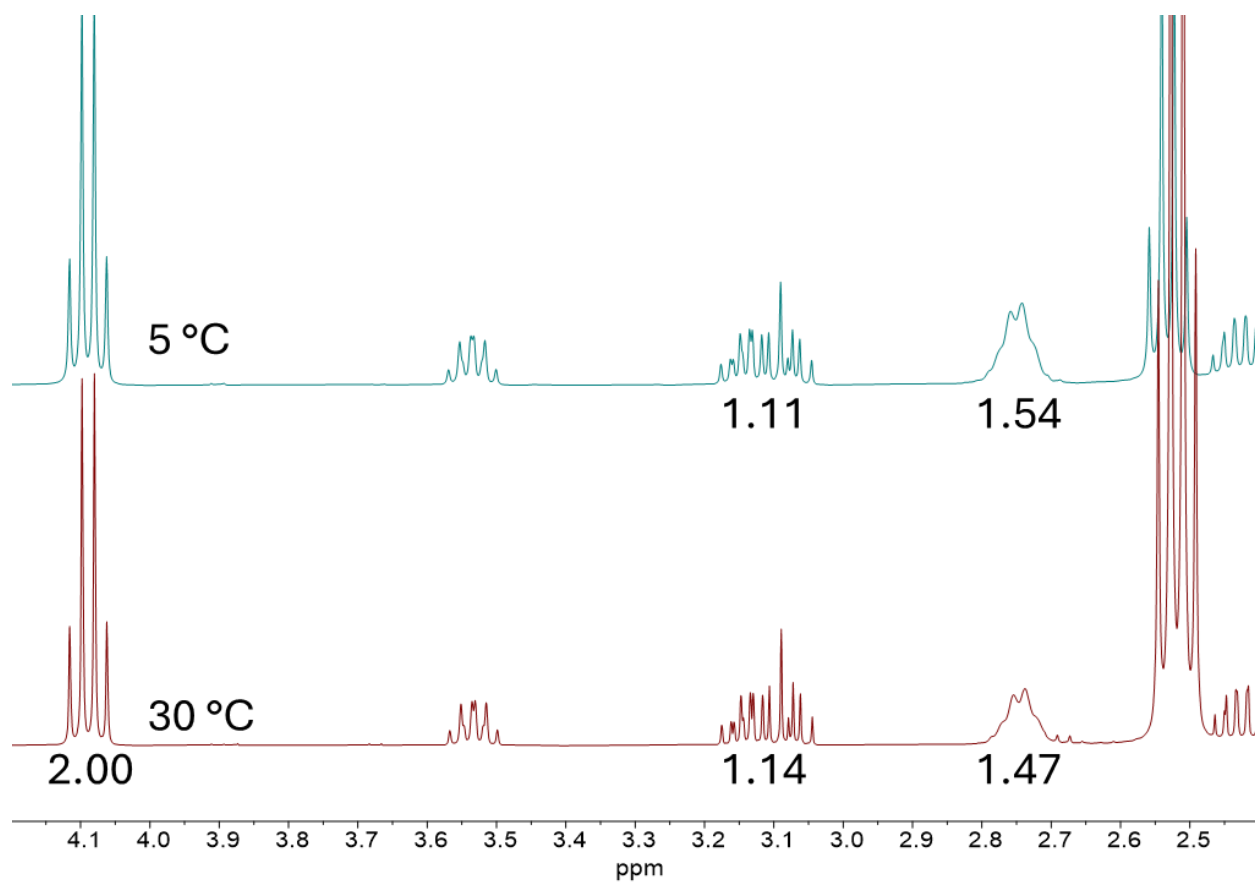


Figure S47.

Crude ¹H-NMR of polymer from the CROP of EtLp at 5 °C and 30 °C sampled at 24 h. The reaction was performed under a nitrogen atmosphere with dried reagents. Integration values are shown. [EtLp]₀ = 4.0 M, 0.05 equiv. DPP, acetone, 1 atm N₂. ¹H-NMR spectrum – 400 MHz, 298 K, CDCl₃.

Table S5.

Summary of Relevant Thermodynamic data for the CROP of EtLp with DPP in acetone.

Entry	[EtLp] ₀ (M)	mol(%) DPP	Solvent	T (°C)	Equilibrium Conversion (%)	<i>M_w</i> (kDa)	<i>M_n</i> (kDa)	<i>Đ</i>
1	4.0	5	DCM	30	64 ± 4	1000 ± 100	400 ± 300	1.8 ± 0.2
2	1.0	5	Acetone	30	13.1 ± 0.9	200 ± 20	68 ± 4	3.1 ± 0.3
3	4.0	1	Acetone	30	60 ± 4	1400 ± 400	700 ± 200	2.1 ± 0.3
4	4.0	5	Acetone	30	67.7 ± 0.3	1040 ± 70	350 ± 70	2.9 ± 0.3
5	4.0	10	Acetone	30	64 ± 1	710 ± 60	300 ± 30	2.4 ± 0.1
6	4.0	5	Acetone	5	78 ± 2	2300 ± 600	1400 ± 500	1.6 ± 0.2
7	4.0	5	Acetone	40	59 ± 1	910 ± 700	430 ± 80	2.2 ± 0.2
8 ^a	4.0	5	Acetone	5	49	2300	1300	1.8
9 ^a	4.0	5	Acetone	30	46	1800	1100	1.7

n = 3 for entries 1–7, n = 1 for entries 8, 9; % conversion was calculated using the equation in Figure S4; ^a reactions were performed under a nitrogen atmosphere using dried solvents and reagents.

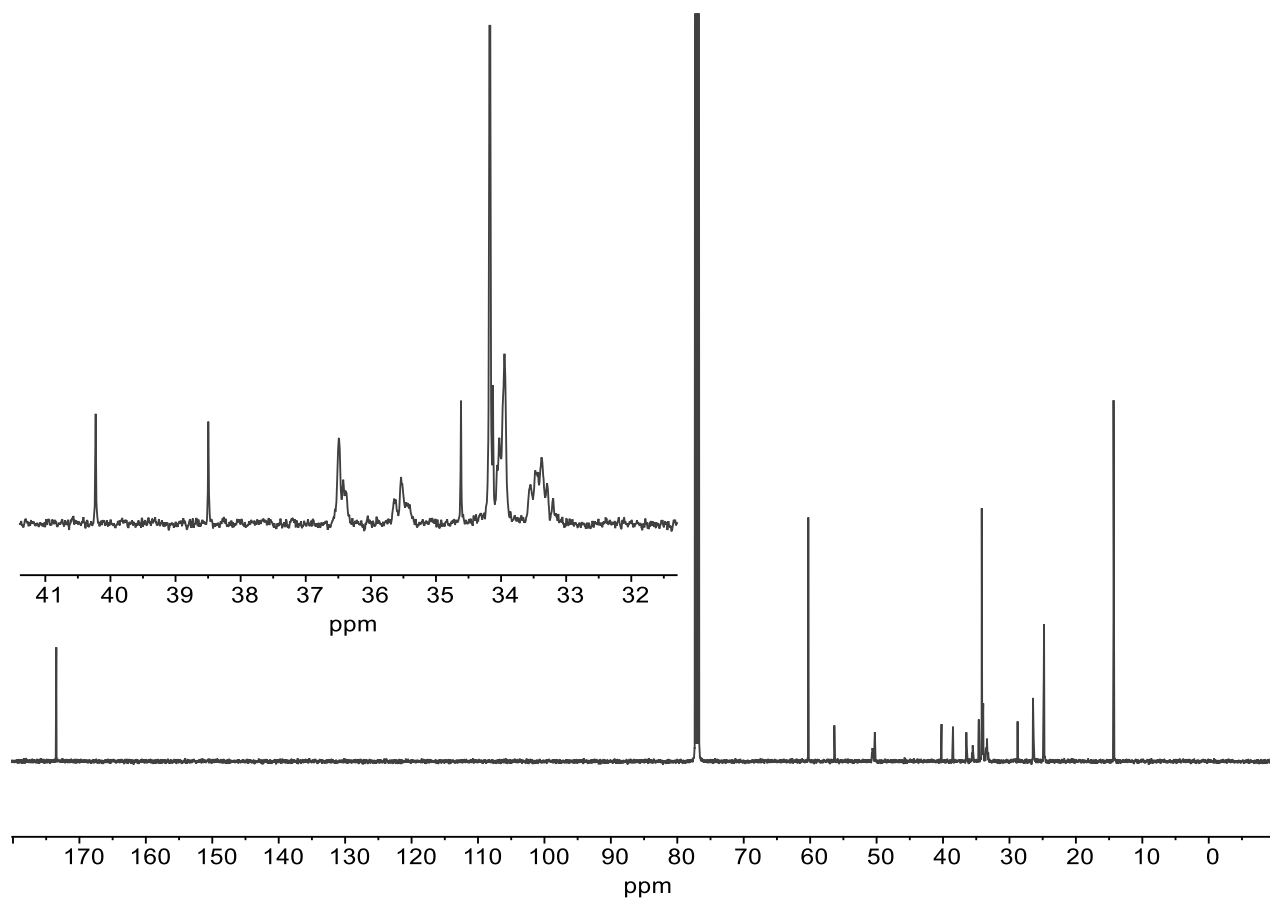


Figure S48.

^{13}C -NMR of polymer from Table S5, entry 6. Our polymer regioselectivity appears mostly random as evidenced by a similar abundance of four triads ($\delta = 35.5\text{--}36.5$). A regioselective polymerization was recently reported by Liu & co-workers; the poly(methyl lipoate) displayed ~85% H–T orientation as evidenced by a high relative abundance of ss triad ($\delta = 36.5$).²

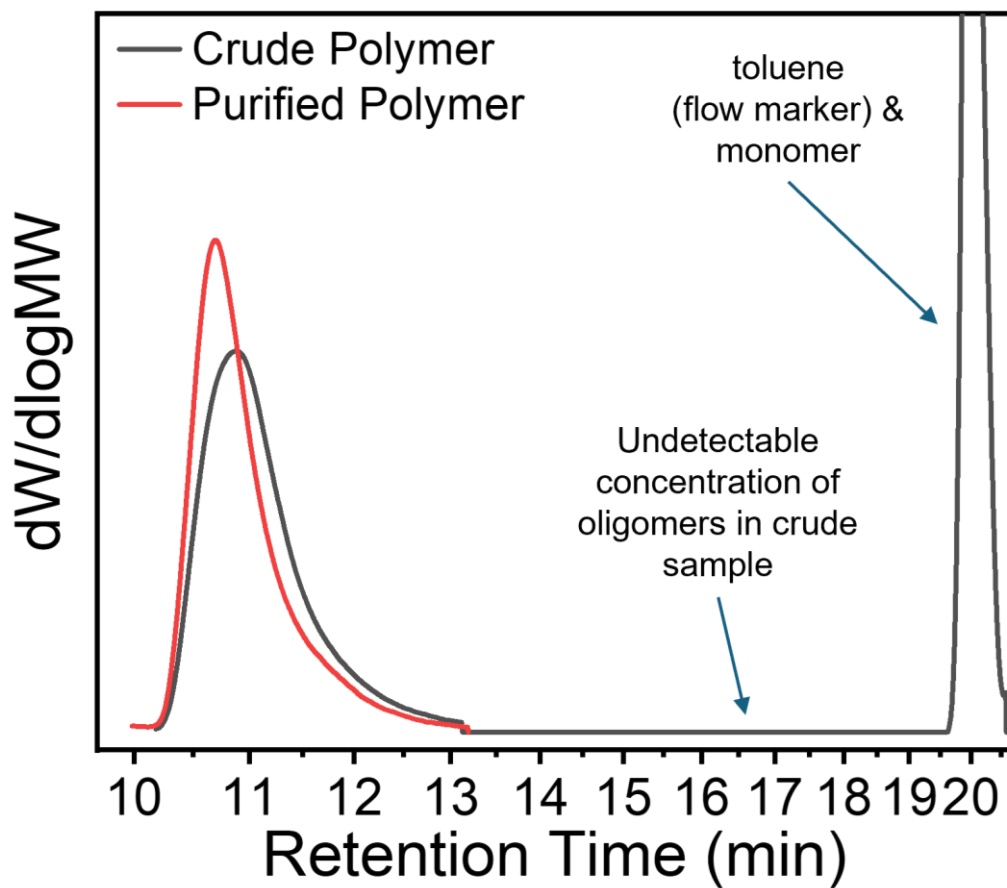


Figure S49.

Full GPC chromatogram of polyEtLp from Table S5. entry 6. Any low molecular weight species, including catalyst and residual monomer, are removed during the purification.

Thermomechanical data

PolyEtLp Film

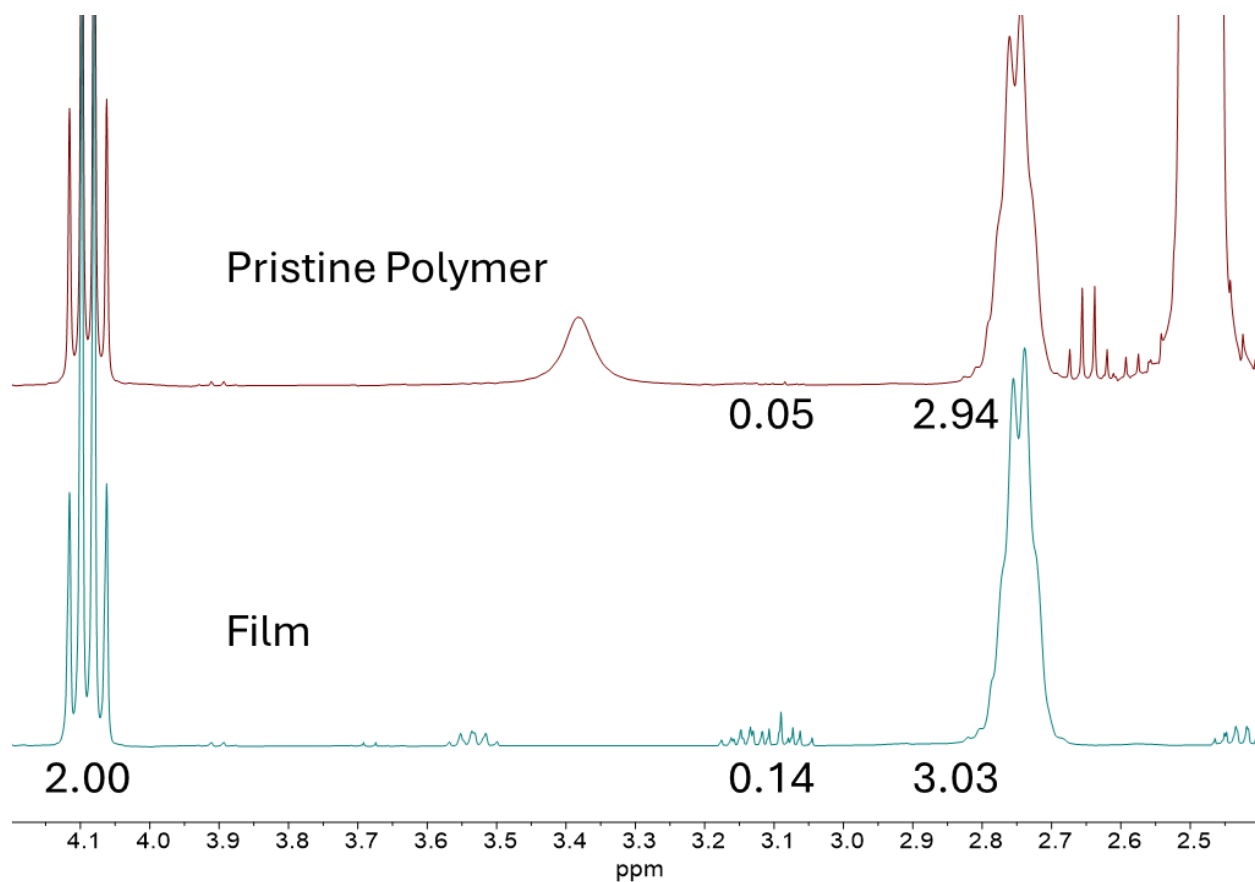


Figure S50.

¹H-NMR of polymer before and after 1 heat press cycle. Integration values are shown. ¹H-NMR spectrum – 400 MHz, 298 K, CDCl₃. *Corresponds to polymer shown in Figure 4b.

Mechanical Recycling

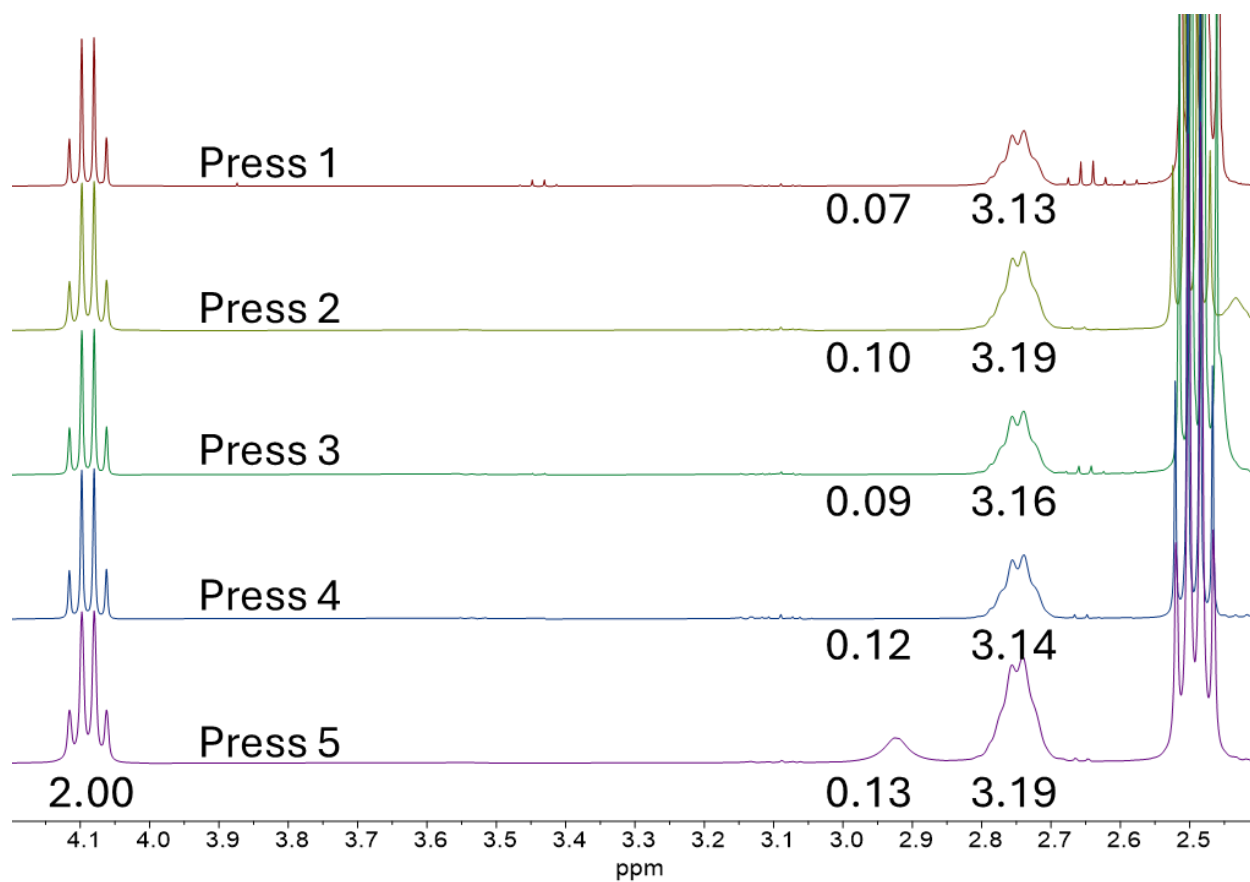


Figure S51.

^1H -NMR of film after multiple hot-press cycles. Integration values are shown for polymer and monomer signals. ^1H -NMR spectrum – 400 MHz, 298 K, CDCl_3 .

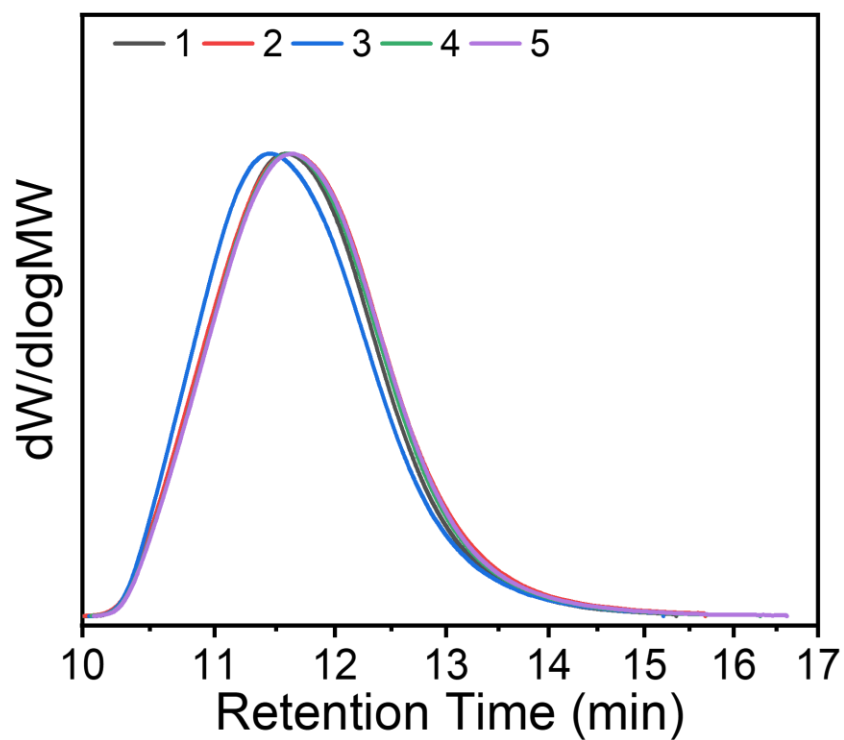


Figure S52.

GPC chromatogram of polyEtLp film after multiple hot-press cycles. Legend indicates the number of unique molding cycles that the film underwent.

Table S6.

Summary of Remolding Data for polyEtLp.

Press	Conversion (%)	M_p (kDa)	M_w (kDa)	M_n (kDa)	$\bar{D}$
1	97	1400	1470	700	2.1
2	96	1350	1450	620	2.4
3	96	1660	1550	740	2.1
4	95	1370	1450	610	2.4
5	94	1360	1440	570	2.5

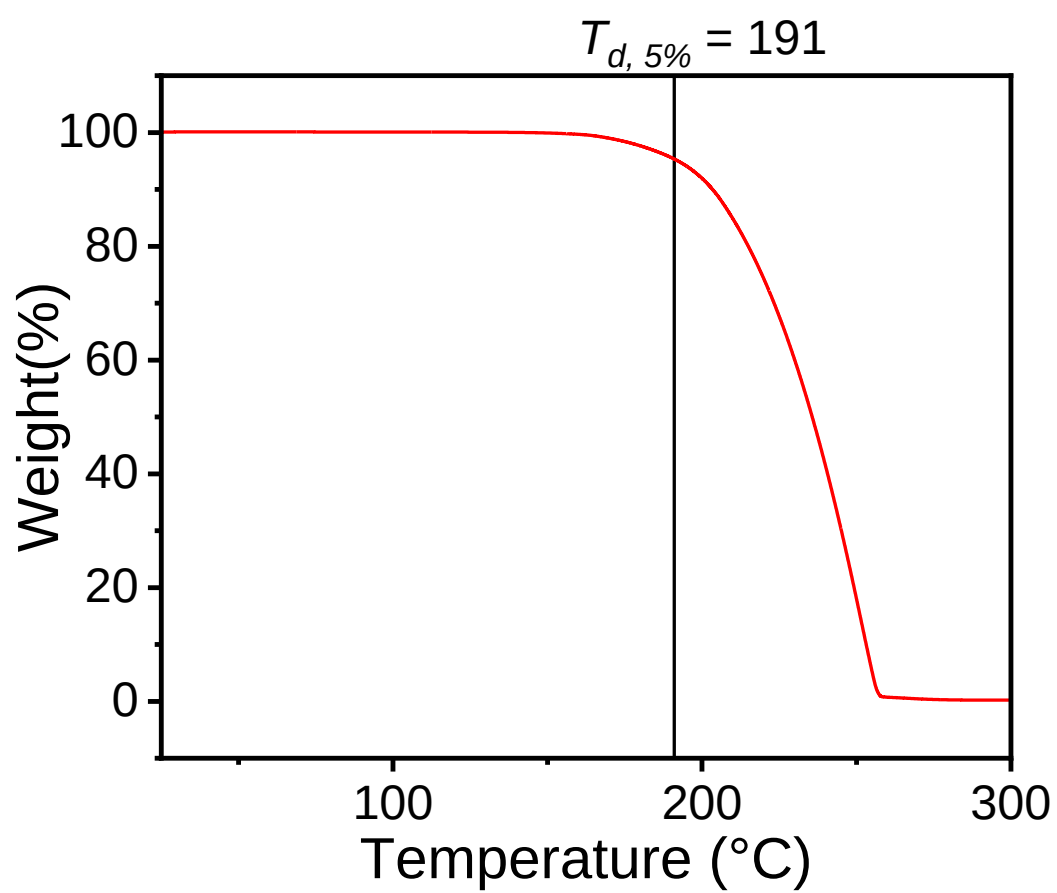


Figure S53.
TGA thermogram of polyEtLp film.

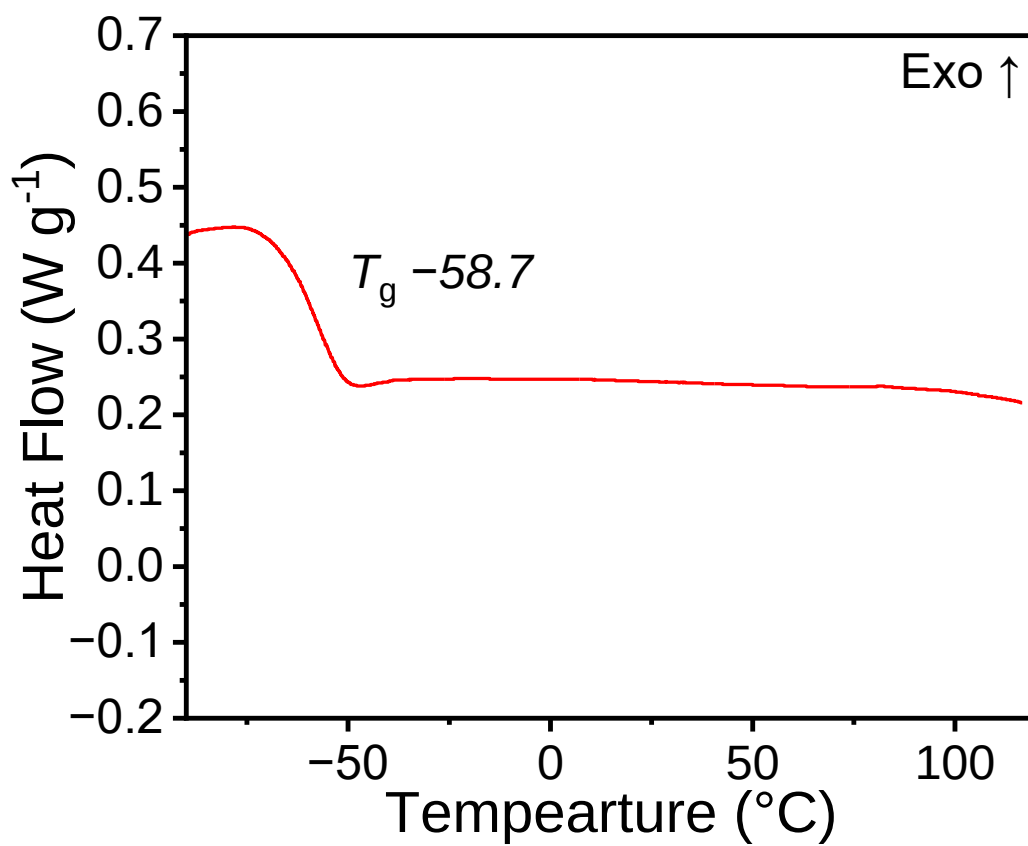


Figure S54.
DSC thermogram of polyEtLp film.

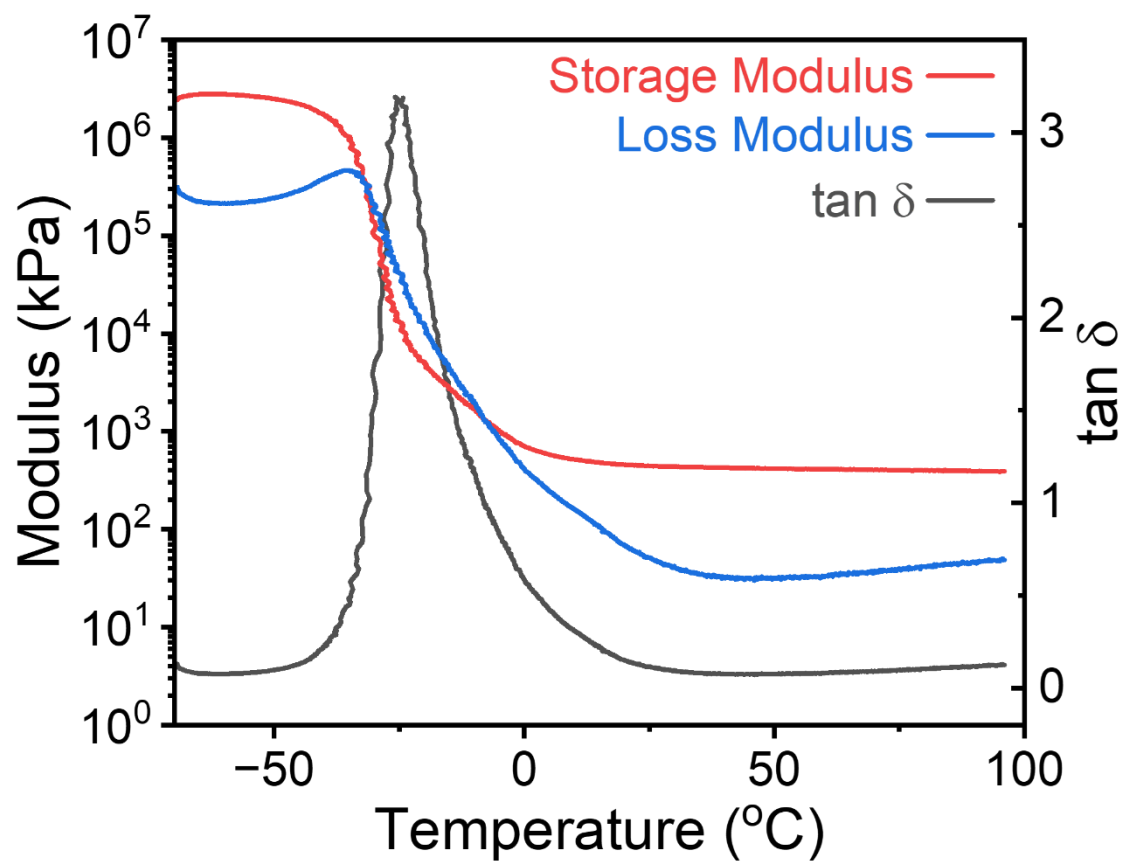


Figure S55.

DMA thermogram (5 °C min⁻¹) plotting storage modulus, loss modulus, and tan δ vs. temperature for polyEtLp film made from control polymer sample under Schlenk conditions. Polymer data listed in Table S4 entry 8.

RCD

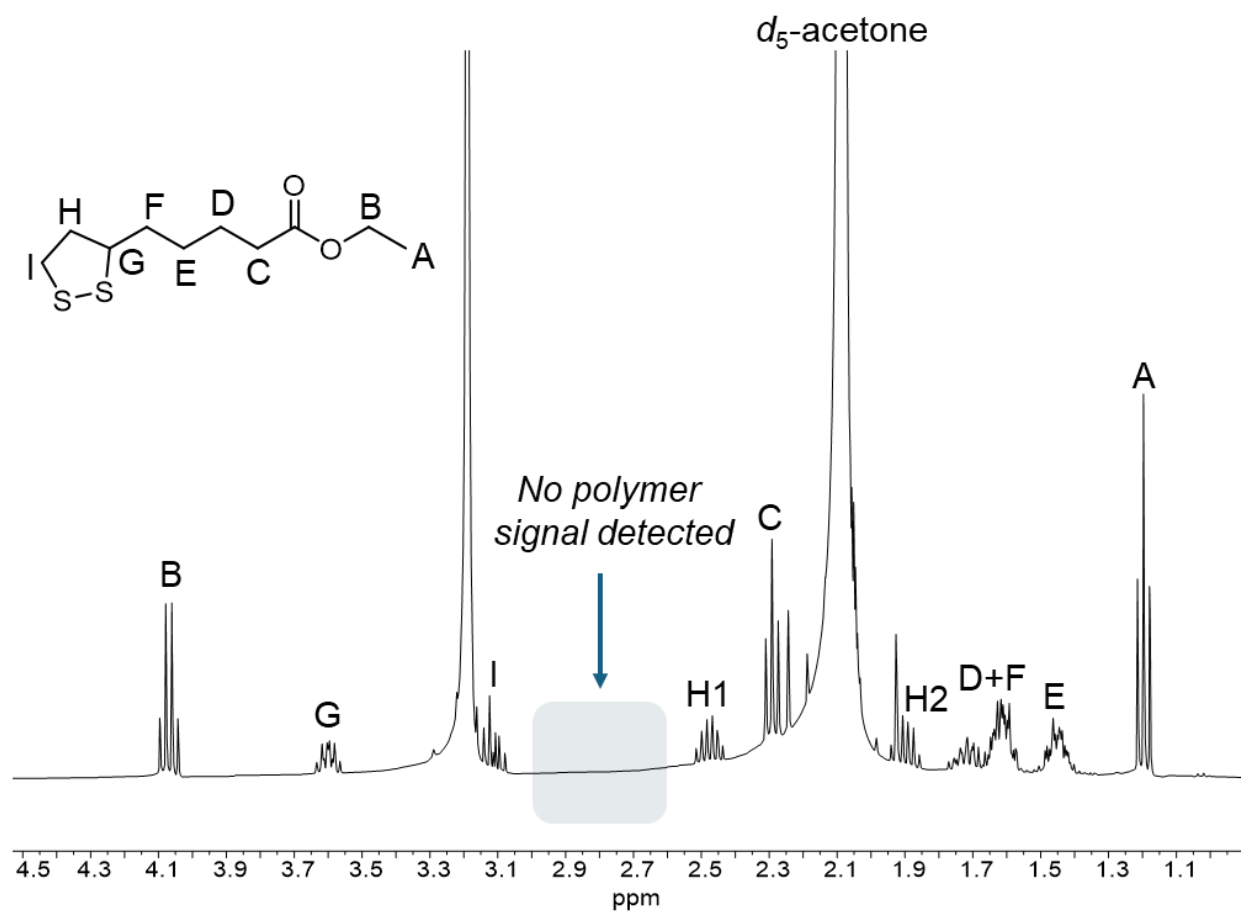


Figure S56.

Crude ^1H -NMR spectrum of EtLp made by RCD in d_6 -acetone using HCl_{aq} as a catalyst. No polymer signal was detected, thus indicating quantitative depolymerization. ^1H -NMR spectrum – 400 MHz, 298 K, d_6 -acetone.

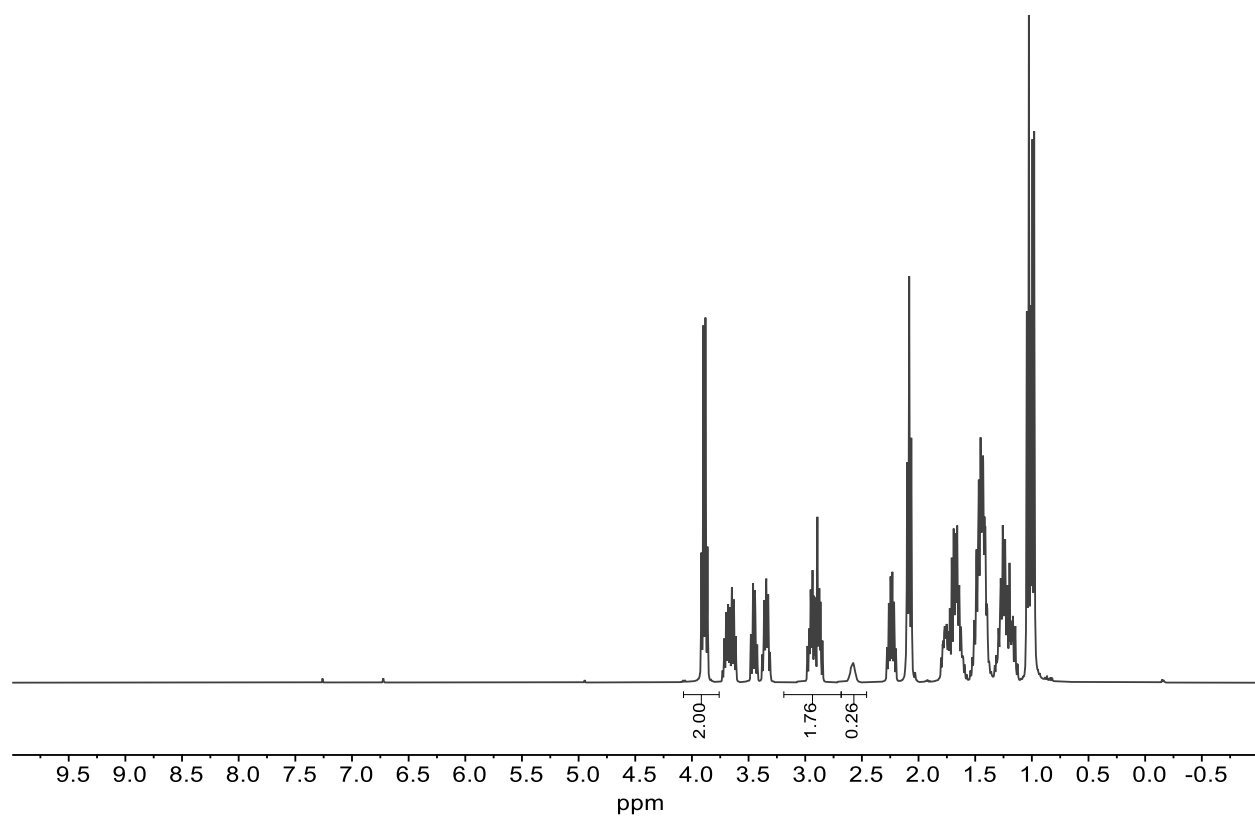


Figure S57.

¹H-NMR spectrum of EtLp obtained after RCD in 2-MeTHF at 70 °C. A minor signal from polymer (~10 mol%) is visible near $\delta = 2.75$. ¹H-NMR spectrum – 400 MHz, 298 K, CDCl₃.

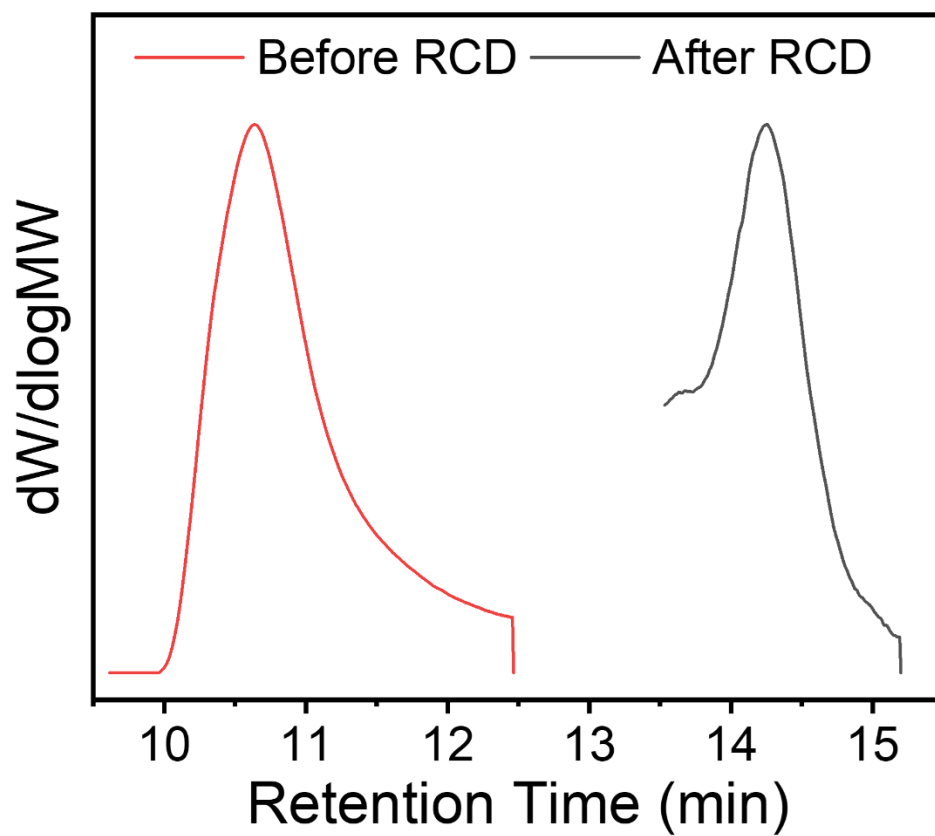


Figure S58.

GPC chromatogram of polyEtLp film after RCD in DMI. Polymer is still seen in the chromatogram, small molecules eluted at 20 min.

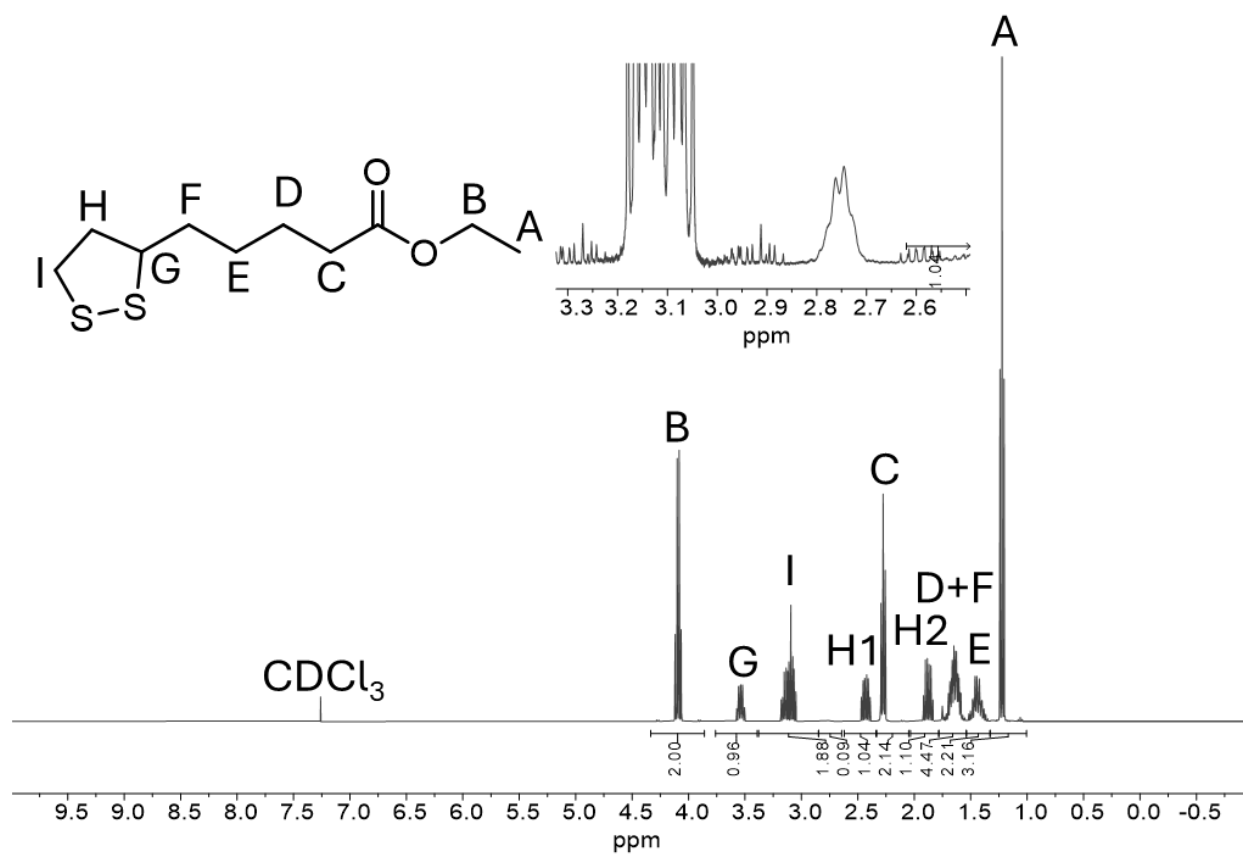


Figure S59.

¹H-NMR spectrum of EtLp obtained from RCD in chloroform. A minor signal from polymer is visible near $\delta = 2.75$ (see inset). ¹H-NMR spectrum – 400 MHz, 298 K, CDCl₃.

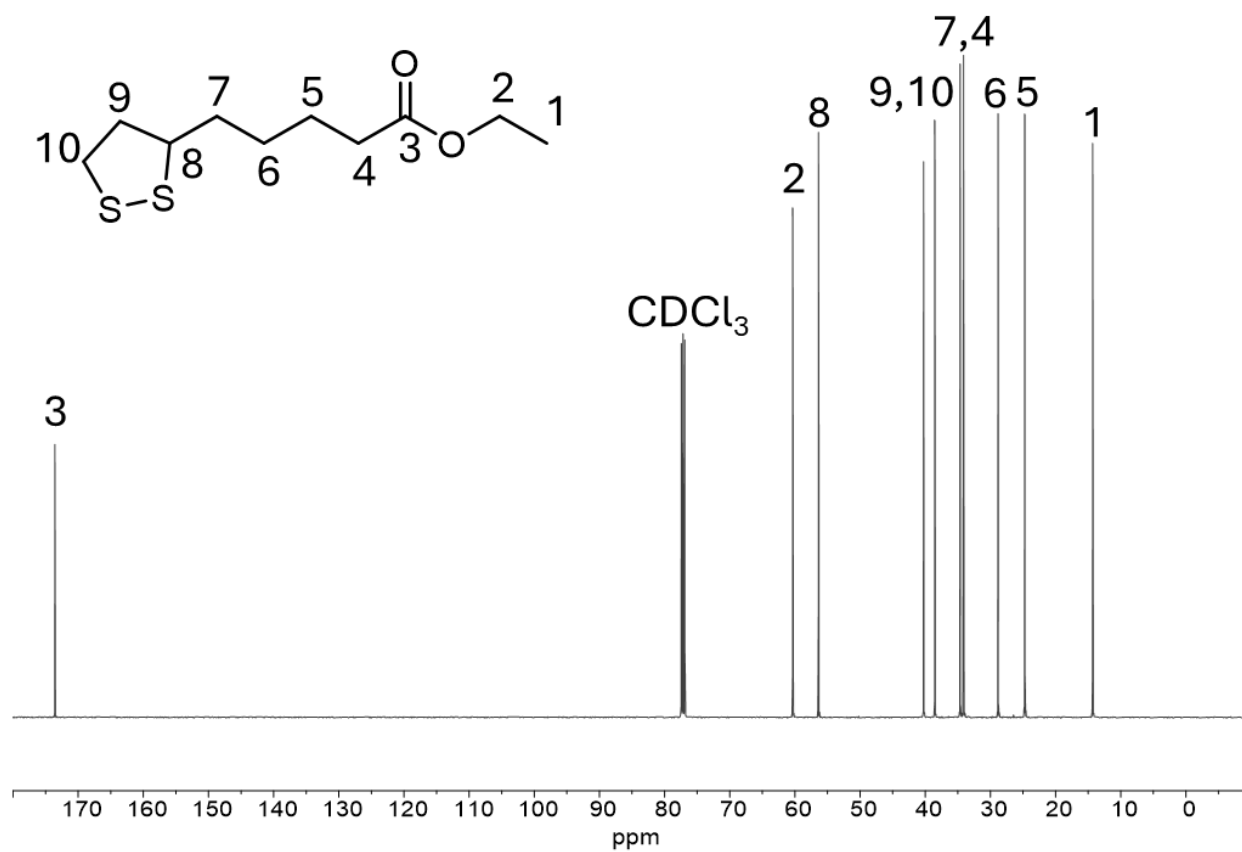


Figure S60.

^{13}C -NMR of EtLp obtained from RCD in chloroform ^{13}C -NMR spectrum – 100.57 MHz, 298 K, CDCl_3 .

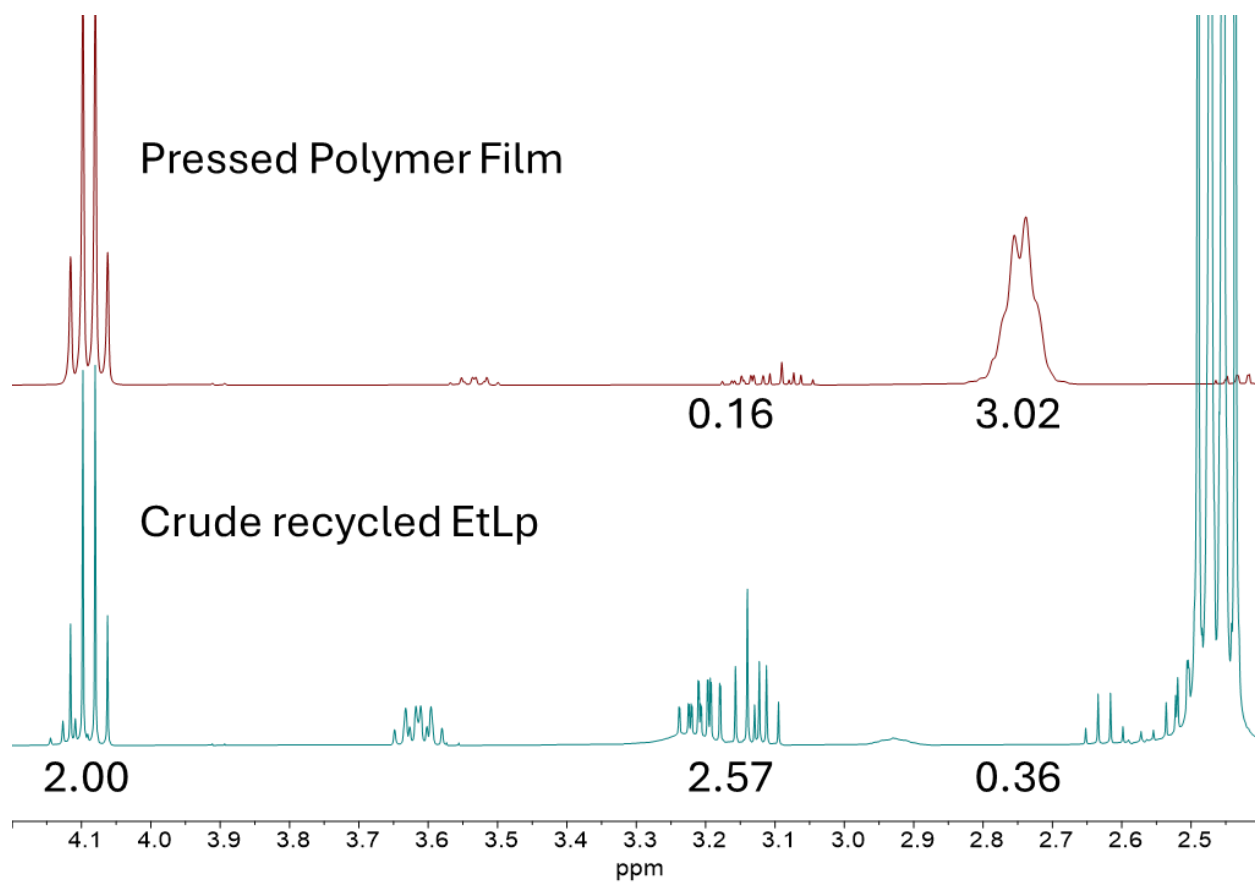


Figure S61.

¹H-NMR of crude EtLp made by RCD in acetone. A minor signal from polymer is visible near $\delta = 2.75$. The monomer subsequently polymerized during isolation and purification steps. ¹H-NMR spectrum – 400 MHz, 298 K, CDCl₃.

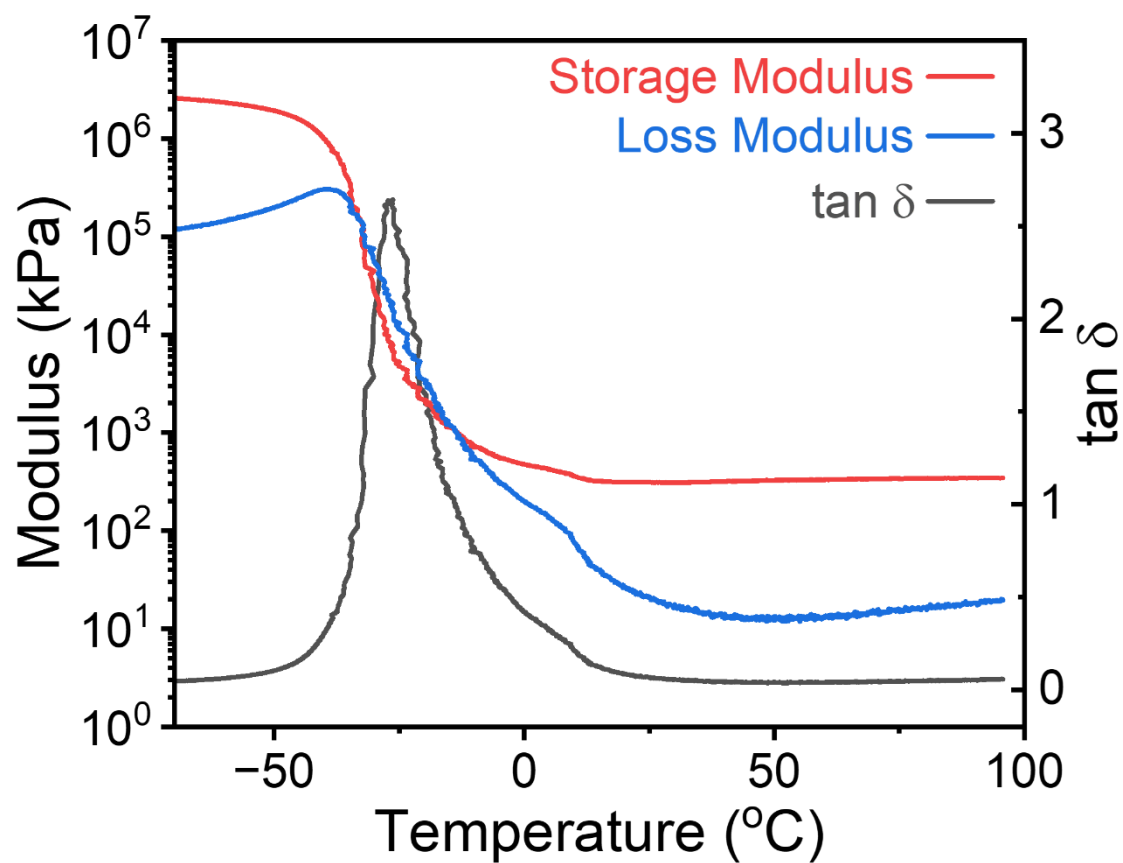


Figure S62.

DMA thermogram (5 °C min⁻¹) plotting storage modulus, loss modulus, and tan δ vs. temperature for polyEtLp film that was synthesized from recycled EtLp (obtained from RCD).

Life Cycle Analysis

EtLp Synthesis

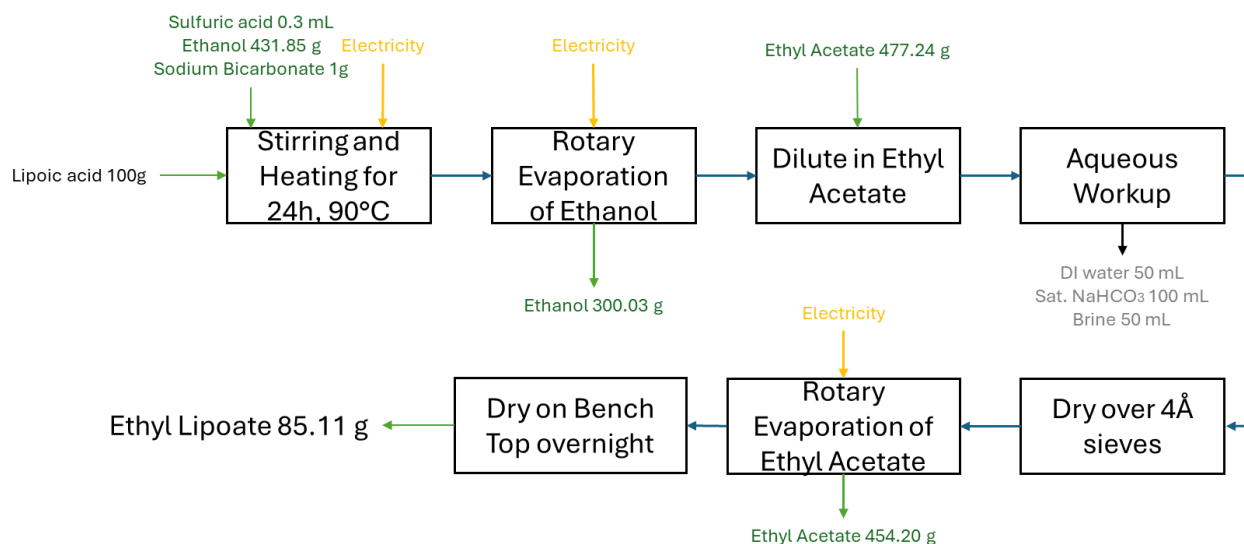


Figure S63. Flow chart for the synthesis of EtLp from lipoic acid. Materials are highlighted in green; electricity use is highlighted in yellow; waste materials are highlighted in grey. Boxes contain process withing the procedure, blue arrows indicate the next step in the procedure.

Table S7.

Material and energy inventory for EtLp from lipoic acid. Waste flows are orange highlighted.

Item	Amount (g)	Source/provider
Lipoic acid ^a	68.92	market for fatty acid fatty acid Cutoff, U - GLO
	31.08	market for sulfur sulfur Cutoff, U - GLO
Ethanol	132.217	market for ethanol, without water, in 99.7% solution state, from ethylene ethanol, without water, in 99.7% solution state, from ethylene Cutoff, U - RoW
Sulfuric acid	0.549	market for sulfuric acid sulfuric acid Cutoff, U - RoW
Sodium bicarbonate	1.000	market for sodium bicarbonate sodium bicarbonate Cutoff, U - RoW
Ethyl acetate	23.040	market for ethyl acetate ethyl acetate Cutoff, U - GLO
Electricity for heating + stirring ^b	10.164 kWh	market for electricity, medium voltage electricity, medium voltage Cutoff, U - US-SERC
Electricity for rotary evaporator (ethanol) ^c	2.422 kWh	market for electricity, medium voltage electricity, medium voltage Cutoff, U - US-SERC
Water	50.000	market for water, deionized water, deionized Cutoff, U - RoW
Saturated sodium bicarbonate	98.000	market for water, deionized water, deionized Cutoff, U - RoW
	2.050	market for sodium bicarbonate sodium bicarbonate Cutoff, U - RoW

Brine (sodium chloride in water)	31.880	market for water, deionized water, deionized Cutoff, U - RoW
	18.120	market for sodium chloride, powder sodium chloride, powder Cutoff, U - GLO
Waste treatment ^d	393.200	market for wastewater, average wastewater, average Cutoff, U - RoW
Electricity for rotary evaporator (ethyl acetate) ^c	2.422 kWh	market for electricity, medium voltage electricity, medium voltage Cutoff, U - US-SERC
EtLp	85.110	Output material

^a Lipoic acid was replaced by a fatty acid, a carboxylic acid with a long aliphatic chain. According to Ref. <https://www.science.org/doi/10.1126/sciadv.abf0187>, “The life cycle GHG footprints of TREN and dimedone were estimated using neural network models that identified compounds with similar molecular characteristics and existing production data because no data specific to the production of these compounds were available. When these models could not identify any similar compounds, as was the case for DCC, we obtained their industrial synthesis route through available patents and created a separate process simulation to estimate material and energy use, which was then used to calculate life cycle GHG emissions”. Accordingly:

- We were unable to find organosulfur compounds similar to lipoic acid, such as thiophene fatty acids, thiomalic acid, mercaptopropionic acid, thiolactic acid, or cysteine.
- Therefore, we modeled lipoic acid as:
 1. 31.08 wt% sulfur and
 2. 68.92 wt% caprylic acid (octanoid acid), which is indeed a precursor of lipoic acid. As this compound is not found in the utilized database, it has been replaced by “fatty acid” from coconut oil, which contains 6-10% of fatty acids.

^b using a Fisherbrand SP88850200 Isotemp Stirring Hot Plate with a power of 605 W. 24 h use, considered to function at a 70% workload.

^c using a BUCHI Rotavapor™ with a power of 1730 W. 2 h use, considered to function at a 70% workload.

^d it includes the ethanol, sulfuric acid, sodium bicarbonate, DI water, brine and others. Estimated pH value is 7–8.

Sample calculation of E-factor

$$\begin{aligned}
 \text{E-factor} &= \frac{\text{mass waste}}{\text{mass product}} = \frac{m_{\text{LpOH}} + m_{\text{EtOH}} + m_{\text{H}_2\text{SO}_4} + m_{\text{NaHCO}_3} + m_{\text{EtOAc}} + m_{\text{NaCl}}}{m_{\text{EtLp}}} \\
 &= \frac{\left(100\text{g} - \left(\frac{206.33\text{g mol}^{-1} \times 85.11\text{g}}{234.079\text{g mol}^{-1}}\right)\right) + 132.217\text{g} + 0.549\text{g} + 1.000\text{g} + 23.040\text{g} + 10.000\text{g}}{85.11\text{g}} \\
 &= \frac{193.78\text{g}}{85.11\text{g}} = 2.276
 \end{aligned}$$

Equation S1. Example E-factor calculation for the synthesis of ethyl lipoate from lipoic acid. The mass of LpOH waste was calculated by taking the initial mass and subtracting the amount that was converted into recovered product. The mass of ethanol and ethyl acetate waste was calculated by taking the initial mass of solvent used and subtracting the mass of solvent recovered. The mass of aqueous salts used for workup were estimated to be 10 g.

CROP of pristine EtLp

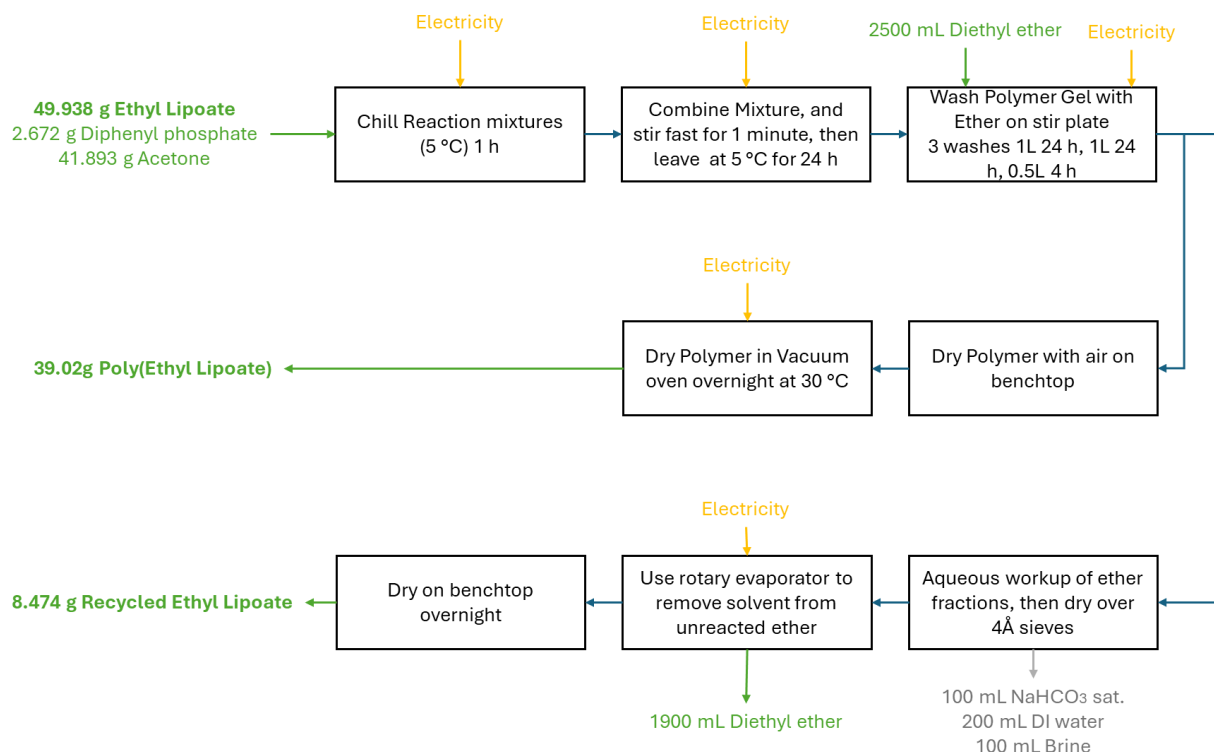


Figure S64. Flow chart for the synthesis of polyEtLp. Materials are highlighted in green; electricity use is highlighted in yellow; waste materials are highlighted in grey. Boxes contain process within the procedure, blue arrows indicate the next step in the procedure.

Table S8. Material and energy inventory for synthesis of polyEtLp.

Item	Amount (g)	Source/provider
EtLp	49.938	<i>From previous process</i>
Acetone	41.893	market for acetone, liquid acetone, liquid Cutoff, U - RoW
Diphenyl phosphate ^a	2.672	market for triphenyl phosphate triphenyl phosphate Cutoff, U - GLO
Electricity for chilling _b	0.025 kWh	market for electricity, medium voltage electricity, medium voltage Cutoff, U - US-SERC
Electricity for washing ^c	21.175 kWh	market for electricity, medium voltage electricity, medium voltage Cutoff, U - US-SERC
Electricity for drying _d	2.768 kWh	market for electricity, medium voltage electricity, medium voltage Cutoff, U - US-SERC
polyEtLp	39.020	Output material
Water	200.000	market for water, deionized water, deionized Cutoff, U - RoW
Saturated sodium bicarbonate	98.000	market for water, deionized water, deionized Cutoff, U - RoW
	2.050	market for sodium bicarbonate sodium bicarbonate Cutoff, U - RoW

Brine (sodium chloride in water)	63.760	market for water, deionized water, deionized Cutoff, U - RoW
	36.240	market for sodium chloride, powder sodium chloride, powder Cutoff, U - GLO
Electricity for rotary evaporator (ether) ^e	0.605 kWh	market for electricity, medium voltage electricity, medium voltage Cutoff, U - US-SERC
Waste treatment ^f	393.200	
EtLp ^g	8.474	Output material

^a Diphenyl Phosphate is replaced by a compound that shares a common phosphate group structure. Triphenyl phosphate was selected as the most suitable replacement found in the database, replacing (C₆H₅O)₂PO₂H by (C₆H₅O)₃PO. However, it should be noted that diphenyl phosphate is a less hazardous substance than triphenyl phosphate based on the MSDS of each.

^b using a refrigerator with a power of 300 W. 25 h use, considered to function at a 70% workload, and the amount of materials that could be refrigerated is 20 kg.

^c using a Fisherbrand SP88850200 Isotemp Stirring Hot Plate with a power of 605 W. 50 h use, considered to function at a 70% workload.

^d using an oven with a power of 1400 W with capacity to dry 0.5 kg (OMH 60), so energy consumption is reduced by 5 times considering a sample of 100 grams. A laboratory UTGE-3510 vacuum pump is added, with a power of 248 W. 12 h use, considered to function at a 70% workload.

^e using a BUCHI RotavaporTM with a power of 1730 W. 0.5 h use, considered to function at a 70% workload.

^f it includes acetone, sodium bicarbonate, DI water, brine, ether, and others. Estimated pH value is 7-to-7.5.

^g of similar purity but also contains lower molecular weight oligomers. This resin can be used for polymerization and the lower molecular weight does not affect the overall conversion of monomer to polymer.

RCD of polyEtLp

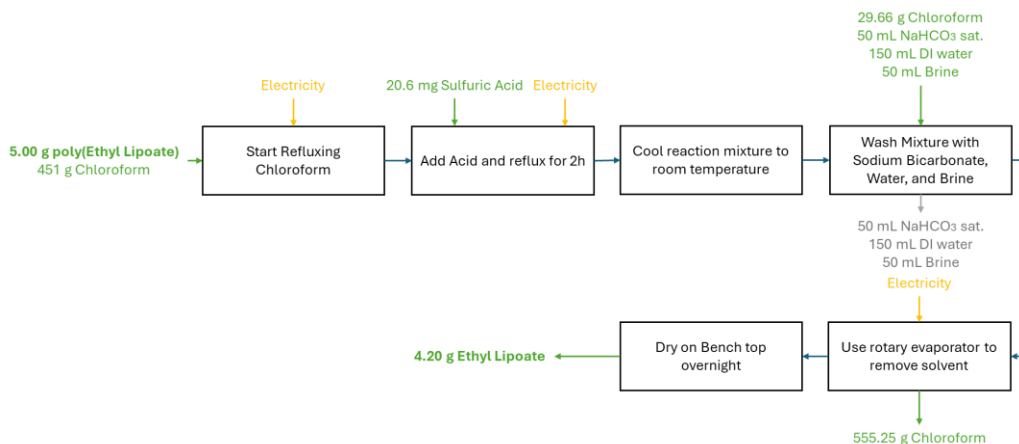


Figure S65. Flow chart for the cationic RCD of polyEtLp to EtLp monomer. Materials are highlighted in green; electricity use is highlighted in yellow; waste materials are highlighted in grey. Boxes contain process withing the procedure, blue arrows indicate the next step in the procedure.

Table S9. Material and energy inventory for the cationic RCD of P-LpEt to LpEt monomer.

Item	Amount (g)	Source/provider
polyEtLp	5.000	<i>From previous process</i>
Chloroform	25.890	market for chloroform chloroform Cutoff, U – RoW.
Electricity for heating + stirring ^a	0.848 kWh	market for electricity, medium voltage electricity, medium voltage Cutoff, U - US-SERC
Sulfuric acid	0.021	market for sulfuric acid sulfuric acid Cutoff, U - RoW
Electricity for heating + stirring ^a	0.848 kWh	market for electricity, medium voltage electricity, medium voltage Cutoff, U - US-SERC
DI water	150.000	market for water, deionized water, deionized Cutoff, U - RoW
Saturated sodium bicarbonate	49.000	market for water, deionized water, deionized Cutoff, U - RoW
	1.025	market for sodium bicarbonate sodium bicarbonate Cutoff, U - RoW
Brine (sodium chloride in water)	31.880	market for water, deionized water, deionized Cutoff, U - RoW
	18.120	market for sodium chloride, powder sodium chloride, powder Cutoff, U - GLO
Electricity for rotary evaporator (chloroform) ^b	0.606 kWh	market for electricity, medium voltage electricity, medium voltage Cutoff, U - US-SERC
Waste treatment ^c	25.890	market for spent solvent mixture spent solvent mixture Cutoff, U - RoW
Waste treatment ^d	279.886	market for wastewater, average wastewater, average Cutoff, U - RoW
Recycled EtLp resin	4.200	Output material

^a using a Fisherbrand SP88850200 Isotemp Stirring Hot Plate with a power of 605 W. 2 h use, considered to function at a 70% workload.

^b using a BUCHI Rotavapor™ with a power of 1730 W. 0.5 h use, considered to function at a 70% workload.

^c includes chloroform.

^d includes ethyl lipoate, sodium bicarbonate, DI water, and brine.

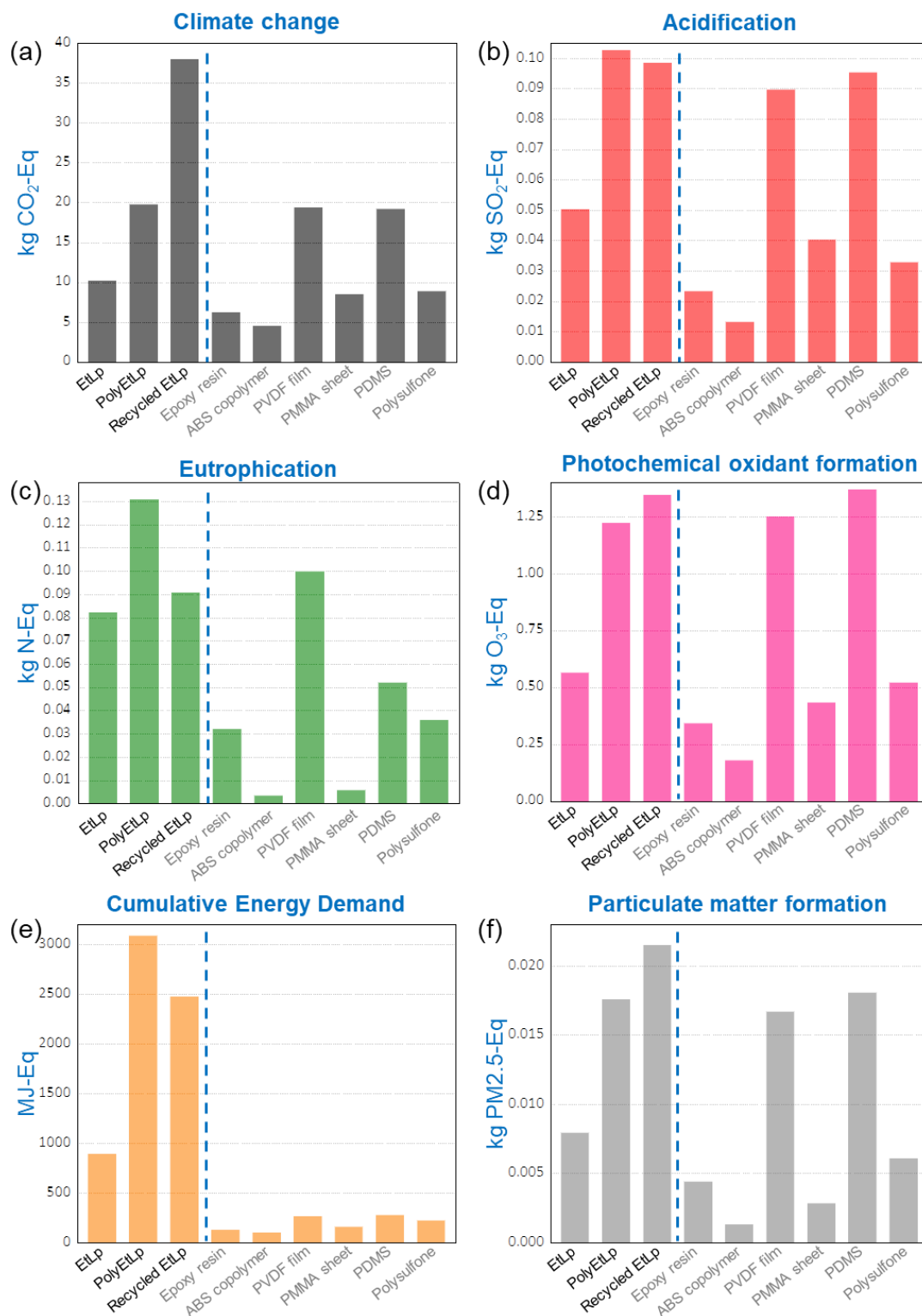


Figure 66. Environmental impacts (a) climate change, (b) acidification, (c) eutrophication, (d) photochemical oxidant formation, (e) cumulative energy demand, (f) particulate matter formation of monomer synthesis, polymer synthesis, and depolymerization back to monomer according to TRACI 2.1.

Table S10. Environmental impacts of EtLp, polyEtLp, and RCD of polyEtLp to monomer according to TRACI 2.1 and Cumulative Energy Demand (CED).

	Impact category	EtLp	polyEtLp	RCD to monomer	Unit
TRACI v2.1	Acidification	0.05026	0.10277	0.09856	kg SO ₂ -Eq
	Climate change	10.20564	19.79019	37.96137	kg CO₂-Eq
	Ecotoxicity: freshwater	578.33374	1,912.31304	1,624.69660	CTUe
	Eutrophication	0.08225	0.13086	0.09085	kg N-Eq
	Human toxicity: carcinogenic	0.00000	0.00002	0.00001	CTUh
	Human toxicity: non-carcinogenic	0.00000	0.00001	0.00001	CTUh
	Ozone depletion	0.00000	0.00000	0.00450	kg CFC-11-Eq
	Particulate matter formation	0.00794	0.01758	0.02149	kg PM _{2.5} -Eq
Cumulative energy demand (CED)	Photochemical oxidant formation	0.56613	1.22376	1.34523	kg O ₃ -Eq
	Energy resources: non-renewable	141.94251	254.92374	294.74170	MJ-Eq
	Energy resources: non-renewable, biomass	2.81038	2.96465	0.01808	MJ-Eq
	Energy resources: non-renewable, fossil	135.22909	241.94695	275.25555	MJ-Eq
	Energy resources: non-renewable, nuclear	3.90304	10.01214	19.46807	MJ-Eq
	Energy resources: renewable	750.19436	2,832.68505	2,177.91360	MJ-Eq
	Energy resources: renewable, biomass	62.00570	94.04874	33.38168	MJ-Eq
	Energy resources: renewable, geothermal	0.18635	0.41739	1.09931	MJ-Eq
	Energy resources: renewable, geothermal, solar, wind	18.91263	74.00078	60.78236	MJ-Eq
	Energy resources: renewable, solar	0.01218	0.03107	0.02656	MJ-Eq
	Energy resources: renewable, water	669.27603	2,664.63553	2,083.74956	MJ-Eq
	Energy resources: renewable, wind	18.71410	73.55232	59.65649	MJ-Eq
	Total	892.13687	3,087.60879	2,472.65530	MJ-Eq

Table S11.

Environmental impacts correspond to industrial-scale manufacturing of conventional petroleum-derived polymers.

	Impact category	Epoxy resin	ABS copolymer	PVDF film	Unit
TRACI v2.1	Acidification	0.02330621	0.01318410	0.08976072	kg SO ₂ -Eq
	Climate change	6.22623095	4.50626341	19.36954724	kg CO₂-Eq
	Ecotoxicity: freshwater	126.90832314	16.63463247	320.37246415	CTUe
	Eutrophication	0.03199625	0.00353530	0.10000078	kg N-Eq
	Human toxicity: carcinogenic	0.00000164	0.00000032	0.00000336	CTUh
	Human toxicity: non-carcinogenic	0.00000171	0.00000019	0.00000580	CTUh
	Ozone depletion	0.00000022	0.00000001	0.00000023	kg CFC-11-Eq
	Particulate matter formation	0.00440702	0.00131884	0.01669808	kg PM _{2.5} -Eq
Cumulative energy demand (CED)	Photochemical oxidant formation	0.34283513	0.18114938	1.24951707	kg O ₃ -Eq
	Energy resources: non-renewable	123.35525723	102.10832947	244.38682391	MJ-Eq
	Energy resources: non-renewable, biomass	0.00580248	0.00173228	0.00656787	MJ-Eq
	Energy resources: non-renewable, fossil	117.69426029	98.10138610	220.06387668	MJ-Eq
	Energy resources: non-renewable, nuclear	5.65519446	4.00521110	24.31637936	MJ-Eq
	Energy resources: renewable	5.16774934	0.51217444	18.40473021	MJ-Eq
	Energy resources: renewable, biomass	1.37294799	0.18423451	3.53437252	MJ-Eq
	Energy resources: renewable, geothermal	0.28478968	0.00773715	1.28171030	MJ-Eq
	Energy resources: renewable, geothermal, solar, wind	1.42061338	0.05072157	5.61340581	MJ-Eq
	Energy resources: renewable, solar	0.00516837	0.00080070	0.01999182	MJ-Eq
	Energy resources: renewable, water	2.37418797	0.27721836	9.25695188	MJ-Eq
	Energy resources: renewable, wind	1.13065534	0.04218372	4.31170369	MJ-Eq
	Total	128.52300657	102.62050391	262.79155413	MJ-Eq
	Impact category	PMMA sheet	PDMS	Polysulfone	Unit
TRACI v2.1	Acidification	0.04027678	0.09542933	0.03278926	kg SO ₂ -Eq
	Climate change	8.48564525	19.18369674	8.89140708	kg CO₂-Eq
	Ecotoxicity: freshwater	22.88867331	271.14072982	196.21029064	CTUe
	Eutrophication	0.00583245	0.05203944	0.03600320	kg N-Eq
	Human toxicity: carcinogenic	0.00000030	0.00000347	0.00000233	CTUh
	Human toxicity: non-carcinogenic	0.00000023	0.00000478	0.00000237	CTUh
	Ozone depletion	0.00000000	0.00153145	0.00000033	kg CFC-11-Eq
	Particulate matter formation	0.00285132	0.01808220	0.00608231	kg PM _{2.5} -Eq
Cumulative energy demand (CED)	Photochemical oxidant formation	0.43526832	1.36917625	0.52243125	kg O ₃ -Eq
	Energy resources: non-renewable	148.49149300	251.00013745	217.71116587	MJ-Eq
	Energy resources: non-renewable, biomass	0.00118533	0.01160826	0.00589597	MJ-Eq
	Energy resources: non-renewable, fossil	136.07175816	222.95681787	209.15101518	MJ-Eq
	Energy resources: non-renewable, nuclear	12.41854951	28.03171132	8.55425471	MJ-Eq
	Energy resources: renewable	1.36043422	29.53909580	7.09719372	MJ-Eq
	Energy resources: renewable, biomass	0.49943991	12.20656481	1.83411754	MJ-Eq
	Energy resources: renewable, geothermal	0.00218276	1.05574879	0.37554900	MJ-Eq
	Energy resources: renewable, geothermal, solar, wind	0.01229884	5.12279432	2.12650362	MJ-Eq
	Energy resources: renewable, solar	0.00064007	0.01247056	0.00809995	MJ-Eq
	Energy resources: renewable, water	0.84869546	12.20973667	3.13657256	MJ-Eq
	Energy resources: renewable, wind	0.00947601	4.05457497	1.74285467	MJ-Eq
	Total	149.85192722	280.53923324	224.80835959	MJ-Eq

One-pot ROP

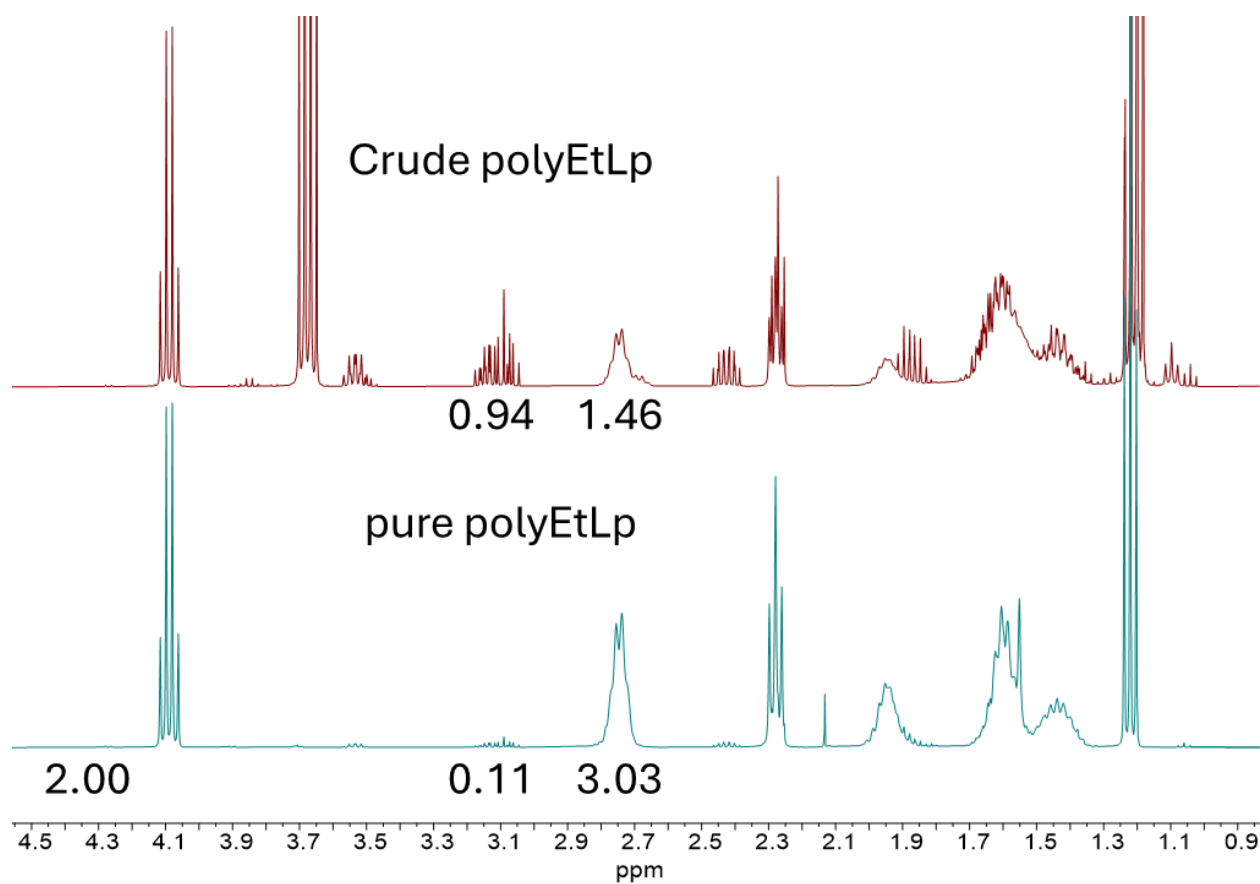


Figure S67.

¹H-NMR of crude polyEtLp and purified polyEtLp synthesized from the one-pot esterification-ROP from lipoic acid (small amount of monomer is visible near $\delta = 3.1$). ¹H-NMR spectrum – 400 MHz, 298 K, CDCl₃.

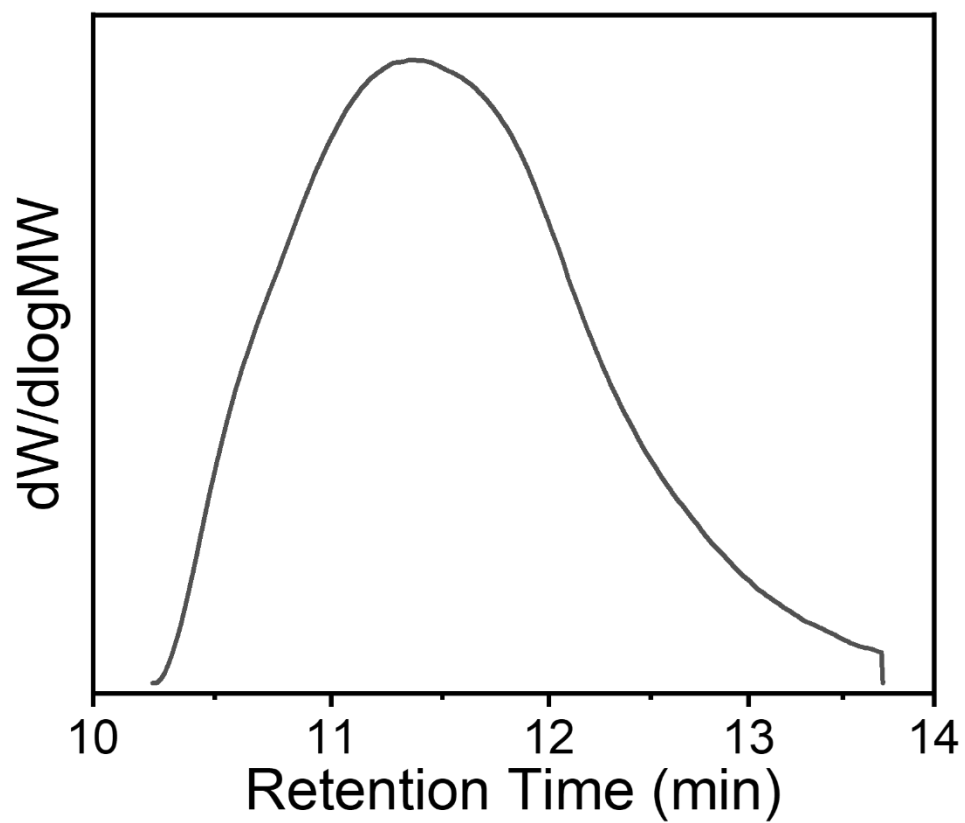


Figure S68.

GPC chromatogram of polyEtLp synthesized from the one-pot esterification-ROP from lipoic acid.

Reduced-pressure ROP

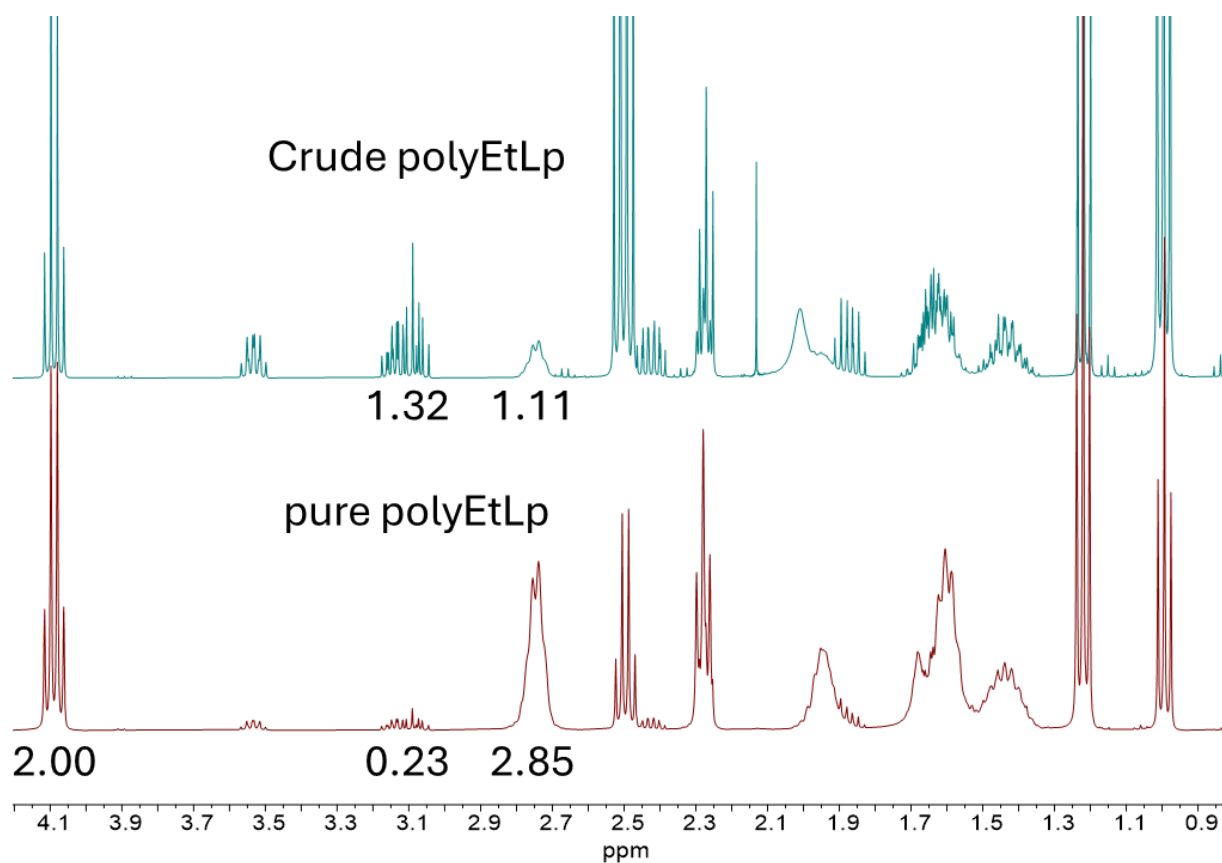


Figure S69.

^1H -NMR of crude polyEtLp and purified polyEtLp synthesized from ROP at reduced pressure using a rotary evaporator. ^1H -NMR spectrum – 400 MHz, 298 K, CDCl_3 .

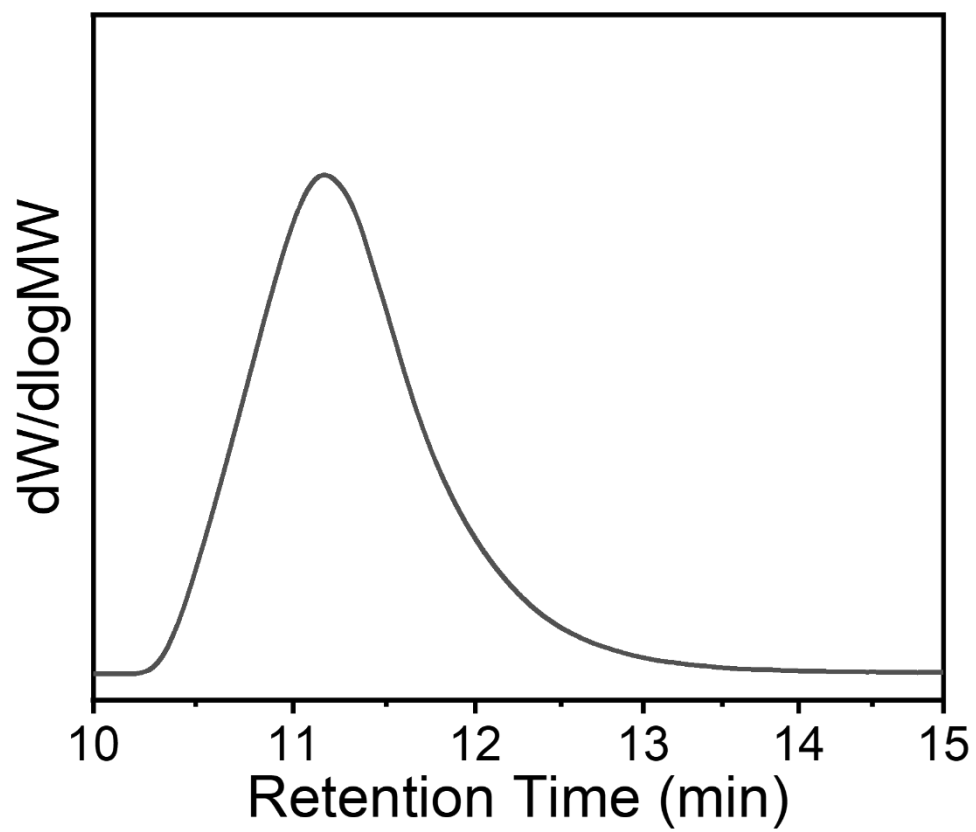


Figure S70.

GPC chromatogram of polyEtLp synthesized from ROP at reduced pressure using a rotary evaporator.

One-pot RCD to ROP

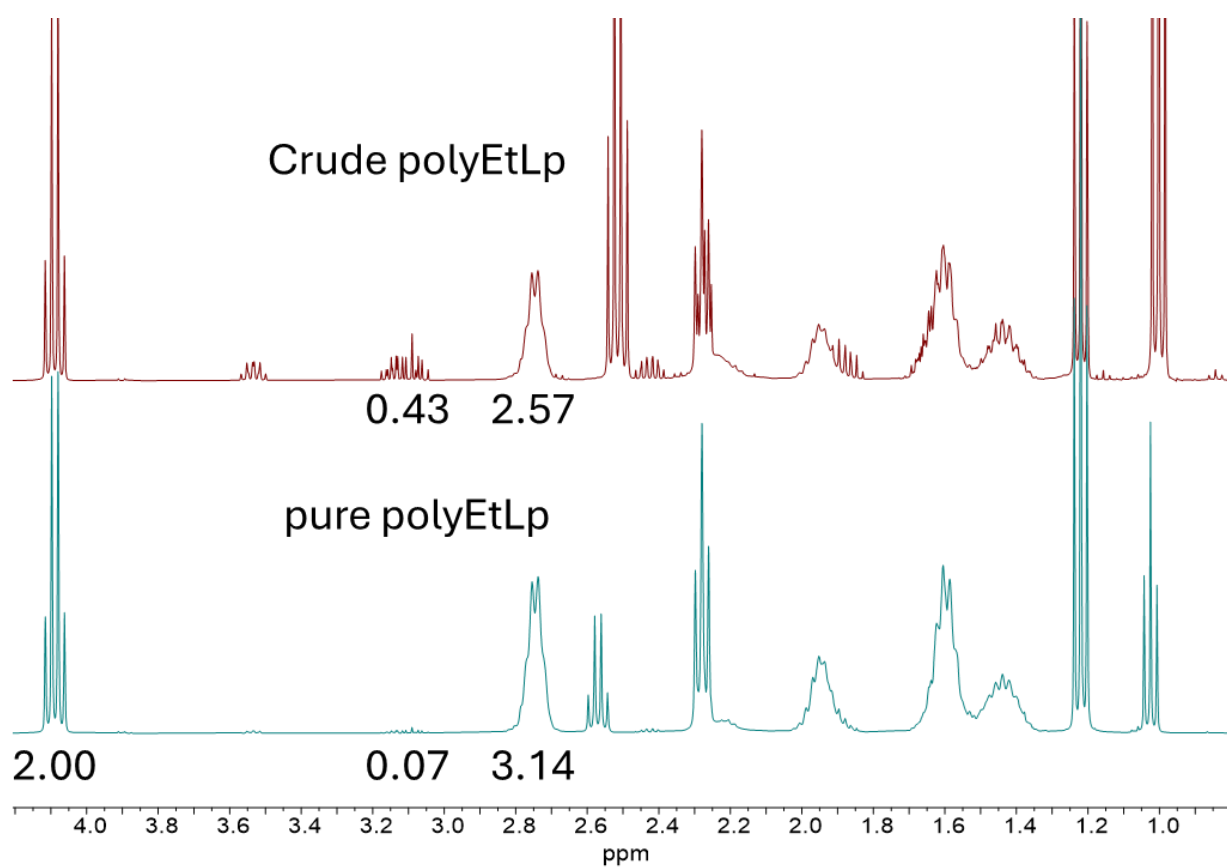


Figure S71.

¹H-NMR of crude polyEtLp and purified polyEtLp synthesized from one-pot RCD to ROP. ¹H-NMR spectrum – 400 MHz, 298 K, CDCl₃.

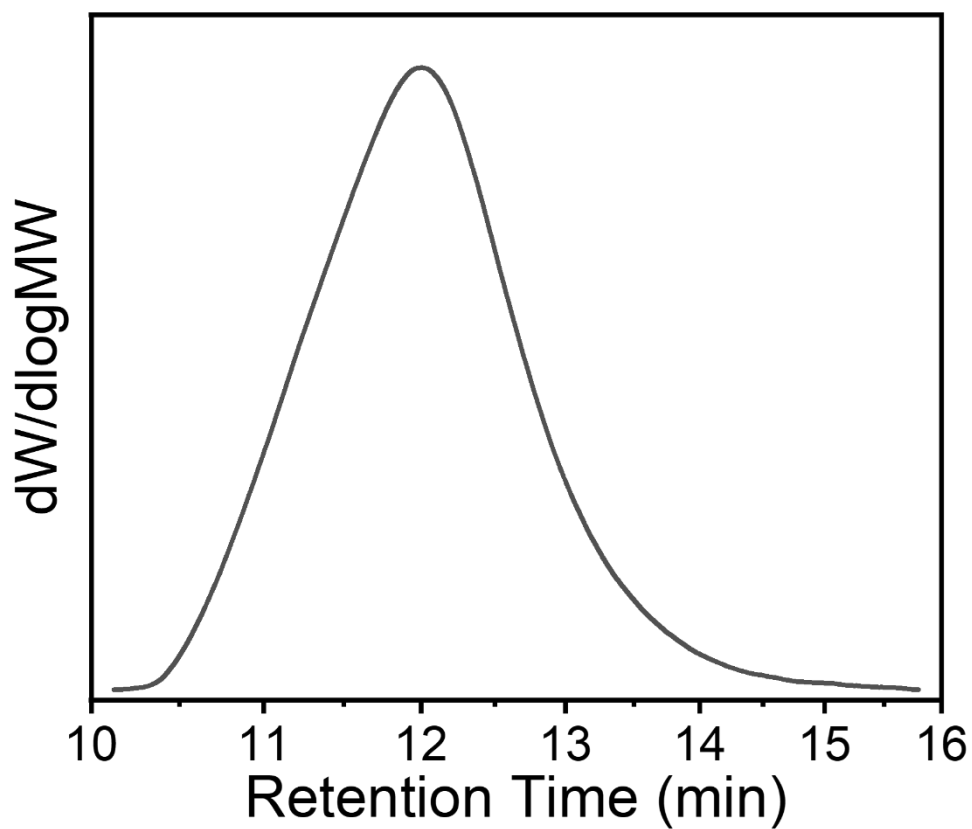


Figure S72.

GPC chromatogram of polyEtLp synthesized from one-pot RCD to ROP.

Table S12.

Summary of alternative CROP methods.

Experiment	Conversion (%)	M_w (kDa)	M_n (kDa)	$\bar{D}$
One-pot ROP	51	1900	1100	1.7
Reduced Pressure ROP	36	2800	1700	1.7
One-pot RCD to ROP	80	1200	600	2.1

$n = 1$ for each experiment; conversion was calculated using the equation in Figure S4.

Controlled CROP Experiments

CROP of EtLp with reversible proton trap

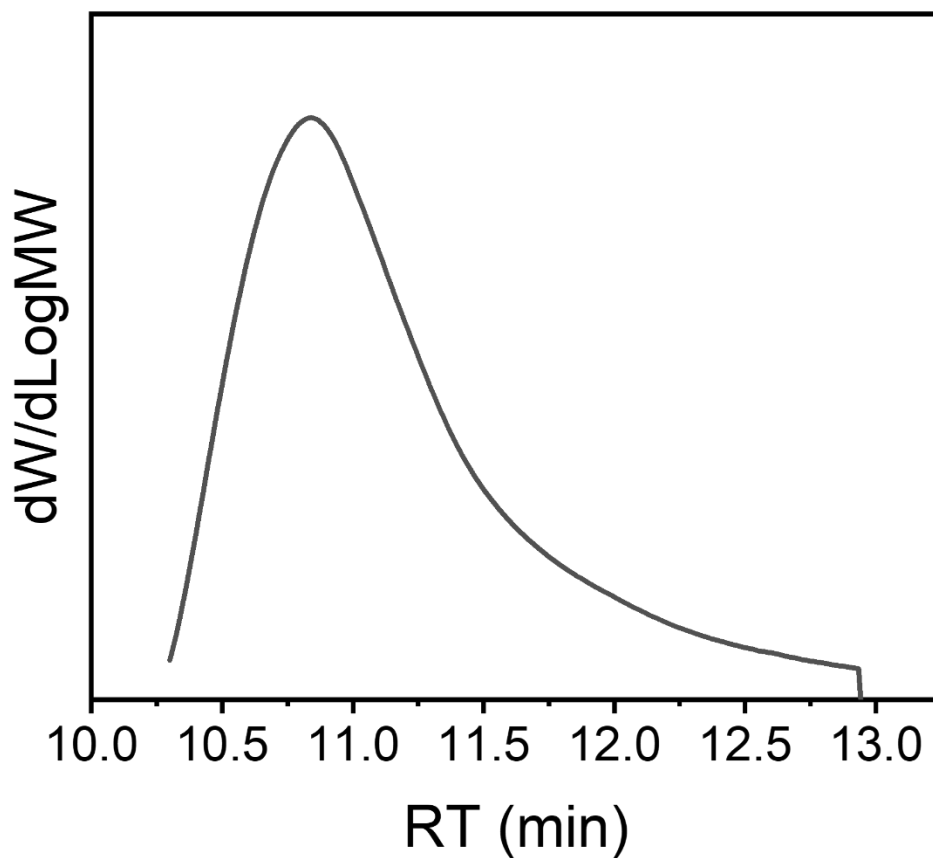


Figure S73.

GPC chromatogram of crude polymer from the CROP of EtLp with a proton trap. $[\text{EtLp}]_0 = 4.0$ M, 0.05 equiv. DPP, 0.05 equiv. 2,6-ditertbutylpyridine, acetone, 24 h. Molecular weight data listed in Table S13 entry 1.

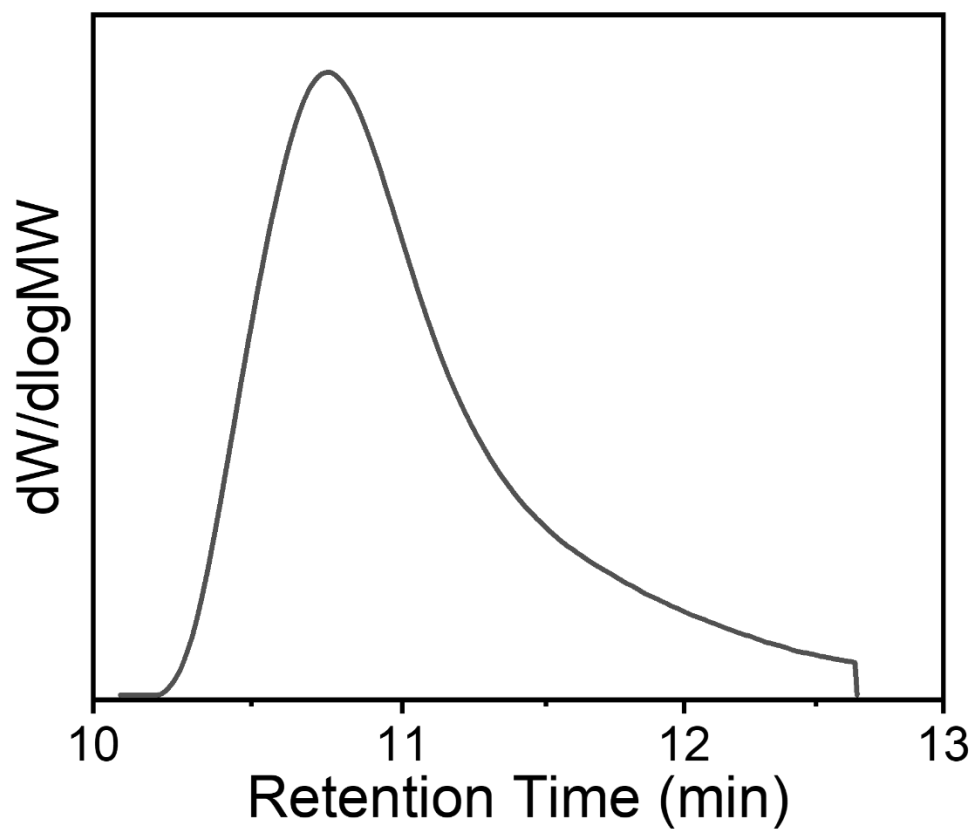


Figure S74.

GPC chromatogram of crude polymer from the CROP of EtLp with a proton trap. $[EtLp]_0 = 4.0$ M, 0.05 equiv. DPP, 0.1 equiv. 2,6-di-tertbutylpyridine, acetone, 24 h. Molecular weight data listed in Table S13 entry 2.

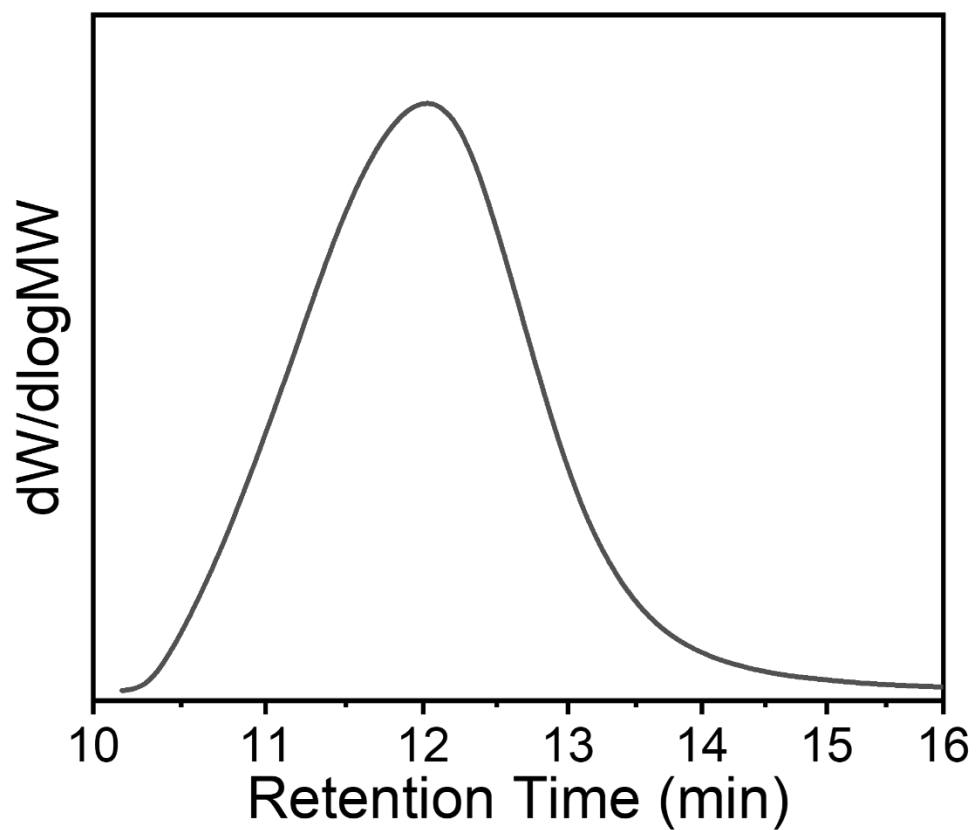


Figure S75.

GPC chromatogram of crude polymer from the CROP of EtLp with a proton trap. $[EtLp]_0 = 4.0$ M, 0.05 equiv. DPP, 0.05 equiv. dibenzyl disulfide, acetone, 24 h. Molecular weight data listed in Table S13 entry 3.

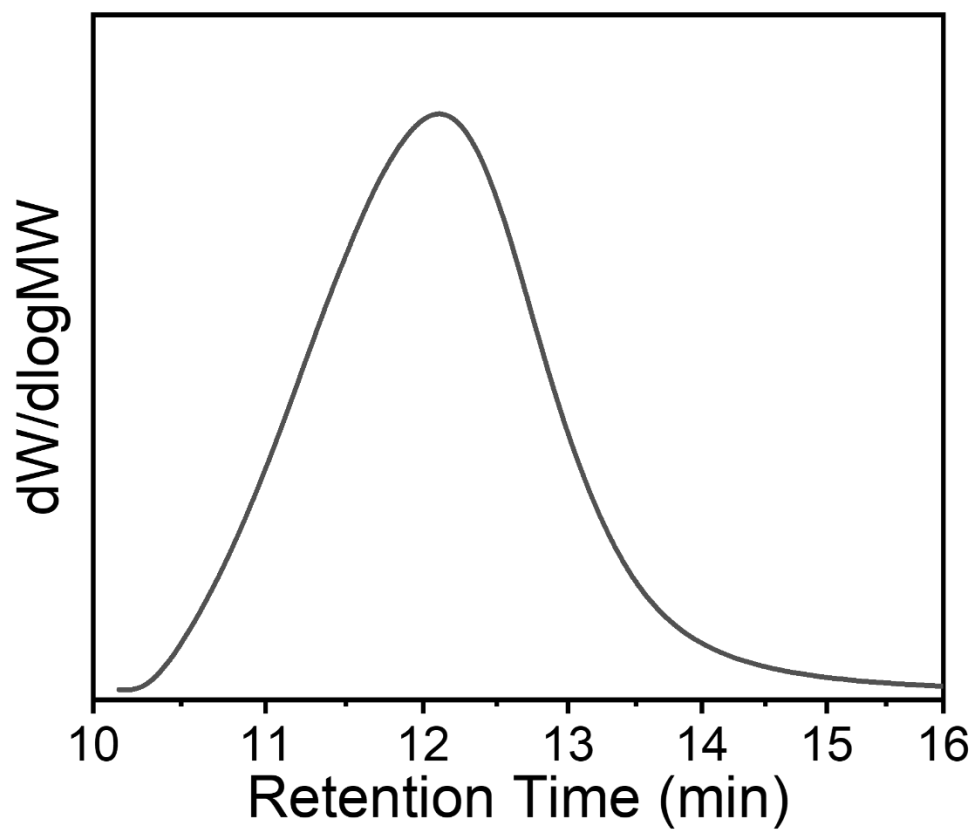


Figure S76.

GPC chromatogram of crude polymer from the CROP of EtLp with a proton trap. $[\text{EtLp}]_0 = 4.0$ M, 0.05 equiv. DPP, 0.1 equiv. dibenzyl disulfide, acetone, 24 h. Molecular weight data listed in Table S13 entry 4.

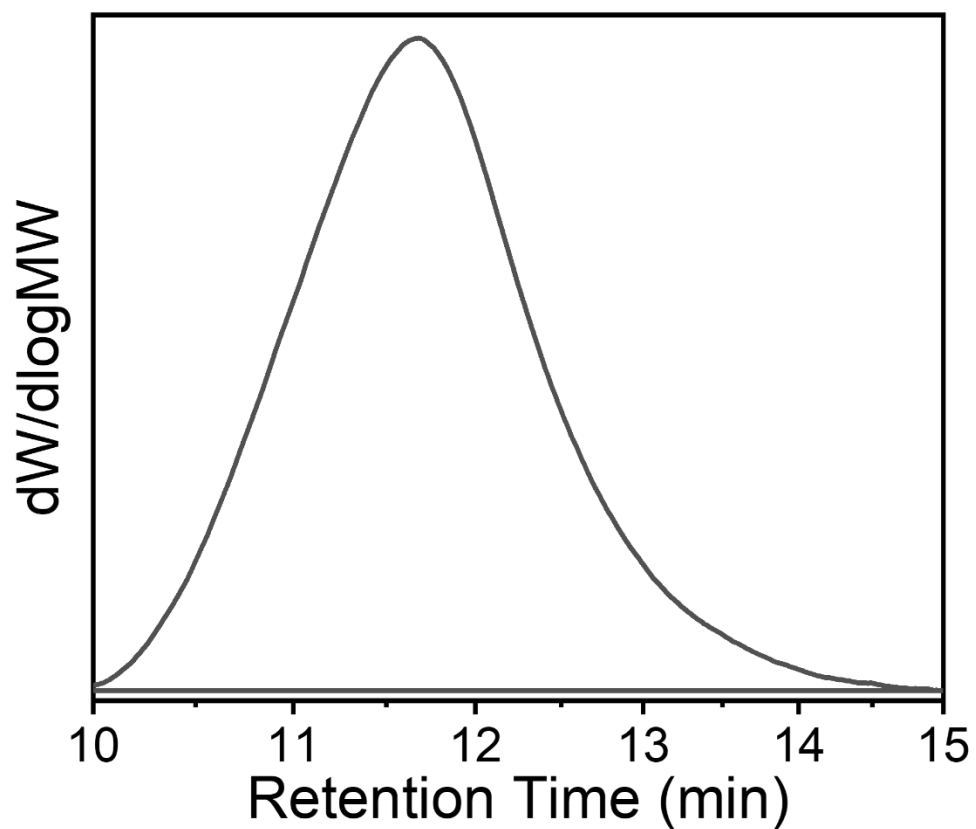


Figure S77.

GPC chromatogram of crude polymer from the CROP of EtLp with a proton trap. $[EtLp]_0 = 4.0$ M, 0.1 equiv. HCl, 0.01 equiv. diphenyl disulfide, DCM, 24 h. Molecular weight data listed in Table S13 entry 5.

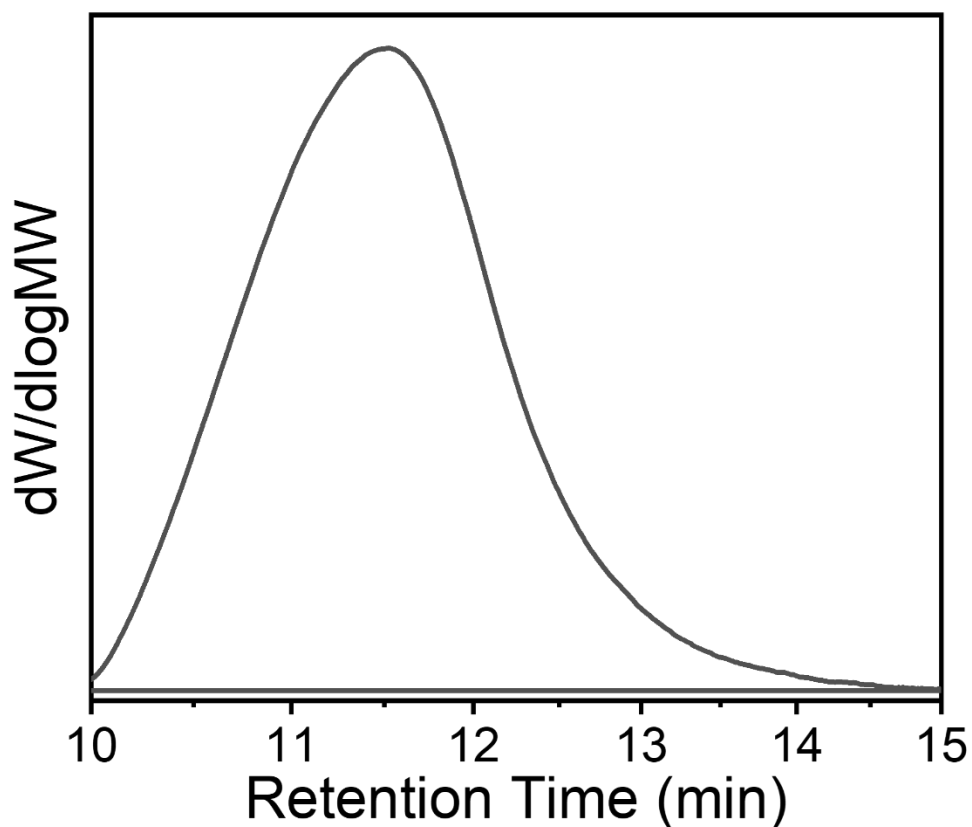


Figure S78.

GPC chromatogram of crude polymer from the CROP of EtLp with a proton trap. [EtLp]₀ = 4.0 M, 0.1 equiv. HCl, 0.01 equiv. diisopropyl disulfide, DCM, 24 h. Molecular weight data listed in Table S13 entry 6.

Table S13.

Summary of controlled CROP methods.

Entry	Additive	(mol%)	M_w (kDa)	M_n (kDa)	$\bar{D}$
1 ^a	1,6-ditertbutylpyridine	0.05	2720 ± 50	1800 ± 100	1.51 ± 0.08
2 ^a	1,6-ditertbutylpyridine	0.1	2990 ± 80	2200 ± 100	1.38 ± 0.06
3 ^a	Dibenzyl disulfide	0.05	1440 ± 90	570 ± 40	2.53 ± 0.01
4 ^a	Dibenzyl disulfide	0.1	1300 ± 100	500 ± 40	2.58 ± 0.03
5	Diphenyl disulfide	0.01	1500	750	2.0
6	Diisopropyl disulfide	0.01	1200	600	2.0

^a n = 3 for each experiment.

Photographs of reactions

Fischer Esterification

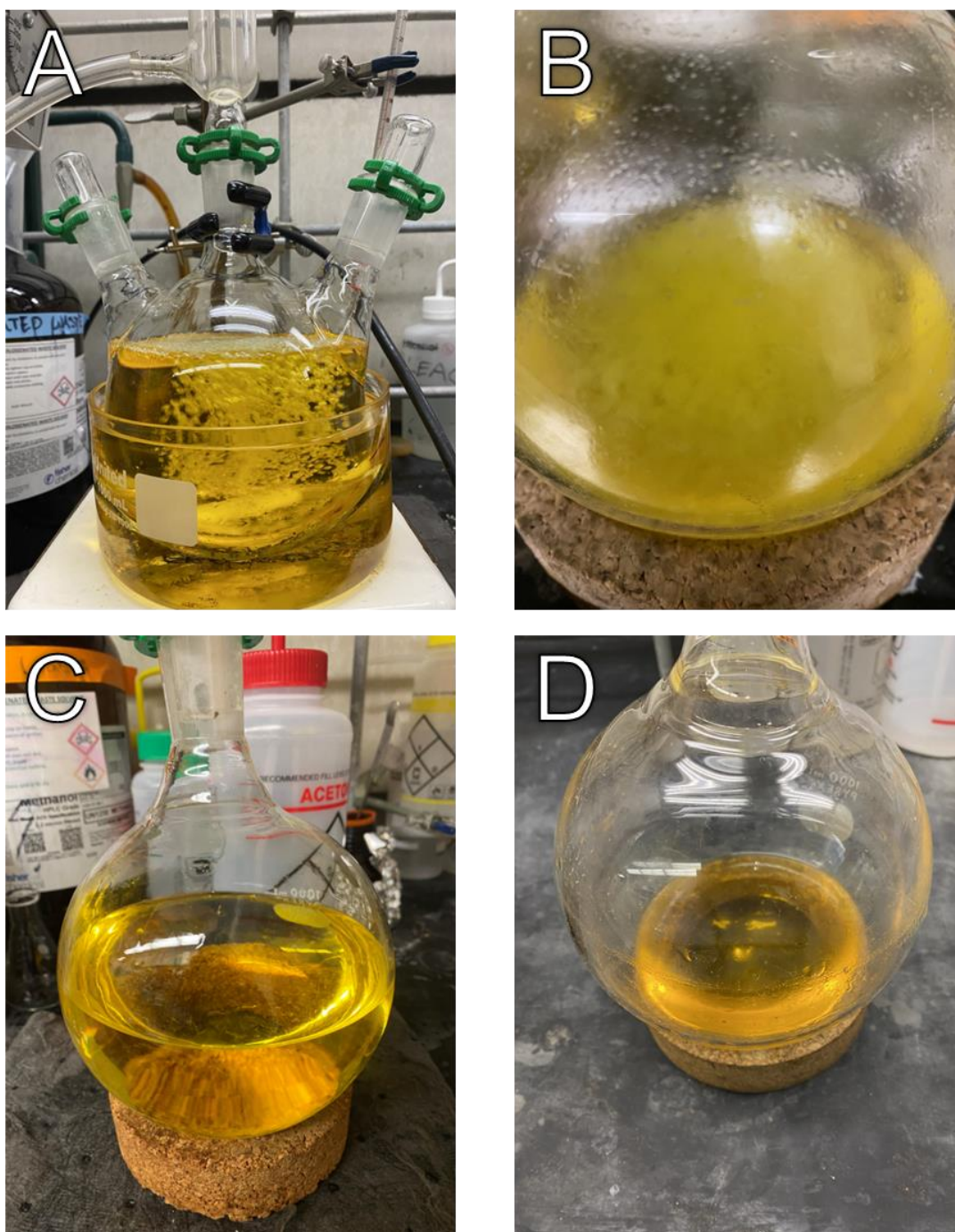


Figure S79.

Photos from the 100-gram scale Fischer esterification of lipoic acid to synthesize EtLp. (a) reaction mixture refluxing in EtOH. (b) Precipitation of salts after quenching and distillation of ethanol. (c) crude mixture in EtOAc after work-up. (d) crude EtLp before drying under ambient conditions.

CROP of EtLp at 50-gram scale using DPP as a catalyst

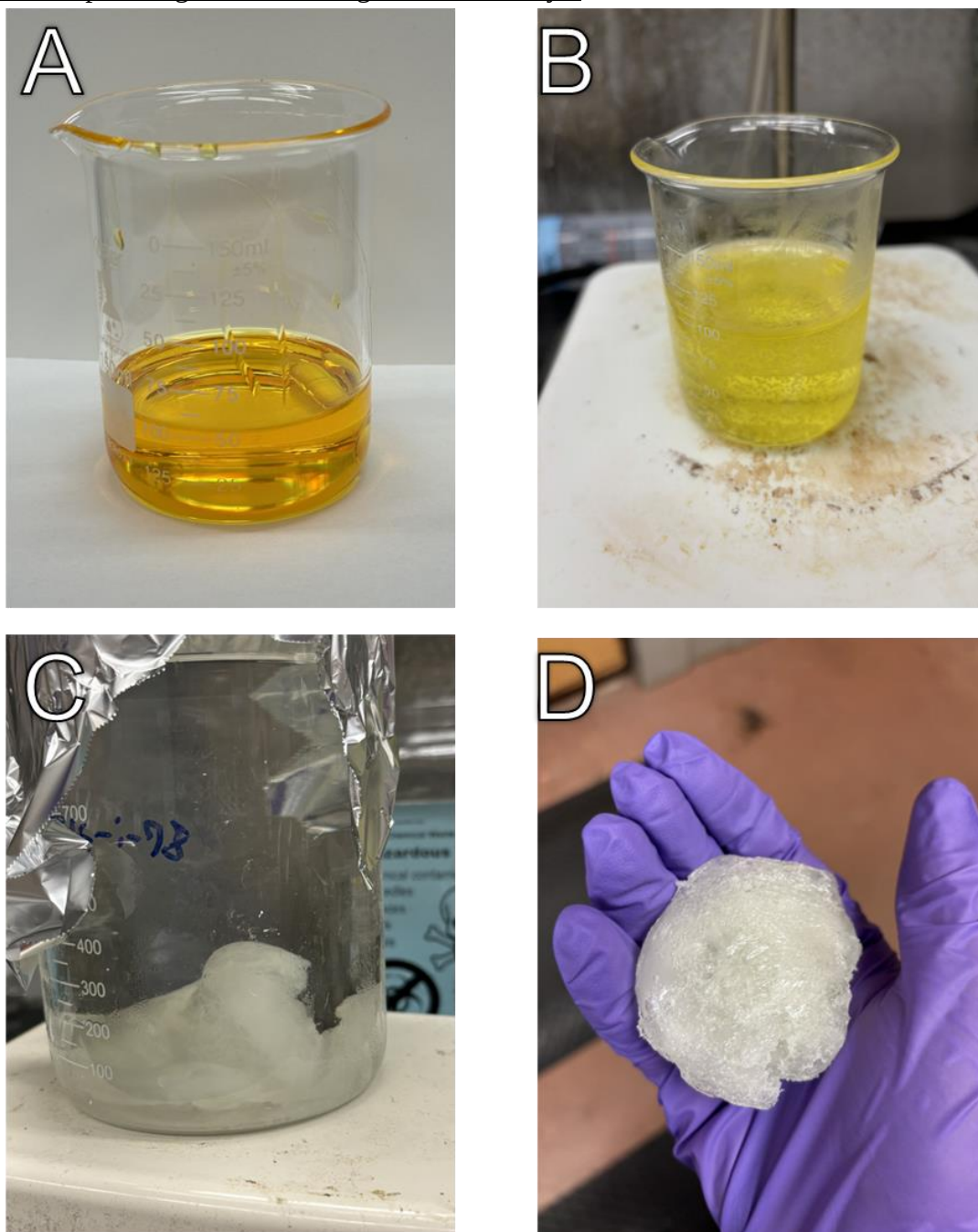


Figure S80.

Photos from the CROP of EtLp. (a) 50 g of EtLp. (b) polyEtLp gel after 24 h. (c) polymer gel being washed in diethyl ether. (d) pure polyEtLp after vacuum drying at 40 °C for 12 h.

Cationic RCD of polyEtLp

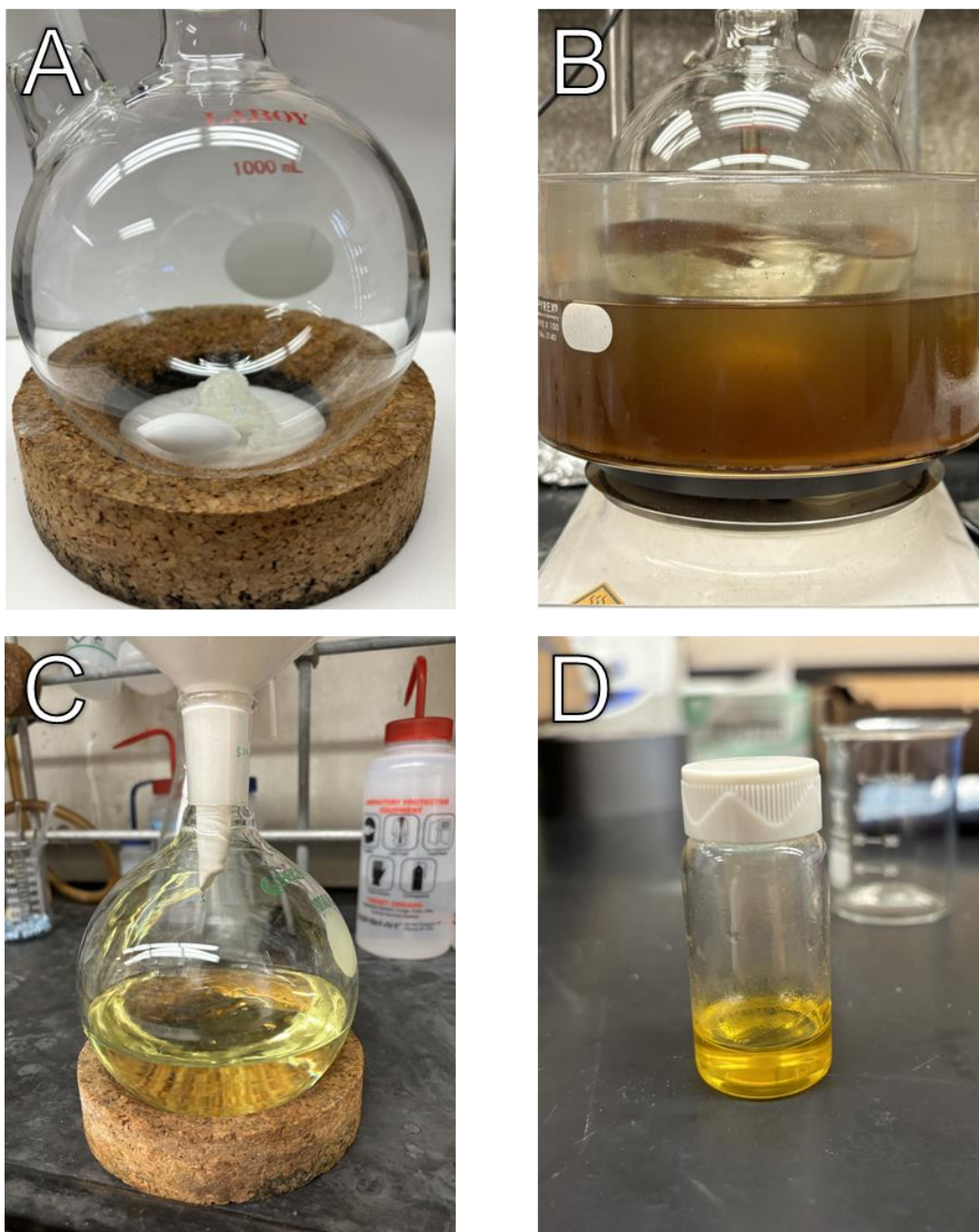


Figure S81.

Photos from the Cationic RCD of polyEtLp. (a) 5 g of polyEtLp. (b) polyEtLp being depolymerized in chloroform. (c) washed and dried chloroform EtLp mixture. (d) pure EtLp.

One-pot esterification-ROP from lipoic acid.

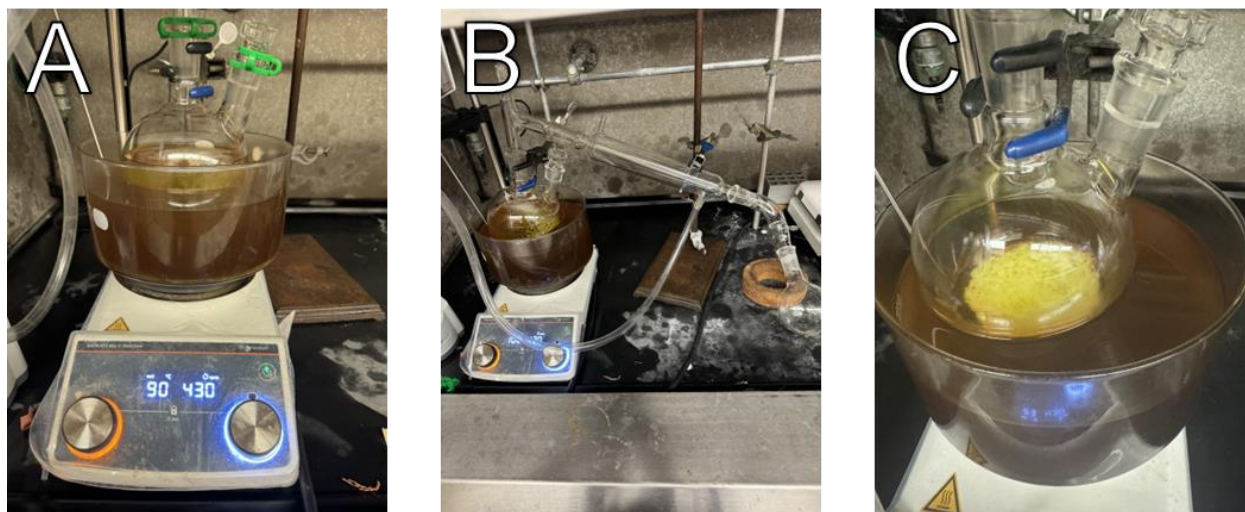


Figure S82.

Photos from the from one-pot esterification-ROP to yield polyEtLp from lipoic acid. (a) Fischer esterification of LpOH with ethanol. (b) the distillation setup for removing EtOH from EtLp. (c) the crude polyEtLp product.

ROP at reduced pressure using a rotary evaporator

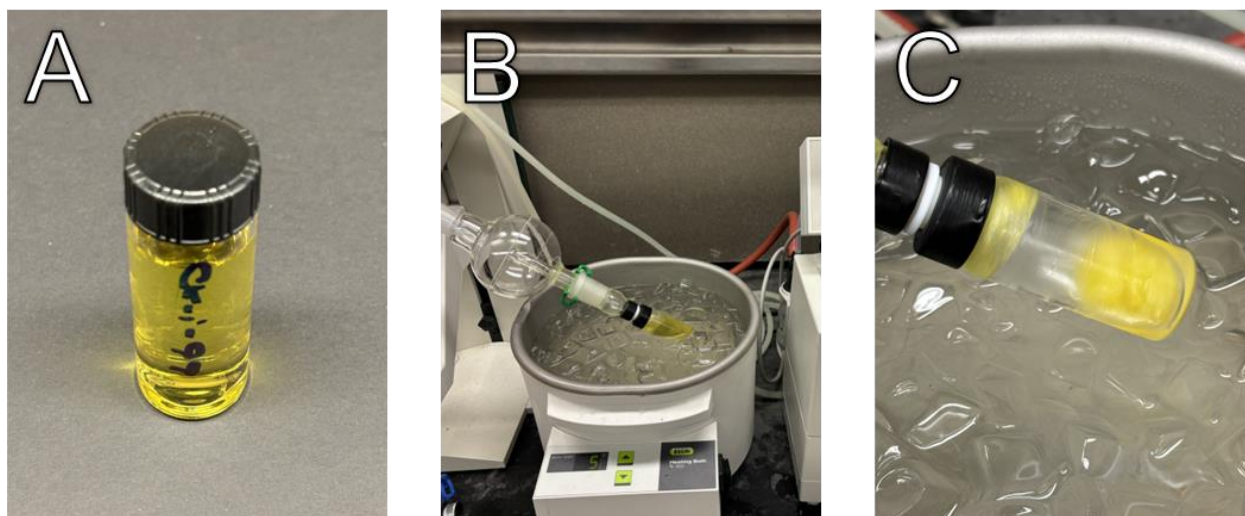


Figure S83.

Photos from the from ROP of EtLp at reduced using a rotary evaporator. (a) reaction mixture in acetone. (b) distillation setup for removing acetone from EtLp. (c) crude polyEtLp product.

One-pot RCD to ROP

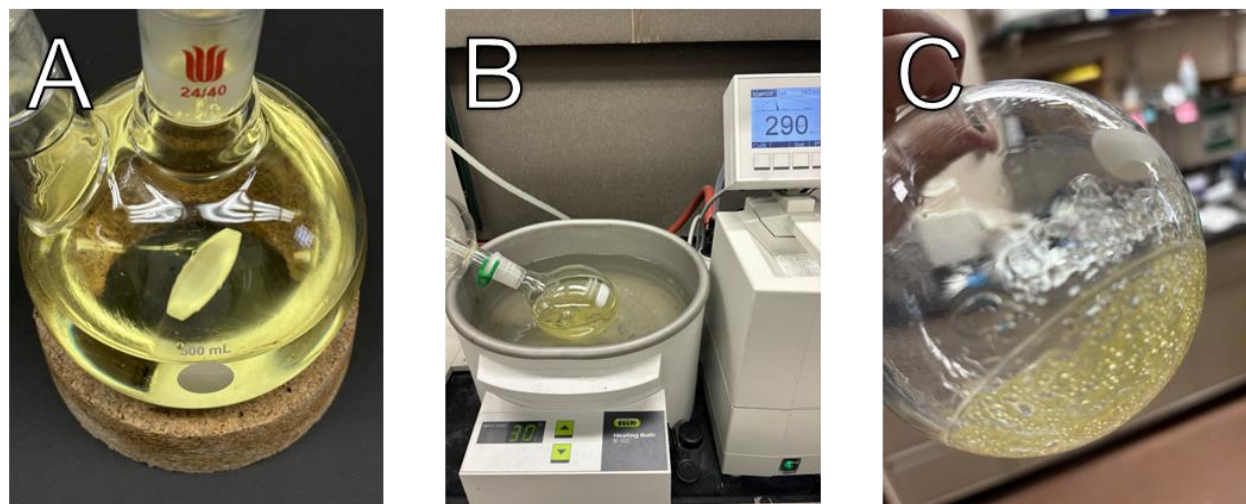


Figure S84.

Photos from the one-pot RCD of polyEtLp to ROP of EtLp. (a) recycled EtLp in chloroform from RCD of polyEtLp. (b) distillation setup for removing acetone from EtLp. (c) the crude polyEtLp product.

References

1. B. S. Wang, Q. Zhang, Z. Q. Wang, C. Y. Shi, X. Q. Gong, H. Tian and D. H. Qu, *Angew. Chem. Int. Ed.*, Acid-catalyzed Disulfide-mediated Reversible Polymerization for Recyclable Dynamic Covalent Materials, 2023, **62**, e202215329.
2. T. Du, B. Shen, J. Dai, M. Zhang, X. Chen, P. Yu and Y. Liu, *J. Am. Chem. Soc.*, Controlled and Regioselective Ring-Opening Polymerization for Poly(disulfide)s by Anion-Binding Catalysis, 2023, **145**, 27788-27799.